The "Depression Trap": Genetic Risk, Labor Supply, and the Case for Precision Psychiatry*

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Abstract

Why is depression linked to persistent non-employment, and how can mental health care be better targeted to improve mental health and employment outcomes? Using longitudinal UK panel data, we provide evidence for the "depression trap": a self-reinforcing cycle of poor mental health and non-employment, using event studies of depression recurrences and redundancies. Early treatment could disrupt this cycle, but it is challenging to predict who will enter the trap. We show that a Polygenic Index for depressive symptoms constructed using molecular genetics data is a uniquely powerful predictor of the depression trap and its persistence. We embed the depression trap mechanism in a structural life-cycle model of health, labor supply and treatment, and evaluate using genetics data to allocate treatment. We find that targeting therapy towards individuals with high genetic risk for depression improves the clinical efficacy of treatment by 29%. Furthermore, we estimate that by increasing employment through improved treatment allocation, the information provided by genetic risk generates a net fiscal surplus sufficient to finance the cost of universal genotyping.

Keywords: Mental Health, Depression, Polygenic Index, Genoeconomics, Labor Supply, Life-Cycle Model, Disability Benefits, Precision Psychiatry.
JEL Codes: D15, I14, I18, J22, J24

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1 Introduction

Mental health issues are prevalent and costly. In England, 22.6% of adults aged 16 to 64 have a common mental health condition such as anxiety or depression (Morris et al., 2025). Mental ill health is estimated to cost the UK economy about 5% of GDP, mainly through lost employment (McDaid and Park, 2022). The number of working-age adults in the UK out of the labor force because of long-term sickness is near its record high: 2.8 million people, or 6.4% of the working-age population, and over half of them report depression or anxiety (Office for National Statistics, 2026, 2023).

Why is depression associated with such persistent non-employment? One possibility is a self-reinforcing cycle: depression reduces employment, *and* non-employment in turns worsens depression. This could generate a "depression trap", a persistent state of depression and non-employment, and would suggest that early, targeted treatment would be particularly effective. As genetics data becomes increasingly predictive for health conditions such as depression, can we use molecular genetics data to predict who is at risk of entering the "depression trap"? And, as the National Health Service in England moves towards universal genomic testing of newborns and population-based polygenic risk scoring (Department of Health and Social Care, 2025), could genetics data be used to target limited therapeutic resources more effectively, improving mental health and employment outcomes?

This paper studies the joint dynamics of labor supply, mental health and treatment decisions of working-age men in the UK, and how they interact with genetic risk for depression.[1] Using the British Household Panel Survey and its successor, the UK Household Longitudinal Study (*Understanding Society*), we first use event studies of the recurrence of depressive episodes and redundancies (layoffs) to provide suggestive evidence for the depression trap, a self-reinforcing relationship between non-employment and depression. Second, we link our panel data to molecular genetic data and show that a Polygenic Index for depressive symptoms is a powerful predictor of the depression trap and its persistence. Third, we develop and estimate a structural life-cycle model of labor supply and mental health treatment that replicates these empirical patterns and evaluate using genetics data to prioritize treatment allocation. We find that prioritizing treatment towards men with high genetic risk for depression helps them to avoid the depression trap, increasing the clinical efficacy of treatment by 29% and generating a fiscal return sufficient to finance the cost of universal genotyping.

In our empirical analysis, we first provide suggestive evidence, consistent with the literature, that depression reduces employment and non-employment in turn, worsens mental health. We identify the relationship between depression and employment using event studies of two quasi-random shocks: the recurrence of depressive episodes and redundancies (layoffs). We find that

[1] We defer studying women to subsequent work as genetic risk for depression predicts women's completed fertility, which may interact with mental health and employment outcomes (Appendix C).

the recurrence of depressive are associated with a reduction in employment of 2.7 percentage points, broadly consistent with evidence from RCTs of psychotherapy and longitudinal studies (Smith et al., 2025; Oparina et al., 2024; Bryan et al., 2022). We also find that involuntary redundancies are associated with an increase in depressive symptoms of 0.2 s.d., consistent with prior evidence from plant closures, longitudinal analyses and RCTs (Marcus, 2013; Picchio and Ubaldi, 2024b; Sterud et al., 2025; Hussam et al., 2022). This suggests that temporary mental health shocks can develop into a persistent, self-reinforcing cycle of poor mental health and non-employment, and that early treatment targeted towards individuals at high risk of entering the trap could be particularly effective.

Second, we show that genetic risk for depression predicts both who enters the trap and how long they stay. We measure genetic risk by constructing a Polygenic Index (PGI) for depressive symptoms, a weighted average of one's genotype, using weights estimated in an external 'discovery' sample (Baselmans et al., 2019). We find that a one s.d. increase in the PGI for depressive symptoms is associated with a 6.9 percentage point increase in the probability of depression and a 3 percentage point reduction in employment.[2] We find that men with above median genetic risk for depression experience significantly larger drops in employment following redundancies and depression recurrences, and they are 70 percent more likely to enter the "depression trap", the joint state of depression and non-employment and 40 percent less likely to leave it in a given year.[3] Timely treatment, particularly for high risk individuals, could therefore prevent transitory shocks from developing into a persistent 'depression trap'.

Third, we develop and estimate a structural life-cycle model that embeds the "depression trap" mechanism, to evaluate a precision psychiatry framework that uses genetics data to prioritize access to treatment. Agents choose consumption, saving, labor supply and whether to apply for therapy. Poor mental health lowers the wage offer, adds a time cost to working and raises benefits out of work.[4] Non-employment in turn worsens future health and erodes human capital. Therapy improves mental health and is easier to take up out of work, but it is rationed, so an application does not guarantee treatment. Genetic risk shifts wage offers and health transitions. We estimate the model by the method of simulated moments. To identify the bidirectional relationship between health and work, we target employment rates by health status and health transitions by employment status, and we reproduce both event studies as untargeted moments. Randomizes control trial evidence pins down the effect of therapy. It delivers three main findings.

First, the depression trap has a scarring effect on employment which persists after the depression itself has passed and that is stronger for high-risk men. We place every agent in the

[2]Causal designs comparing siblings deliver similar point estimates but wide confidence intervals; estimates are robust to controls for socioeconomic status and PGIs of physical health and educational attainment (Appendix E).

[3]Over short time horizons, depressive symptoms are more predictive for first entry into the trap, but genetic risk for depression outperforms depressive symptoms over longer time horizons, and generally outperforms socioeconomic indicators.

[4]We calibrate in-work health benefits from our panel; receipt is concentrated among the non-employed.

trap, simulate the next ten years and compare the outcomes with each agent's baseline path. Depression stays elevated for several years, particularly at high genetic risk, but returns to baseline after four to five years. Non-employment does not: after ten years it is still 15 percent above baseline. As a result, agents are 16 percent more likely to be in the trap at the end of the decade, and high-risk men 25 percent more likely. This is because human capital lost out of work lowers the wage offers that would end the spell, and being out of work raises the risk of relapse. Treatment that prevents entry into the trap could therefore have lasting effects.

Second, giving high-risk men priority for therapy improves their mental health and employment without adding to the supply of therapy. The NHS offers psychotherapy free at the point of use, but applicants face uncertain access because supply is limited.[5] We raise the probability of access for high-risk applicants while holding the total number of courses fixed. In the model, therapy is no more effective for high-risk men.[6] Shorter waits nevertheless help them avoid the trap, so each course averts 29 percent more years of severe depression. Higher income tax revenue and lower benefit payments improve the public finances by £335 per course reallocated.

Third, genotyping and a larger therapy workforce both pay for themselves. Without genetic data, a planner could target therapy on observed histories of employment, earnings, health and treatment, but these predict a man's genetic risk only weakly (an area under the ROC curve of 0.65[7]). Targeting on genetic risk instead raises the fiscal return by about £18 per person, more than the cost of genotyping the working-age male population.[8] However, the recent rise in mental ill health calls for more than reallocation.[9] We estimate that this rise will reduce net fiscal contributions by £804 per person. A 10 percent increase in the therapist workforce, as proposed by the UK government,[10] is self-financing: each additional therapist raises income tax revenue by £226,000 and cuts benefit payments by £146,000, more than covering their training and wages.

This paper contributes to three literatures. First, we introduce genetics to the empirical literature on mental health and work, which finds that poor mental health reduces employment (Ettner et al., 1997; Frijters et al., 2014; Peng et al., 2016; Biasi et al., 2021) and that job loss worsens mental health (Paul and Moser, 2009; Kuhn et al., 2009; Marcus, 2013; Sterud et al., 2025); Picchio and Ubaldi (2024a) review this evidence. Genoeconomics has mostly studied the polygenic index for educational attainment (Papageorge and Thom, 2020; Barth et al., 2020; Bolt et al., 2025). We show that a polygenic index for depression predicts who enters the depression

[5] In 2023–24, NHS Talking Therapies received 1.83 million referrals and 1.26 million people entered treatment (NHS England, 2024).

[6] Depression polygenic scores do not predict response to cognitive behavioral therapy (Bäckman et al., 2025).

[7] Given the histories of one randomly chosen high-risk and one randomly chosen low-risk individual, the classifier ranks the high-risk individual higher 65% of the time; 50% is chance.

[8] SNP-array genotyping costs about $20 per person (Raben et al., 2022).

[9] In our data, severe depression among working-age adults rose from 14.0 to 17.3 percent after the pandemic, and from 9.1 to 22.2 percent at ages 25–29 (Section 7).

[10] The NHS Long Term Workforce Plan for England commits over £600 million over three years to train about 15,000 more psychological therapists and practitioners (NHS England, 2023).

trap, how long they stay and how much employment they lose after the same shock. To the best of our knowledge, we are the first to link such an index to labor market dynamics and to embed it in a structural model.

Second, we join two lines of structural work on mental health and labor supply. In one, mental-health shocks propagate through schooling, occupation and employment (Jolivet and Postel-Vinay, 2025; Diaz-Campo et al., 2025; Drozd, 2026). In the other, treatment is a choice, but non-employment does not itself worsen mental health (Cronin et al., 2024; Abramson et al., 2024; Sauer et al., 2025)[11]. Our model has both: shocks persist because depression and non-employment reinforce each other, and therapy is chosen but rationed. This makes the timing of treatment matter: therapy that arrives early can stop a temporary episode from becoming a persistent spell of depression and non-employment.

Third, we contribute to the public economics of targeting and rationing in health care (Akerlof, 1978; Logue and Slemrod, 2008; Kleinberg et al., 2015; Gravelle and Siciliani, 2009). Prioritizing patients in a health-care queue need not pay (Aakvik et al., 2015), although longer waits raise disability receipt (Godøy et al., 2024) and psychotherapy can pay for itself (Serena, 2025). Because our model includes the institutions that determine the fiscal return to treatment (health-related benefits, income taxes and free but rationed therapy), we provide what is, to the best of our knowledge, the first structural evaluation of the fiscal return to allocating rationed therapy by genetic risk and to expanding its supply.

The paper is organized as follows. Section 2 discusses how this paper relates to the existing literature in more detail. Section 3 presents the data and our reduced-form evidence. Section 4 presents our model and Section 5 explains our multi-step identification and estimation procedure. Section 6 presents the counterfactual and policy analyses, Section 7 simulates the post-pandemic decline in mental health and evaluates expanding mental health care in response, and Section 8 concludes. Various robustness checks and additional results are presented in Appendices A–H.

2 Related Literature

This paper bridges the gap between the genetic psychiatry and structural labor and health literatures by incorporating genetics, health, labor supply and endogenous treatment in a rich life-cycle labor supply model.

This paper is most closely related to the burgeoning literature incorporating mental health into structural models of labor supply and treatment. One strand studies how mental health shocks propagate through labor supply and human capital decisions. Jolivet and Postel-Vinay

[11] In Cronin et al. (2024), mental-health transitions depend on treatment but not on employment, which instead raises the time cost of treatment. In Abramson et al. (2024), labour-market outcomes affect mental health only through exogenous productivity shocks and the resources available for treatment. In Sauer et al. (2025), work is harmful to mental health, so the employment response to a shock dampens it.

(2025) show how employment in stressful occupations feeds back into future mental health, while Diaz-Campo et al. (2025) and Drozd (2026) provide evidence that endogenous schooling choices amplify the long-run consequences of early mental health shocks. A second strand endogenizes mental health treatment. Cronin et al. (2024) explain why psychotherapy is underused relative to antidepressants despite its larger treatment effects, Cozzi et al. (2024) study psychotherapy alongside smoking in a model of women's employment, marriage and fertility decisions, and Abramson et al. (2024) show how the pessimism associated with poor mental health depresses investment in risky assets and sorting into demanding occupations. As well as introducing genetics data to this literature, we bridge the two strands by endogenizing treatment in a model that features rich propagation of mental health shocks through dynamic feedback effects between work and mental health. We also improve upon the identification of the two-way relationship between work and health using event study approaches, which our model reproduces as untargeted moments. Furthermore, we incorporate institutional features of the UK that allow us to assess the public-finance case for treatment: health-dependent benefits, income taxation, and publicly funded but supply-constrained psychotherapy. This allows us to estimate the fiscal returns to mental health treatment and ask whether publicly funded genotyping and an expansion of therapy could pay for themselves.

This paper also relates to the larger literature on physical health and labor supply. A branch of the literature focuses on the effect of physical health and retirement decisions (French, 2005; French and Jones, 2011) or on long-term earnings and welfare (Hosseini et al., 2021; De Nardi et al., 2024). This literature generally does not consider the feedback effects of employment on health, or consider mental health dynamics. Exceptions include Salvati (2025), who incorporates a bidirectional relationship between work and health for women near retirement, and Sauer et al. (2025) and Diaz-Campo et al. (2025) who jointly physical and mental health dynamics, as well as employment feedback effects for women and men, respectively. We take a similar approach to Diaz-Campo et al. (2025) in jointly modeling physical and mental health, in addition to treatment. In line with evidence from RCTs, exogenous plant closures and longitudinal analyses (Picchio and Ubaldi, 2024b; Hussam et al., 2022; Sterud et al., 2025), we allow employment to affect physical health in addition to mental health. Furthermore, we provide evidence that genetic variants linked to depression also play a role in explaining the association between poor physical health and non-employment.

We also contribute to the 'Genoeconomics' literature, which links molecular genetic data to economic surveys to study the interaction between genetic and environmental effects. We show how genetic risk for depression predicts economic outcomes and demonstrate how it reinforces health and economic shocks. Biroli et al. (2022) provides a useful introduction to this literature. The literature relies on Genome-Wide Association Studies (GWAS) to estimate the association between genetic variants and a trait of interest in a large population (Uffelmann et al., 2021). These estimated associations can then be used to generate individual-level predictors of a trait

of interest, called Polygenic Scores (PGS) or Indices (PGI). Thus far, this literature has almost entirely focussed on educational attainment (EA) as a trait of interest (Okbay et al., 2022), which has been shown to be a strong predictor of asset accumulation (Barth et al., 2020), labor market outcomes (Papageorge and Thom, 2020), as well as parental investment and skill formation (Bolt et al., 2025). There have been limited attempts to incorporate genetics in structural models, and PGIs other than EA have thus far received very little attention in economics. Barth et al. (2022) incorporate the EA PGI in a consumption-savings model of asset accumulation. Houmark et al. (2024), Bolt et al. (2025) and Agostinelli and Weingarten (2025) model how genetics affect parental investment and child skill formation, with the latter incorporating genetic predictors of ADHD. Jeong et al. (2024) use reduced-form approaches to explore a genetic predictor of Alzheimer's, while De Nardi et al. (2024) briefly discusses how PGIs for educational attainment, BMI, smoking and longevity predict physical health. While Amin et al. (2019) explore gene-environment associations for a depressive symptoms PGI with education as the outcome, ours is the first study to link it to labor market outcomes, mental health treatment and dynamics, and to explore the role of targeted interventions.

This paper also bridges the gap between labor economics and precision psychiatry literatures. We demonstrate that using genomic information to decide the order in which patients are treated improves both clinical and employment outcomes, and that the resulting labor supply response and reduced benefit expenditure make the intervention fiscally self-sustaining. Where capacity rather than price is the binding constraint, it requires no expenditure at all. While there is a large, long-established body of evidence for the effectiveness of psychotherapy (Baranov et al., 2020; Cuijpers et al., 2014), recent research has highlighted how PGIs can be used to identify high-risk patients and eventually develop personalized interventions (Torkamani et al., 2018). To date, there remains little evidence of how treatments could be *adapted* depending on an individual's genetics, but a recent RCT suggests that the treatment effect of a Cognitive Behavioural Therapy intervention for depressed patients does not appear to vary depending on their depression PGI (Bäckman et al., 2025). This suggests that high-risk individuals are not 'predetermined' to be depressed, but that treatments can be effective. The take-up of psychotherapy, however, remains low. Cronin et al. (2024) consider various mechanisms and conclude that the under-utilization of psychotherapy is not primarily due to financial costs or time constraints, but likely relates to stigma or difficulties opening up to a stranger. To overcome these constraints Breza et al. (2026) conduct an RCT in which they offer financial incentives to attend therapy, which increases take-up only among symptomatic patients, consistent with the broader evidence that financial incentives can increase mental health treatment engagement (Khazanov et al., 2022). Where therapy is already free at the point of use, however, price is not the margin that binds, and our policy exercise therefore works on the one that does, which is the order of the queue.

3 Empirical Insights

This section establishes four facts that motivate the structural model of Section 4. First, genetic risk for depression carries a substantial earnings penalty that particularly operates through withdrawal from the labour market into disability. Second, the penalty is dynamic: men at high genetic risk are more likely to develop depression and to leave employment. They are less likely to recover from either, and particularly struggle to leave the joint state of non-employment and depression, which we term the "depression trap". Third, each leg of that trap is causal. We identify the effect of employment on mental health from involuntary job loss, and the effect of mental health on employment from the recurrence of a depressive episode. Fourth, genetic risk exacerbates the effects of shocks to mental health and employment: individual with high risk for depression experience larger declines in mental health following redundancy and larger declines in employment following the recurrence of depression. These facts motivate our structural model, which aims to reproduce the dynamics of the depression trap and explore policy solutions.

3.1 Data and Measurement

We use the UK Household Longitudinal Study and its predecessor the British Household Panel Survey, spanning 1991–2023, which link molecular genetic data to an annual panel of health and labor market histories. Mental health is measured by the "caseness" scoring of the GHQ-12, a well-validated screen for depression (Wojujutari et al., 2024); following the psychiatric literature we use thresholds of two and four symptoms to denote mild depressive symptoms and probable major depressive disorder (Anjara et al., 2020). Of 9,220 genotyped individuals, our reduced-form sample is the roughly 2,000 men aged 25–55; model estimation extends the age range to 85. Women are excluded throughout, and not for want of data: they contribute more genotyped person-waves than men on every margin. Appendix B reports what each exercise below gives for women, in the subsection corresponding to each, and Appendix C sets out the reasons for the restriction, which turn on fertility.

3.2 Genetics Data

To measure genetic predisposition to depressive symptoms, we use a polygenic index (PGI). A PGI is an individual-level predictor constructed from genotype data. It can be interpreted as a weighted sum of genetic variants. Let $SNP_{i,j} \in \{0, 1, 2\}$ denote the number of effect alleles carried by individual i at variant j, where the allele may increase or decrease depression risk. The PGI aggregates many such variants with small individual effects into a single summary measure:

$$PGI_i = \sum_{j=1}^{J} \beta_j^{GWAS} SNP_{i,j}. \tag{1}$$

The weights β_j^{GWAS} are estimated in an external GWAS discovery sample that is separate from our UKHLS target sample. In the GWAS, the association between each variant j and the trait of interest is estimated using regressions of the form

$$Y_k = \beta_j^{GWAS} SNP_{k,j} + \gamma \mathbf{X}_k + u_{k,j}, \tag{2}$$

where k indexes individuals in the discovery sample, Y_k is depressive symptoms, and $\mathbf{X}_k$ includes standard GWAS controls (sex, age polynomials, and genetic principal components) to adjust for population structure.

In our main analysis, we use weights from Baselmans et al. (2019), which studies depressive symptoms. The GWAS discovery sample does not overlap with the UKHLS target sample. As additional controls in some specifications, we also use PGIs constructed from GWAS of educational attainment (Okbay et al., 2022) and self-reported health (Harris et al., 2017). Both the discovery and target samples are restricted to individuals of European-like White ancestry. We impose this restriction because PGIs do not transport well across ancestry groups and because non-European GWAS sample sizes remain more limited. See Price et al. (2006) for discussion of population structure and related concerns. Details on the genetic data and PGI measurement are provided in Appendix A.

Men above the median of genetic risk are worse off on every health and labor market margin (Appendix Table 14), but they do not differ in parental background: neither paternal occupation ($p = 0.31$) nor maternal education ($p = 0.26$) differs across the groups, so the index does not appear to proxy dynastic advantage. Appendix E shows that controlling for parental background and for the educational-attainment and physical-health polygenic indices, and estimating family fixed-effects models on linked relatives, leaves the estimates stable, as Okbay et al. (2022) find for depressive symptoms in a much larger sample.

3.3 The Genetic Penalty

To identify the relationship between genetic risk for depression and subsequent labor market outcomes, we estimate the following specification for individual i at time t:

$$Y_{i,t} = \alpha + \beta G_i + \mathbf{X}'_{i,t}\gamma + \mathbf{M}'_{i,t}\theta + \epsilon_{i,t}, \tag{3}$$

where Y_i is standardized gross earnings (with earnings coded as 0 for the non-employed), G_i is the standardized depression polygenic index and $\mathbf{X}_{i,t}$ collects the ten genetic principal components with age, wave and interview-year fixed effects; standard errors are clustered on the individual. To locate the penalty in the life cycle we add mediators $\mathbf{M}_{i,t}$ in four chronological blocks: educational attainment, first occupation, contemporaneous labor force status, and current occupation. We report the penalty per standard deviation of the index; Appendix B

reports the median-split version, which gives the same picture in units directly comparable to the descriptive and transition tables.

Table 1 reports the ladder. A one standard deviation increase in genetic risk lowers gross labor income by 0.123 standard deviations (s.e. 0.022). Education absorbs 19 percent of that association and first occupation a further 6 percent, consistent with genetic risk impairing human capital accumulation early in life. The largest single block, however, is contemporaneous labor force status, which absorbs 27 percent on its own, more than education and first occupation combined. The penalty is therefore not primarily about what high-risk men are paid, but about whether they are working at all.

Table 1: The Genetic Earnings Penalty and Its Channels

Mediators included	Effect on labor income (s.d.)	Share of baseline (%)	Explained by this block (%)
Baseline	-0.1229***	100.0	–
	(0.0222)		
+ Education	-0.0992***	80.7	19.3
	(0.0201)		
+ First occupation	-0.0921***	75.0	5.8
	(0.0196)		
+ Labor force status	-0.0584***	47.5	27.4
	(0.0185)		
+ Current occupation	-0.0420***	34.2	13.3
	(0.0163)		

Notes: Each row is the coefficient on genetic risk in a regression of standardized gross labor income, adding one block of mediators at a time. The index is continuous and standardised, oriented so that higher values denote greater liability, so each coefficient is the effect of a one standard deviation increase in genetic risk. Appendix B reports the continuous per-standard-deviation version. All specifications control for the first ten genetic principal components and for age, survey-wave and interview-year fixed effects. Column (2) expresses each coefficient as a percentage of the baseline, column (3) the share absorbed by the block added in that row. Contemporaneous mental health is deliberately excluded: the blocks above are predetermined relative to current earnings whereas mental health is a contemporaneous outcome of genetic liability, so conditioning on it here would net out part of the effect being decomposed; its share is reported in Table 2. Standard errors clustered on the individual. $N = 19{,}995$ person-waves. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 2: Which Margin Carries the Genetic Penalty, and How Much Runs Through Mental Health

Outcome	Effect of +1 s.d. of genetic risk	Share explained by mental health (%)	95% CI
Depressed (pp)	6.908***	–	
	(0.530)		
Employed (pp)	-3.098***	48	[33, 75]
	(0.525)		
Long-term sick or disabled (pp)	1.900***	53	[40, 81]
	(0.375)		
Labor income (s.d.)	-0.123***	23	[15, 32]
	(0.022)		
Log gross pay, if employed (log pts.)	-0.054***	7	[0, 15]
	(0.013)		
Holds a degree (pp)	-2.785**	≈ 0	
	(1.119)		

Notes: Column (1) is the effect of genetic risk on the outcome named,in the units given in the row label. The index is continuous and standardised, oriented so that higher values denote greater liability, so each coefficient is the effect of a one standard deviation increase in genetic risk. Column (2) is the percentage of that gap absorbed when contemporaneous GHQ-12 distress is added, with a cluster-bootstrap interval in column (3); ≈ 0 denotes no attenuation. All specifications control for the ten genetic principal components and for age, survey-wave and interview-year fixed effects, with standard errors clustered on the individual. The depression row has no share because it is the mediator itself. Appendix B reports every outcome and the continuous per-standard-deviation version. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 2 takes each margin as a separate outcome, and the first row fixes the scale of everything below it: a one standard deviation increase in genetic risk raises the probability of probable major depression by 6.9 percentage points (s.e. 0.5). It lowers the probability of employment by 3.1 points (s.e. 0.5), and that withdrawal is concentrated in long-term sickness and disability, at 1.9 points (s.e. 0.4). Most of the employment gap is accounted for by this one destination: these men leave the labor force through ill health rather than search for work. Conditional on working, gross pay falls by only 0.054 log points (s.e. 0.013), and the degree gap of 2.8 points (s.e. 1.1) is smaller in relative terms.

The third column reports how much of each association operates through contemporaneous mental health, and the pattern is orderly. About half of the labor-supply effects run through it: 48 percent for employment and 53 percent for long-term sickness. A fifth of the income effect does, 7 percent of the intensive margin, and essentially none of the degree effect, which is what one would expect of an outcome settled before the distress we observe. This is the pathway the structural model is built around: genetic predisposition manifests as poor mental health, which triggers withdrawal from the labor force through disability rather than through unemployment or lower wages.

The penalty is not specific to men. Estimated on women, it is if anything larger on the extensive margin, at 4.0 percentage points of employment against 2.6 for men on a common sample, with comparable mediation through mental health. The difference is where the withdrawal goes: women also move into caring for home and family, a state men essentially do not occupy,

which accounts for about thirty percent of their employment penalty. Appendix B.4 reports the full comparison, and Appendix C explains why that channel, and the fertility behind it, is what keeps women out of the structural model.

3.4 The Depression Trap

The relationship between employment and mental health runs in both directions. Meta-analyses of studies exploiting exogenous variation in employment find that employment improves men's mental health by about 0.1 standard deviations (controlling for income) (Picchio and Ubaldi, 2024b), and RCT evidence puts the non-pecuniary benefit at 0.23 s.d. (Hussam et al., 2022). In the other direction, Frijters et al. (2014) instrument mental health with the death of a close friend and find that a one standard deviation deterioration reduces employment by 30 percentage points, while panel estimates put the effect of disorder onset on exit from employment at 7–18 points (Mitra and Jones, 2017). Appendix D sets out the clinical channels the structural model incorporates: cognitive impairment that lowers the wage offer, fatigue and anhedonia that raise the disutility of effort, and physical-health spillovers. Both legs appear in our own panel, conditional on permanent heterogeneity and past income (Appendix Table 15), but they are associations, which is why Section 3.5 identifies each leg from a shock instead.

A relationship that runs in both directions admits a self-reinforcing state: out of work and depressed, with each condition sustaining the other. The question this paper asks is whether genetic predisposition governs how easily men enter that state and how long they remain. Table 3 addresses all three sets of transitions at once, splitting the genotyped sample at the median of the polygenic index, estimating the transition model separately within each group, and averaging both fits over the same pooled covariate distribution, so that column (3) reflects a difference in behavior rather than in composition.

The gradient appears on every margin. Men at high genetic risk are 3.5 percentage points more likely to become depressed in a given year, on a base of 9.3 percent, and 15.3 points less likely to recover if they are. They are 0.7 points more likely to leave employment, on a base of 1.6 percent, and 13.7 points less likely to re-enter it. Panel C combines the two: high-risk men are 0.7 points more likely to enter the trap, a 72 percent higher entry rate, and 24.4 points less likely to leave it, so the expected spell lasts 1.7 years at low genetic risk against 2.8 years at high genetic risk. It is the exit margin that accounts for most of this. Entry rates differ by less than a percentage point in absolute terms while persistence differs by more than a year, which is why we describe the phenomenon as a trap rather than as a difference in incidence, and it is the feature the structural model must reproduce. The gradient is not a restatement of the permanent heterogeneity that model already carries: adding the five permanent types of Appendix F as controls leaves all six margins intact (Table 17).

Every one of these gradients replicates for women, and entry into the trap is more strongly graded for them than for men. Persistence is not: the exit differential is a third the size, and it

is sensitive to whether spells of caring for home and family are counted as non-employment. Appendix B.3 reports the women's transitions and Appendix C argues that the same observed state means something different when it coincides with caring for a young child.

Table 3: Mental Health, Employment and "Depression Trap" Dynamics by Genetic Risk

Annual transition probability (%)	Low genetic risk (1)	High genetic risk (2)	Difference (2)−(1)
Panel A: Mental health dynamics			
Onset (healthy → depressed)	9.33	12.84	3.51***
	(0.49)	(0.61)	(0.78)
Recovery (depressed → healthy)	64.39	49.07	-15.31***
	(2.13)	(1.69)	(2.72)
Panel B: Employment dynamics			
Exit (employed → non-employed)	1.56	2.27	0.71**
	(0.18)	(0.21)	(0.28)
Entry (non-employed → employed)	38.76	25.06	-13.70***
	(3.89)	(2.19)	(4.47)
Panel C: "Depression trap" dynamics			
Entry into trap	0.96	1.65	0.69***
	(0.15)	(0.17)	(0.23)
Exit from trap	60.21	35.77	-24.44***
	(6.67)	(3.81)	(7.68)

Notes: Each entry is an average predicted annual transition probability, in percentage points, from a dynamic logit on genotyped UKHLS men aged 25–55 observed for at least five waves. Genetic risk is a median split of the depression polygenic index. The model is estimated separately within each risk group and both fits averaged over the same pooled covariate distribution, so column (3) is a difference in behaviour, not composition; its standard error is the square root of the sum of the two squared standard errors, valid because the subsamples are disjoint. Each cell conditions on the origin state only, so entries are marginal with respect to the other lagged state. Controls: lagged earnings, number of children, a quadratic in age, wave fixed effects and the ten genetic principal components. Depression is GHQ-12 caseness ≥ 4; the "depression trap" is non-employment jointly with caseness ≥ 4. Standard errors clustered on the individual. All panels use 15,206 person-waves on 1,534 individuals. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

3.5 Event Study Evidence for the Depression Trap

The evidence so far is associational. This subsection provide evidence using event study analyses of shocks to employment and mental health. We view these results as suggestive, but convincing in conjunction with the evidence from the existing literature. Specifications, sample construction and the full battery of robustness checks are reported in Appendices B.5 and B.6.

From job loss to mental health: redundancy

To identify the effect of employment on mental health we use *redundancy*. In UK employment law a redundancy is a dismissal for which the employee is not at fault: it arises when an employer closes a site, ceases an activity, or reduces its headcount, so that the *position* is eliminated rather than the worker replaced.[12] It is therefore a firm-side event, driven by the employer's

[12]Dismissal for poor performance, misconduct or ill health is legally distinct and is recorded separately in both surveys, as is voluntary quitting. The natural North American analogue is a permanent layoff arising from a plant closing or a firm downsizing, rather than a discharge for cause.

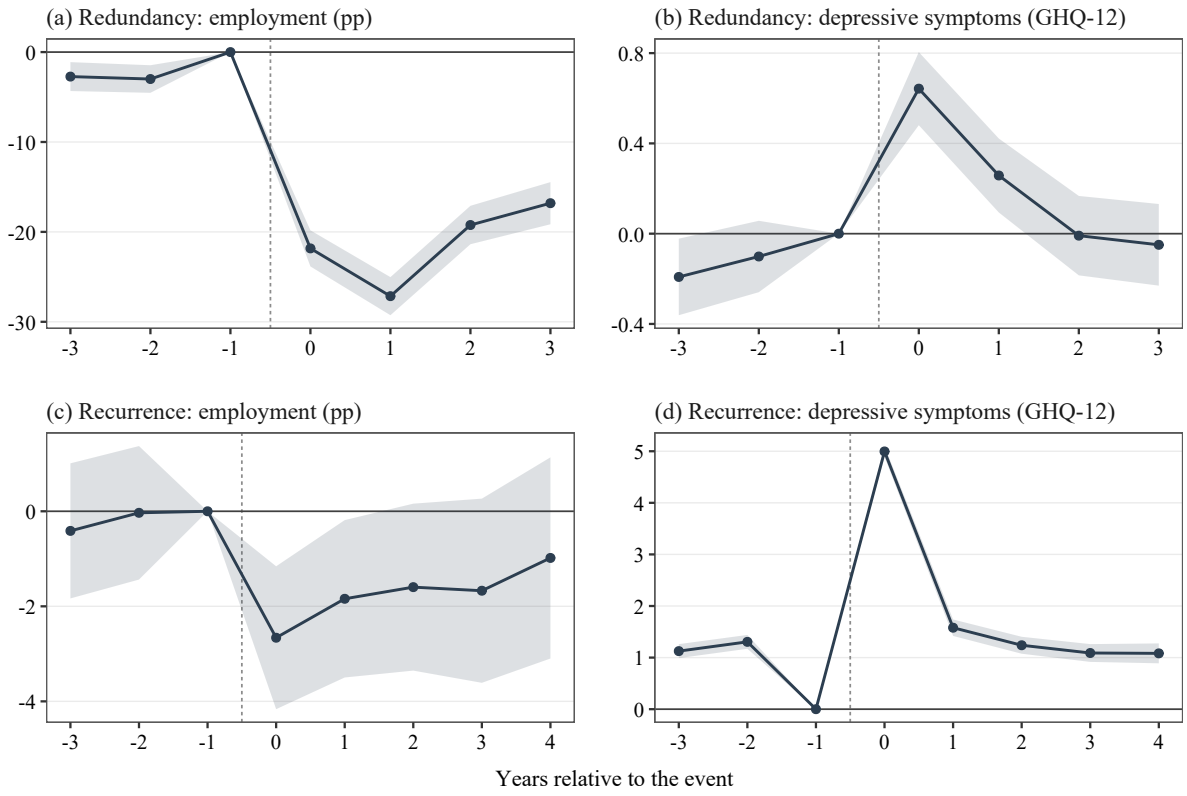
demand for labour rather than by anything about the individual worker, which is what makes it a candidate for exogenous variation in employment in the tradition of the plant-closure literature (Picchio and Ubaldi, 2024b). We date it from the reason-for-job-ending questions in the two surveys. We estimate an interaction-weighted event study (Sun and Abraham, 2021) comparing men made redundant to men not yet made redundant, on 3,515 treated men aged 25–55. Employment falls by 21.8 percentage points on impact (s.e. 1.0), reaches −27.2 a year later, and is still 16.9 points below baseline after three years; monthly earnings fall by £640 on impact and £800 a year later. GHQ-12 caseness rises by 0.64 points on impact (s.e. 0.08) and remains 0.26 points higher a year later. Panels (a) and (b) of Figure 1 plot the two paths. Both estimates are insensitive to specification: across twelve combinations of estimator, comparison group and anticipation window, the mental-health effect lies between 0.45 and 0.83 points and is significant in every one. A placebo event assigned to men never made redundant gives 0.06 (s.e. 0.07), and randomization inference over four hundred permutations of the event date gives $p < 0.001$. Appendix B.5 reports the battery.

The magnitude is in line with the literature this design speaks to. Scaled by the pre-event dispersion of GHQ-12 caseness among the treated, the impact is 0.22 standard deviations, and 0.16 to 0.22 across the heterogeneity-robust estimators. Meta-analyses of employment shocks report about 0.10 standard deviations for men (Picchio and Ubaldi, 2024b) and the experimental estimate of the non-pecuniary benefit of work is 0.23 (Hussam et al., 2022); for scale, antidepressants achieve about 0.30 and psychotherapy about 0.60 (Cipriani et al., 2018; Hofmann et al., 2012).

The central threat here is reverse causality: if deteriorating mental health causes job loss, the event study attributes to redundancy what mental health caused. Panel (b) is the direct answer. The pre-event coefficients are −0.19 and −0.10 points three and two years out, small against an impact of 0.64, and almost all of the movement occurs at the event and after it. They are not jointly zero, however, and the direction is what matters: caseness is *lower* three and two years before the event than in the year immediately preceding it, so distress was rising into the job loss rather than away from it. That is the signature of anticipation of an announced redundancy, not of selection, and it makes the estimate conservative, because the reference year is already contaminated. Treating that year as treated confirms it: moving the reference back one year raises the impact from 0.64 to 0.73, and back two years to 0.83. Appendix B.5 reports the anticipation and placebo evidence in full.

The employment path does have a pre-trend, 2.7 points below the reference year three years out (s.e. 0.8), but its sign is favorable: employment was rising into the job loss, so extrapolating that trend through the event would predict a larger drop than we measure.

Figure 1: Two Shocks, Two Legs of the Depression Trap



Notes: Event-study coefficients relative to the year before the event ($k = -1$, omitted); bands are 95 percent confidence intervals with standard errors clustered on the individual. Both use the interaction-weighted estimator of Sun and Abraham (2021), and both are estimated on men aged 25–55. Panels (a) and (b) compare men made redundant to men not yet made redundant, on 3,515 treated men observed over a ±3-year window, controlling for age, its square, and interview-year and education fixed effects. Panels (c) and (d) compare men who suffer a recurrence of a depressive episode to men who have not yet done so, on 3,114 treated men in panel (c) and 3,121 in panel (d), which does not require the employment measure; these control for age, its square, and wave fixed effects. An episode is a crossing of the GHQ-12 caseness threshold of four and a recurrence is a second crossing after an intervening spell below it. Depressive symptoms are measured by the GHQ-12 caseness score, which runs from 0 to 12, with higher values worse. In panel (d) the pre-event coefficients are positive by construction, because the reference year is mechanically a below-threshold wave whereas earlier years are not; the informative content is the size of the jump at the event and its decay. Appendices B.5 and B.6 report specifications and robustness.

This design transfers to women. Estimated on 2,803 treated women, employment falls by 22.4 points on impact (s.e. 1.2) and caseness rises by 0.54 (s.e. 0.10), against 21.8 and 0.64 for men, with pre-event paths of the same size; Appendix B.5 plots the two series together.

From depression to employment: recurrence

To estimate the reverse leg, the effect of depression on employment, we exploit quasi-random variation in the timing of depression recurrence among individuals with a history of the illness. A first episode cannot supply this variation, because onset is provoked by adverse circumstances, often the very labor market shock whose effect one is trying to estimate: in a population-based twin study the odds of onset in the month of a severe life event are 5.64, and 3.58 within monozygotic pairs, which nets out shared genes and environment, and the qualifying events include job loss and marital breakdown (Kendler et al., 1999).

Recurrence is different. Under the *kindling* hypothesis (Post, 1992), a first onset typically requires a major stressor but a recurrence does not: the depressogenic effect of severe life events is far weaker for recurrences than for first onsets (Kendler et al., 2000), a pattern confirmed meta-analytically (Stroud et al., 2008), and conditional on clinical history the timing of a recurrence is close to unpredictable, with the best prognostic models discriminating at only 0.59–0.72 (Moriarty et al., 2021). Recurrence timing is as close as the natural history of depression comes to a clock that runs independently of an individual's economic circumstances.

Both parts of that premise hold in our data. The first, that recurrence is far less provoked than onset, can be tested within an individual rather than across individuals: take the waves in which a person is well, and so at risk of an onset, and compare those falling before the first episode with those falling after recovery from it, absorbing a separate fixed effect for the person in each regime. A separation or divorce raises the probability of a first onset by 28.1 percentage points (s.e. 8.2) but of a recurrence by only 3.0 (s.e. 7.6); a redundancy raises the first by 9.2 points (s.e. 1.8) and the second by 4.9 (s.e. 1.3). Both differences are statistically significant at the five percent level ($t = 2.4$ and $t = 2.7$): the two events Kendler et al. (1999) identify as canonical triggers of an onset are substantially weaker triggers of a recurrence. The second part, that timing is unpredictable from an individual's economic position, holds directly: a redundancy in the previous year does not predict a recurrence ($p = 0.61$), and predetermined economic covariates add 0.001 to the R^2 of a recurrence model that already conditions on clinical history.

To our knowledge this is the first use of recurrence timing as identifying variation in economics. The existing literature imports exogeneity from outside the depression process, through bereavement (Frijters et al., 2014), natural disaster (Andersen et al., 2024), variation in treatment access (Biasi et al., 2021; Bütikofer and Salvanes, 2020), randomized psychotherapy (Lund et al., 2024) or Mendelian randomization (Campbell et al., 2022). Each identifies something adjacent to the object of interest: the effect of a bereavement, of a disaster, or of treatment, rather than of an episode. Recurrence timing identifies the employment cost of an episode itself, among the individuals for whom episodes recur, which is the population the structural model concerns.

We estimate an interaction-weighted event study (Sun and Abraham, 2021) on 3,114 male recurrences among men aged 25–55, the same age range as the redundancy design, comparing individuals who recur to those who have not yet recurred. A recurrence reduces employment by 2.7 percentage points on impact (s.e. 0.8); panels (c) and (d) of Figure 1 show the employment path alongside the design's first stage. Because a redundancy does remain a trigger of recurrence, the estimate must not rest on episodes that follow one, and it does not: excluding the 6.3 percent of recurrences falling within a year of a redundancy leaves the impact at 2.58 percentage points (s.e. 0.79). The estimate is also robust to alternative estimators, to how the comparison group is drawn, to sample restrictions, and to the definition of a recurrence; Appendix B.6 reports the battery.

The effect has faded by the second year, so the event study measures the contemporaneous cost of a relapse rather than its persistence, and we use it only for the former. The identifying assumption is likewise conditional rather than unconditional: recurrence timing must be unrelated to an individual's employment path given their clinical history, not random outright. The literature that supplies the kindling result also supplies the leading threat to it, since stress *sensitization* holds that the threshold falls with episode number so that minor stressors acquire triggering power (Monroe and Harkness, 2005), and the discriminating evidence favors it (Stroud et al., 2011); the design therefore trades observable severe triggers for unobservable minor ones. The checks above show that such triggers leave no economic trace, in either observed redundancies or predetermined covariates, and we therefore read the estimate as a lower bound on the contemporaneous employment cost of a relapse.

Unlike the redundancy design, this one does not transfer to women, and the reason is instructive. Estimated on women, employment is *rising* into the event, by 1.7 percentage points three waves before it and 2.3 points two waves before, both significantly different from zero, and the impact is a precise null. With 6,352 women's recurrences against 3,577 men's this is not a power problem: for women in this age range a substantial share of depressive episodes coincide with the employment interruptions around childbearing, so recurrence timing is not independent of the employment path in the way the design requires. Appendix B.6 reports the estimates and Appendix C the size of the interference.

Both shocks bite harder on the genetically vulnerable

The two designs give two independent opportunities to ask whether the same shock does more damage to men at higher genetic risk. Table 4 reports the answer at the year of impact, splitting each treated sample at the median of the polygenic index.

Both panels tell the same story, and the mental-health rows are what make it legible. After redundancy the two genetic groups suffer almost the same deterioration in mental health, 0.75 against 0.47 points of GHQ-12 caseness, a difference of 0.28 that is not distinguishable from zero. After a recurrence they suffer episodes of near-identical clinical severity, 5.79 against 5.67, a difference of a tenth of a point. The shock is the same. The employment response is not. Redundancy costs high-risk men 30.1 percentage points of employment against 18.8 for low-risk men, a gap of 11.3 (s.e. 5.0); a recurrence costs them 7.7 points against 0.8, a gap of 6.9 (s.e. 2.3). The same illness, and the same job loss, cost one group far more work than the other.

What licenses the comparison differs across the two designs, and we are explicit about it. Redundancy is not selected on genotype: neither a cross-sectional linear probability model nor an annual hazard model with fixed effects finds any association between the polygenic index and being made redundant ($p = 0.58$ and $p = 0.79$), and the larger of the two point estimates is 0.52 percentage points per standard deviation. Recurrence *is* selected on genotype, because

Table 4: Gene-by-Environment: Effects at Impact, by Genetic Risk

Outcome at $k = 0$	High genetic risk (1)	Low genetic risk (2)	Difference (1)−(2)
Panel A: Redundancy			
Employment (pp)	-30.06	-18.79	-11.26**
	(3.68)	(3.38)	(5.00)
GHQ-12 caseness (0–12)	0.75	0.47	0.28
	(0.35)	(0.24)	(0.43)
Panel B: Recurrence of a depressive episode			
Employment (pp)	-7.69	-0.84	-6.85***
	(1.62)	(1.66)	(2.32)
GHQ-12 caseness (0–12)	5.79	5.67	0.12
	(0.16)	(0.18)	(0.24)

Notes: Effect in the year of the event relative to the year before, estimated separately within each half of the genetic-risk distribution and differenced. Both panels use Sun and Abraham (2021) and a median split of the depression polygenic index, residualised on the first ten genetic principal components; because the index is oriented towards well-being, high risk is the lower-index half. Panel A is the redundancy design on men aged 25–55; Panel B is the recurrence design on men aged 25–60. Column (3) differences two separate regressions rather than estimating an interaction, so its standard error is the square root of the sum of the two squared standard errors, valid because the subsamples are disjoint. Pre-event paths are close to identical across the two genetic groups in Panel A (-1.34 and -1.06 points of employment), so the employment gap is not a differential trend. The caseness rows show that the two groups experience shocks of near-identical severity, which is what makes the employment gap interpretable as a difference in response rather than in exposure. Standard errors clustered on the individual. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

genetic risk predicts who relapses. That selection biases against the finding rather than towards it: the men who recur at low genetic risk are those with the strongest non-genetic reasons to relapse, and are therefore the more adversely selected of the two groups, so a comparison that finds the low-risk group's employment barely moving is finding it in the group where a larger effect would be expected. Appendix B.6 reports the magnitude of the selection.

Two cautions. The genetic cells in the redundancy panel are far smaller than the pooled sample, so the individual point estimates should not be read at face value; what they support is the pattern rather than any single magnitude. And the difference in each row is constructed from two separate regressions rather than estimated as an interaction. Against that, the pre-event paths of the two genetic groups are close to identical in both designs, so neither gap is a differential trend, and the mental-health rows rule out the obvious alternative reading in which the high-risk group simply receives a bigger shock.

Together the two designs close the loop. Because they draw on entirely different sources of variation, one from the labor market and one from the natural history of an illness, the assumptions they require are largely non-overlapping. What the descriptive transitions could only suggest, these estimates identify: poor mental health and non-employment cause one another, and the same shocks bite harder on the genetically vulnerable across every outcome we can measure. We now turn to a structural life-cycle model in which genetic risk governs the stochastic transitions of both mental and physical health and also affects wage offers.

4 Model

We develop a partial equilibrium life-cycle model of consumption, labor supply, and health dynamics, incorporating endogenous decisions regarding psychotherapy. The life cycle commences at age $t = 25$ and terminates at $T = 80$. Agents remain active in the labor market until the exogenous retirement age $t_{ret} = 65$, after which they withdraw from the labor force and rely on pension income and accumulated savings.

State Space and Timing

Time is discrete and indexed by $t \in \{25, \ldots, T\}$. During working life, an agent is characterized by the observed state vector

$$S_t = \{x, g, t, mh_t, ph_t, hc_t, a_t, \epsilon_t\}.$$

The vector includes endogenous and predetermined variables: x captures permanent heterogeneity (type); g indicates genetic predisposition for depressive symptoms; t denotes age; mh_t and ph_t represent mental and physical health status, respectively; hc_t denotes human capital; a_t denotes assets; and ϵ_t is an idiosyncratic stochastic shock to wages. The timing of events within each period t is as follows:

Shock Realization: At the beginning of the period, stochastic shocks are realized. During working life, these include the wage shock ϵ_t. Health status at the beginning of period t is inherited from the previous period's health transition. These realizations fully resolve the observed state vector S_t.

Decisions and Therapy Receipt: Observing S_t, the agent makes choices conditional on their life-cycle stage. Within working life, the therapy application decision and the labor supply/savings decision are resolved sequentially, in two stages:

- *Working Life (*$25 \leq t < 65$*):* The agent first decides whether to apply for mental health therapy, $d_t \in \{0, 1\}$, by comparing the value of certain non-receipt to the expected value of applying. Conditional on applying, the agent receives therapy with probability λ_τ, which is common across individuals; let $\nu_t \in \{0, 1\}$ denote this therapy receipt shock, with $\Pr(\nu_t = 1) = \lambda_\tau$. The application decision does not mechanically imply receipt: non-applicants cannot receive therapy, while applicants receive it with probability λ_τ. Realized therapy receipt is $\tau_t = d_t \nu_t$. Once τ_t is realized, the agent chooses consumption c_t, savings a_{t+1}, and labor supply $n_t \in \{0, 1\}$[13] separately for whichever value of τ_t has occurred. This timing reflects that the agent cannot finance treatment-contingent consumption or adjust labor supply until the outcome of the therapy application is actually known.

[13]The two labor-supply values denote non-employment and full-time employment, respectively.

- *Retirement* ($65 \leq t \leq 80$)*:* Labor supply is fixed at zero ($n_t = 0$), therapy is unavailable ($d_t = 0$ and $\tau_t = 0$), and the agent chooses only consumption c_t and savings a_{t+1}.

Dynamics: Choices made in the current period endogenously influence the state transition for period $t + 1$:

- *Health Evolution:* During the working life, health dynamics are endogenous; the transition probabilities for future mental and physical health, (mh_{t+1}, ph_{t+1}), depend on current health, genetics, labor supply n_t, and realized therapy receipt τ_t. In retirement, health transitions become exogenous, depending only on previous health status and age.
- *Human Capital:* Human capital evolves with labor supply. It accumulates when the agent works and depreciates during non-employment. This captures the dynamic channel through which health shocks can reduce employment, lower future human capital, and depress future wage offers.
- *Survival:* Survival is assumed to be certain during the working life. Upon entering retirement ($t \geq 65$), agents face an age- and health-dependent survival probability $\zeta_t(mh_t, ph_t)$. The model terminates deterministically at age $T = 80$.

Agents choose decision rules to maximize the expected present discounted value of lifetime utility, subject to the period budget constraint, therapy receipt risk, human capital accumulation, and survival risk in retirement.

Heterogeneity

We characterize agent heterogeneity along two distinct dimensions: genetic predisposition and permanent type.

First, a key feature of our framework is the inclusion of a genetic predisposition for depressive symptoms, denoted by g. Second, beyond genetic factors, we incorporate time-invariant heterogeneity to capture persistent ex-ante differences in endowments (e.g., innate ability or non-genetic health robustness). Accounting for these persistent traits is crucial for identification; neglecting them risks confounding permanent individual characteristics with state dependence in the stochastic processes. For instance, without this correction, the model might misattribute a sub-population's persistently poor health to a high probability of adverse shocks, rather than to their underlying type.

To implement this computationally, we discretize permanent heterogeneity following the classification method of Bonhomme et al. (2022). We postulate that agents belong to a finite set of permanent types, denoted by x, which govern their productivity and health trajectories.

Health Dynamics

We model the joint evolution of physical (ph) and mental (mh) health as a first-order Markov process. Each dimension is discretized into three ordered states, $h \in \{G, M, S\}$, corresponding

to *Good*, *Moderate*, and *Severe* health. In our empirical application, physical health states map to Excellent/Good (G), Average (M), and Poor/Very Poor (S), while mental health states represent the clinical progression of depression: asymptomatic (G), mild-to-moderate symptoms (M), and Major Depressive Disorder (S).

Health transitions are governed by an ordered logit framework, where the observed state h_{t+1} is determined by an underlying latent continuous variable h^*_{t+1} falling within estimated thresholds $\{\kappa_j\}$. Assuming the idiosyncratic shocks $(\iota_{t+1}, \upsilon_{t+1})$ follow a standard logistic distribution, the latent process for physical health evolves as:

$$ph^*_{t+1} = ph_t\pi_1 + mh_t\pi_2 + n_t\pi_3 + S^{exo}_t\pi_4 + \iota_{t+1}, \tag{4}$$

where ph_t and mh_t are vectors of indicators for current health status, $n_t \in \{0,1\}$ is the labor supply choice, and S^{exo}_t collects exogenous covariates including age, permanent type (x), and genetic predisposition (g). The latent mental health process is analogous, except that it also incorporates the realized therapy receipt, $\tau_t \in \{0,1\}$:

$$mh^*_{t+1} = mh_t\mu_1 + \tau_t\mu_2 + n_t\mu_3 + ph_t\mu_4 + S^{exo}_t\mu_5 + \upsilon_{t+1}, \tag{5}$$

where μ_2 captures the mental health returns to psychotherapy. Both processes are estimated as standard ordered logit models under the proportional-odds restriction: each covariate has a single coefficient that is common across health thresholds, while threshold-specific intercepts κ_j determine the cutpoints separating the three health states. The resulting probability of transitioning to state j or better is:

$$P(h_{t+1} \leq j \mid \mathbf{X}_t) = \Lambda\left(\kappa_j - \mathbf{X}_t\beta\right), \tag{6}$$

where $\Lambda(\cdot)$ denotes the logistic cumulative distribution function. The physical health process (Eq. 4) depends only on exogenous and predetermined covariates, so π_1–π_4 and its cutpoints are estimated directly from the UKHLS panel and held fixed as an input to the structural model (Panel B of Table 5). The mental health process (Eq. 5) depends on the endogenous therapy decision τ_t, so μ_1, μ_3–μ_5 and its cutpoints must instead be estimated jointly with the rest of the structural model via SMM (Panel C). We assume that, conditional on the current state S_t and choices (n_t, τ_t), the physical health, mental health, and wage innovations $(\iota_{t+1}, \upsilon_{t+1}, \eta_{t+1})$ are mutually independent. We let $H(\cdot \mid S_t, n_t, \tau_t)$ denote the joint conditional distribution function of next-period health $h_{t+1} \equiv (mh_{t+1}, ph_{t+1})$ implied by these two ordered-logit processes; this is the health transition kernel used in the recursive formulation below.

Flow Utility

Agents derive utility from consumption and effective leisure, and incur a non-pecuniary cost when they receive therapy. Flow utility is

$$U(c_t, n_t, \tau_t; mh_t, ph_t) = \frac{\left(c_t^{\gamma} L_t^{1-\gamma}\right)^{1-\omega}}{1-\omega} - \psi(n_t)\tau_t, \tag{7}$$

where ω is the coefficient of relative risk aversion, γ governs the consumption weight, and $\psi(n_t)$ is the disutility of therapy. We allow this disutility to vary with employment status for two reasons. First, therapy is time-intensive, and the marginal cost of the time it requires is substantially higher for the employed: a large share of their time endowment is already committed to work, so hours diverted to therapy come at the expense of scarce leisure, whereas the non-employed can draw on more abundant discretionary time. Second, seeking therapy while employed may carry greater stigma, for example through disclosure at the workplace or concern about being perceived as less committed. Both channels raise the non-pecuniary cost of therapy for the employed, and we accordingly let $\psi(n_t)$ take a distinct value in each employment state.

Effective leisure is

$$L_t = 1 - \ell_n n_t - \delta(mh_t, ph_t). \tag{8}$$

Here, ℓ_n is the time cost of work, $\delta(mh_t, ph_t)$ is the health-related time cost.[14]

The health time cost is

$$\delta(mh_t, ph_t) = mh_t\delta_1 + ph_t\delta_2. \tag{9}$$

This specification captures the welfare cost of poor health even outside employment. By reducing effective leisure, poor health also increases the marginal utility cost of working.

Human Capital

Human capital evolves according to a parsimonious accumulation-depreciation process:

$$hc_{t+1} = Q(hc_t, n_t) \equiv (1 - \delta_{hc})hc_t + n_t. \tag{10}$$

We let $Q(\cdot, \cdot)$ denote this human capital updating function. Thus, human capital depreciates during non-employment and accumulates while working up to a limit. This mechanism captures the dynamic channel through which health shocks can reduce employment, lower future human capital, and depress future wage offers.

[14] Following Abramson et al. (2024), we set $\ell_n = 8 \times 5/100 = 0.4$, corresponding to the time cost of full-time work.

Budget Constraint, Benefits, and Earnings

Agents face a period-by-period budget constraint:

$$c_t + a_{t+1} = (1 + r)a_t + y_t - P_\tau(n_t)\tau_t, \qquad a_{t+1} \geq 0, \tag{11}$$

where r is the risk-free interest rate, $P_\tau(n_t)$ is the out-of-pocket pecuniary cost of realized therapy, and borrowing is not allowed. We allow this cost to depend on employment status, setting it to zero for the non-employed and to a positive value for the employed, with both calibrated to their empirical means in Table 5.[15]

Disposable income y_t depends on the agent's life-cycle stage, labor supply choice, and health status:

$$y_t = \begin{cases} n_t w_t \left[1 - \mathcal{T}(n_t w_t)\right] + b(n_t, mh_t, ph_t), & \text{if } t < 65 \text{ (Working Age)}, \\ p(x), & \text{if } t \geq 65 \text{ (Retirement)}. \end{cases} \tag{12}$$

During working life, labor earnings are $n_t w_t$, and $\mathcal{T}(n_t w_t)$ denotes the effective tax rate as a function of labor earnings. Agents may also receive benefits $b(n_t, mh_t, ph_t)$, which are allowed to vary by both health and employment status.[16] After retirement, agents receive pension income $p(x)$, which may vary by permanent type.

The market wage offer is specified as follows:

$$\ln w_t = \beta_0 + g\beta_1 + x\beta_2 + mh_t\beta_3 + ph_t\beta_4 + hc_t\beta_5 + hc_t^2\beta_6 + \epsilon_t. \tag{13}$$

Here, wages are determined by genetic predisposition g, permanent type x, current health status mh_t and ph_t, and a polynomial profile of human capital hc_t. The idiosyncratic productivity shock ϵ_t follows an AR(1) process:

$$\epsilon_{t+1} = \rho\epsilon_t + \eta_{t+1}, \qquad \eta_{t+1} \sim \mathcal{N}(0, \sigma_\eta^2), \tag{14}$$

where η_{t+1} represents a transient shock to labor productivity. We let $F(\cdot \mid \epsilon_t)$ denote the conditional cumulative distribution function of ϵ_{t+1} given ϵ_t.[17]

[15] The difference reflects how the two groups access care. In our sample, the non-employed obtain therapy almost entirely through the NHS Talking Therapies programme, which is free at the point of use, so their out-of-pocket expenditure is effectively zero. A nontrivial share of the employed instead pay for therapy privately or supplement public provision with paid sessions. A plausible explanation is that their higher opportunity cost of time makes the delays associated with capacity-constrained public provision more costly, raising their willingness to pay for faster private access.

[16] In particular, $b(n_t, mh_t, ph_t)$ may be positive even when $n_t > 0$, allowing employed individuals to receive benefits.

[17] $F(\epsilon_{t+1} \mid \epsilon_t) = \mathcal{N}(\rho\epsilon_t, \sigma_\eta^2)$.

Recursive Formulation

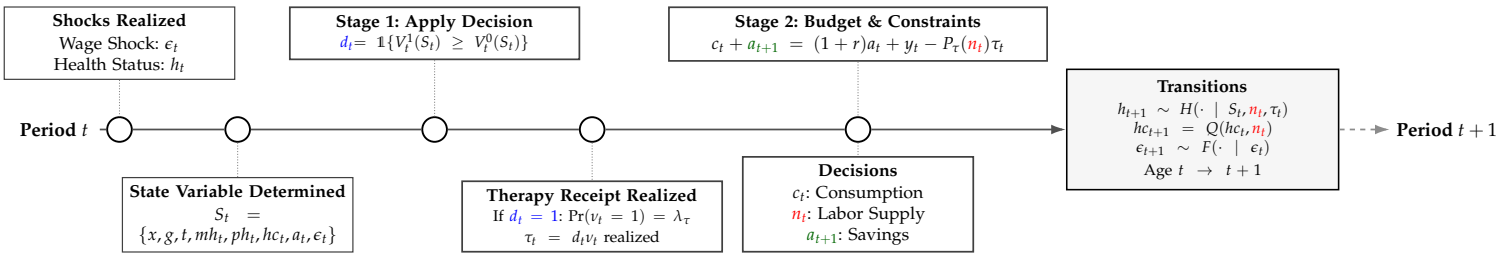
The agent enters the model at age $t = 25$ and lives up to a maximum age of $T = 80$. The life cycle is partitioned into a working stage ($t \in [25, 64]$) and a retirement stage ($t \in [65, 80]$).

Working Age **($25 \leq t < 65$):** In each working-age period, the agent's observed state is summarized by

$$S_t = \{x, g, t, mh_t, ph_t, hc_t, a_t, \epsilon_t\}.$$

Given S_t, the agent first decides whether to apply for therapy, $d_t \in \{0, 1\}$. Conditional on applying, therapy receipt is uncertain: let $\nu_t \in \{0, 1\}$ denote the receipt shock, with $\Pr(\nu_t = 1) = \lambda_\tau$. Realized therapy receipt is $\tau_t = d_t \nu_t$. Only once τ_t is known does the agent choose labor supply $n_t \in \{0, 1\}$, consumption c_t, and next-period assets a_{t+1}. The within-period dynamics are illustrated in Figure 2.

Figure 2: Working Age Within-Period Dynamics



Reflecting the two-stage timing above, we first define, for each realized therapy outcome $\tau \in \{0, 1\}$, the conditional value function

$$V_t^\tau(S_t) = \max_{n_t, a_{t+1}} \left\{ U(c_t, n_t, \tau; mh_t, ph_t) + \beta \iint V_{t+1}(S_{t+1})\, dH(h_{t+1} \mid S_t, n_t, \tau)\, dF(\epsilon_{t+1} \mid \epsilon_t) \right\}, \tag{15}$$

subject to the period budget constraint $c_t + a_{t+1} = (1 + r)a_t + y_t - P_\tau(n_t)\tau$, non-negativity of assets, and the human capital law of motion. The expectation is taken over next-period health $h_{t+1} \equiv (mh_{t+1}, ph_{t+1})$, given labor supply n_t and the realized therapy outcome τ, and over next-period wage shocks.

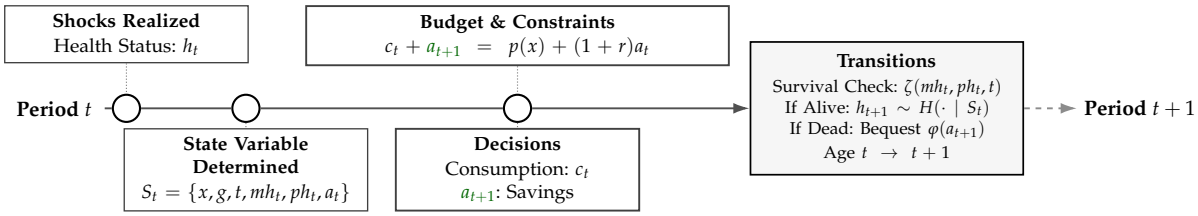
The therapy application decision d_t is then chosen to maximize the value before ν_t is realized:

$$V_t(S_t) = \max_{d_t \in \{0,1\}} \left\{ (1 - d_t)\, V_t^0(S_t) + d_t \left[\lambda_\tau V_t^1(S_t) + (1 - \lambda_\tau) V_t^0(S_t) \right] \right\}. \tag{16}$$

Because non-application guarantees $\tau_t = 0$, applying is weakly preferred whenever $V_t^1(S_t) \geq V_t^0(S_t)$, and the agent applies, $d_t = 1$, exactly in that case. Once ν_t is realized, $\tau_t = d_t \nu_t$ is known, and the agent implements the consumption, savings, and labor supply policy associated with $V_t^{\tau_t}(S_t)$.

Retirement **($t \geq 65$):** Upon reaching age 65, the agent retires exogenously. The retirement state is summarized by $S_t = \{x, g, t, mh_t, ph_t, a_t\}$, since human capital no longer affects earnings or choices. Labor supply is fixed at $n_t = 0$, therapy is unavailable ($d_t = 0$ and $\tau_t = 0$), and income consists of pension income $p(x)$ and returns on assets. Retirement utility is shaped by survival risk. At the end of each period, the agent survives with probability $\zeta(mh_t, ph_t, t)$; if the agent dies, remaining assets generate bequest utility. The retirement timing is illustrated in Figure 3.

Figure 3: Retirement Within-Period Dynamics



If the agent dies, remaining assets generate bequest utility. Following Nardi et al. (2010) and Maccuish (2022), the bequest motive is

$$\varphi(a_{t+1}) = \theta \frac{(a_{t+1} + \kappa)^{(1-\omega)\gamma}}{1-\omega}, \tag{17}$$

where θ governs the strength of the bequest motive and κ controls the luxury-good nature of bequests.

The Bellman equation incorporates survival risk. Writing $\zeta_t \equiv \zeta(mh_t, ph_t, t)$ for brevity,

$$V_t(S_t) = \max_{a_{t+1}} \left\{ U(c_t, 0, 0; mh_t, ph_t) + \beta \underbrace{\zeta_t}_{\text{Survival}} \int V_{t+1}(S_{t+1})\, dH(h_{t+1} \mid S_t) + \beta \underbrace{(1-\zeta_t)}_{\text{Death}} \varphi(a_{t+1}) \right\}. \tag{18}$$

subject to the retirement budget constraint and the non-negativity of assets. At the terminal age $T = 80$, the continuation value is zero.

5 Estimation and Identification

We estimate the model in three steps. First, we calibrate preference, financial, and treatment parameters using external evidence. Second, we estimate permanent individual types, physical health transitions, and institutional inputs outside the structural model. Third, conditional on these inputs, we jointly estimate the remaining parameters by simulated method of moments (SMM), targeting labor supply, therapy use, earnings, saving, and mental health transitions. Table 5 summarizes the parameters, data sources, and estimation targets.

5.1 Calibrated Parameters

We set the annual risk-free interest rate to $r = 4.6\%$ (Jordà et al., 2019), relative risk aversion to $\omega = 3.0$ (Nardi et al., 2010), and the time cost of full-time work to $\ell_n = 0.4$ (Abramson et al., 2024). We take the bequest parameters (θ, κ) from Maccuish (2022) and rescale them to the model's monetary units.

Table 5: Model Parameters, Estimation Targets, and Data Sources

Parameter	Equation	Description	Calibration Values or Empirical Targets	Source
Panel A: Calibrated Parameters				
μ_2	Eq. 5	Effectiveness of therapy	RCT estimate of the effect of therapy on mental health	Cronin et al. (2024)
ω	Eq. 7	Coefficient of relative risk aversion	$\omega = 3.0$	Nardi et al. (2010)
ℓ_n	Eq. 8	Time cost of full-time work	$\ell_n = 0.4$	Abramson et al. (2024)
r	Eq. 11	Annual risk-free interest rate	$r = 4.6\%$	Jordà et al. (2019)
θ, κ	Eq. 17	Bequest parameters	$\theta = 0.4464,\ \kappa = 273$	Maccuish (2022)
Panel B: Inputs Estimated or Calibrated Outside the Model				
$\pi_1 - \pi_4$	Eq. 4	Physical health transition parameters	Physical health transition rates by current health, labor supply, age, type, and genotype	UKHLS
$P_\tau(n_t)$	Eq. 11	Pecuniary cost of therapy	Mean out-of-pocket therapy expenditure per year	UKHLS
$b(n_t, mh_t, ph_t)$	Eq. 12	Benefits schedule by employment and health status	Mean benefits by labor supply, mental health, and physical health	UKHLS
$\mathcal{T}(w_t n_t)$	Eq. 12	Effective tax schedule	Tax rate by labor earnings	UKHLS
$p(x)$	Eq. 12	Pension income	Pension income by type	UKHLS
λ_τ	Eq. 16	Probability of receiving therapy conditional on applying	Ratio of treatment starts to referrals in the national talking-therapies programme	NHS England (2024)
$\zeta(mh_t, ph_t, t)$	Eq. 18	Survival probability	Survival rates by age, mental health, and physical health	UKHLS & UK Life Tables
Panel C: Parameters Estimated Jointly by SMM				
$\mu_1, \mu_3 - \mu_5$	Eq. 5	Mental health transition parameters and cutoffs, excluding therapy effects	Mental health transition rates by current health, labor supply, therapy receipt, age, type, and genotype	UKHLS
γ	Eq. 7	Consumption share in utility	Average working hours	UKHLS
$\psi(n_t)$	Eq. 7	Non-pecuniary disutility of therapy	Therapy take-up by employment status	UKHLS
$\delta(mh_t, ph_t)$	Eq. 8	Health-related time cost, incurred regardless of employment status	Labor supply by mental and physical health groups	UKHLS
δ_{hc}	Eq. 10	Human capital depreciation	Wage growth and labor supply around employment and non-employment spells	UKHLS
$\beta_0 - \beta_6$	Eq. 13	Wage offer process, including human capital returns	Mean wages by age, health, type, genotype, and labor-market history	UKHLS
ρ, σ_η	Eq. 14	Wage shock parameters	Wage dispersion by health, type, genotype, and age group	UKHLS
β	Eq. 18	Discount factor	Aggregate saving ratio	UKHLS

Notes: The table distinguishes externally determined inputs from the parameters estimated jointly to explain individual behavior over the life cycle. UKHLS denotes the UK Household Longitudinal Study; RCT denotes a randomized controlled trial. Panel A reports calibrations from external studies, including the therapy effect on mental health, μ_2. Panel B reports health transitions and institutional inputs estimated or calibrated before structural estimation and then held fixed. Panel C reports parameters estimated by simulated method of moments (SMM), which selects parameters to minimize a weighted distance between empirical and simulated moments. The fourth column lists calibration values or the empirical features most informative about each parameter; these are not one-to-one identification restrictions. The central distinction is that treatment effectiveness is calibrated using experimental evidence, while treatment choices and the remaining mental health dynamics are fitted jointly within the model. Bequest parameters are rescaled to the model's monetary units.

We calibrate the therapy effect, μ_2, using evidence from randomized controlled trials. Treatment choices in the panel may respond to adverse mental health shocks, so observed differences between treated and untreated individuals need not identify treatment effectiveness. Following the approximation in Borenstein et al. (2009), we map the standardized mean difference (SMD) reported in meta-analyses of randomized controlled trials into the log-odds coefficient in the ordered-logit mental health transition:

$$\mu_2 = \text{SMD} \times \frac{\pi}{\sqrt{3}}. \tag{19}$$

This use of external treatment evidence follows recent structural work on mental health (Abramson et al., 2024; Cronin et al., 2024).

5.2 Estimation Outside the Model

Permanent Individual Heterogeneity (Type)

We approximate persistent individual heterogeneity with five discrete types, $x_i \in \{1, \ldots, 5\}$. Following Bonhomme et al. (2022), we construct types by applying k-means clustering to individual wage and health moments. The procedure requires these moments to be informative about permanent heterogeneity; the relevant identifying condition is an injective mapping from permanent heterogeneity to the population moment vector. Our moments summarize mental and physical health histories and wages, with adjustments for age and genetic group described in Appendix F.

Table 6 summarizes the estimated types. Type 1 is a high-wage, high-health group. Type 2 has relatively high wages and good mental health but substantially worse physical health. Type 3 is characterized by very low mental health despite moderate wages and physical health. Type 4 combines high mental health with relatively good physical health at lower wages. Type 5 is a low-wage group with the poorest physical health. We treat the estimated type as a fixed state variable in the structural model. Section 5.5 assesses how accurately the classification procedure recovers known types in simulated data.

Health Transitions and Institutional Inputs

We estimate the ordered-logit physical health process in Equation 4 directly from the UKHLS panel and use the fitted probabilities as the model's transition kernel. This approach requires labor supply and the other included covariates to be predetermined with respect to the physical health innovation. We hold the estimated parameters π_1–π_4 fixed during SMM estimation.

We construct the benefits schedule $b(n_t, mh_t, ph_t)$ using UKHLS mean benefit income by employment status and mental and physical health states. We also estimate the effective tax schedule $\mathcal{T}(w_t n_t)$ by labor earnings and pension income $p(x)$ by type. The pecuniary therapy

Table 6: Characteristics of Estimated Permanent Types

	Share	Mental health (%)			Physical health (%)			Employed	Mean annual earnings
Type	(%)	Good	Moderate	Severe	Good	Moderate	Severe	(%)	(£)
1	20.1	83.1	8.9	8.0	65.9	25.3	8.8	97.3	41,800
2	17.5	64.1	13.1	22.8	68.9	23.6	7.5	96.8	39,300
3	9.1	17.9	11.9	70.2	41.7	30.0	28.3	65.5	28,600
4	33.0	82.5	9.1	8.4	70.0	23.4	6.6	94.2	27,800
5	20.2	74.1	11.6	14.3	20.3	37.1	42.7	80.9	25,700

Notes: The sample comprises 2,651 non-retired men aged 25–64 in the UK Household Longitudinal Study. Types are identified using k-means clustering ($K = 5$; Appendix F). Shares refer to individuals; all other statistics are calculated across person-wave observations within type. Employment requires at least 0.25 full-time-equivalent years of work annually. Earnings are mean gross annual labor earnings among employed observations, in 2015 pounds.

cost $P_\tau(n_t)$ equals average annual out-of-pocket expenditure by employment status: £353 for the employed and zero for the non-employed.[18]

We calibrate the probability of receiving therapy conditional on applying (λ_τ) to 0.689, using the ratio of treatment starts to referrals in the NHS Talking Therapies programme in 2023–24 (NHS England, 2024).

Finally, we combine UKHLS health information with UK life tables to estimate the retirement survival probability $\zeta(mh_t, ph_t, t)$ in Equation 18. Appendix Tables 33–34 report the estimates and construction of the external inputs.

5.3 Estimation of Core Structural Parameters via SMM

Conditional on the calibrated parameters and externally estimated inputs, we estimate the remaining parameter vector ϑ jointly by simulated method of moments (SMM). For each candidate vector, we solve the model, simulate individual life-cycle histories, and compute the simulated counterparts of the empirical moments. The estimator minimizes

$$\widehat{\vartheta} = \arg\min_{\vartheta \in \mathcal{V}} \left[m^S(\vartheta) - m^D\right]' W \left[m^S(\vartheta) - m^D\right], \tag{20}$$

where m^D and $m^S(\vartheta)$ are the empirical and simulated moment vectors, W is the weighting matrix, and $\mathcal{V}$ is the parameter space. Panel C of Table 5 summarizes the parameters and their empirical targets. The discussion below describes the main sources of identifying variation; individual moments can inform multiple parameters.

Preferences, treatment, and labor supply. Therapy take-up rates by employment status inform the nonpecuniary therapy costs $\psi(n_t)$, conditional on health, treatment effectiveness, access, and out-of-pocket costs. Employment rates across joint mental and physical health states (Good, Moderate, and Severe in each dimension) inform the health-related time costs $\delta(mh_t, ph_t)$ in

[18] The zero out-of-pocket cost for the non-employed indicates that, in our sample, they rely entirely on the NHS free-therapy scheme.

Equation 8. These costs reduce available leisure and affect incentives to work. Average working hours inform the consumption share γ in the consumption–leisure composite.

Wages and human capital. Mean wages by age, health, type, genetic group, and labor-market history inform the deterministic wage parameters β_0–β_6 in Equation 13. Wage growth and labor supply around employment and non-employment spells inform the human capital depreciation rate δ_{hc} in Equation 10. Wage dispersion by health, type, genetic group, and age informs the persistent wage shock parameters (ρ, σ_η): ρ governs shock persistence, and σ_η governs the dispersion of innovations.

Mental health dynamics. We estimate μ_1 and μ_3–μ_5, together with the ordered-logit cutoffs, jointly with the other structural parameters, using mental health transition rates by current health, labor supply, therapy receipt, age, type, and genetic group. The therapy effect μ_2 remains fixed at its external calibration.

Saving. The aggregate saving ratio informs the discount factor β, conditional on the calibrated interest rate and bequest parameters and the other estimated parameters.

Appendix Table 35 reports the structural parameter estimates.

5.4 Targeted Model Fit

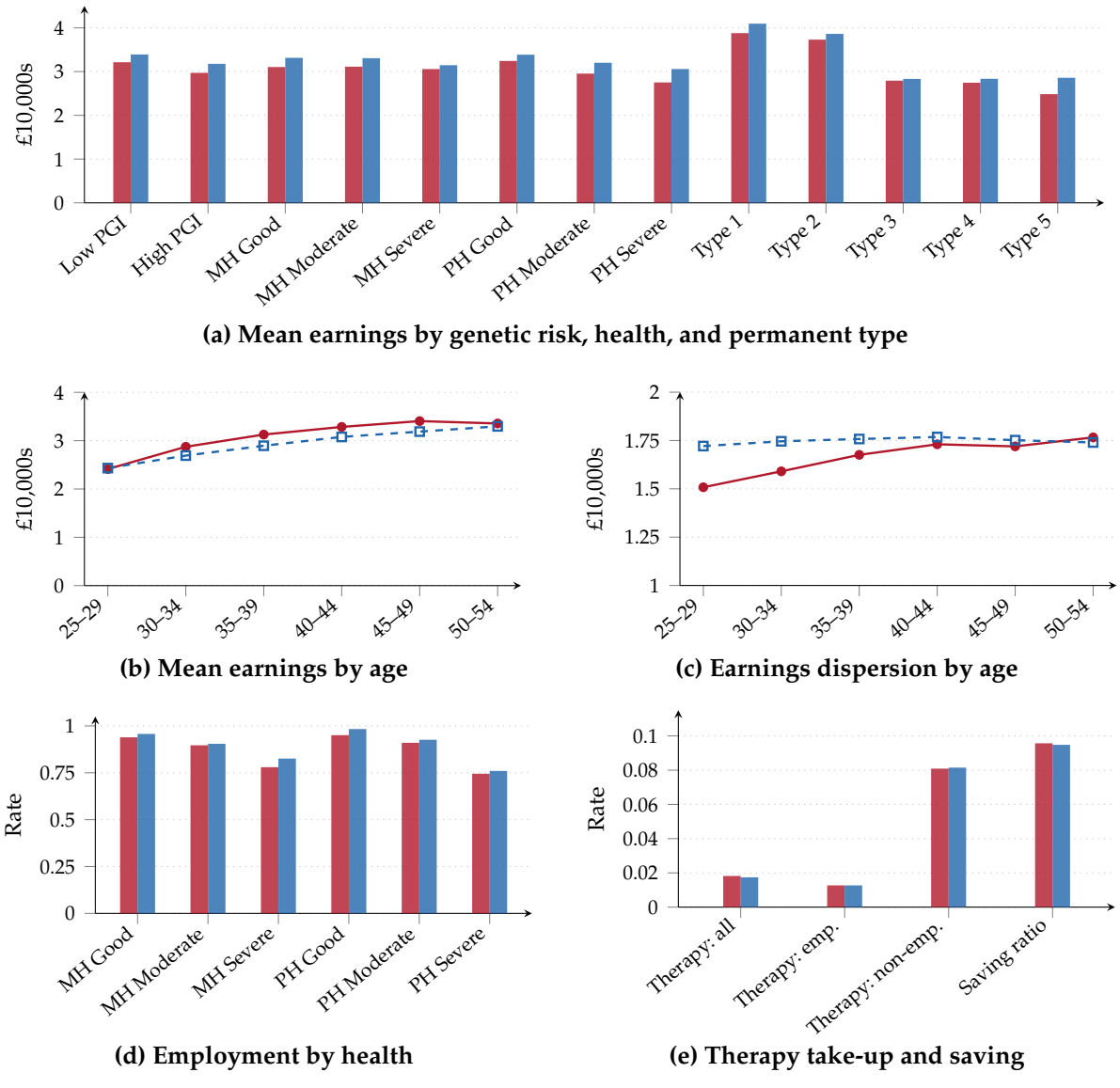
Figure 4 compares the targeted earnings, employment, treatment, and saving moments with their simulated counterparts. The model captures several broad patterns, although the earnings and employment moments show systematic discrepancies.

Earnings. Panel (a) shows that the model reproduces higher earnings among men with lower genetic risk and better physical health, as well as the relatively high earnings of Types 1 and 2. It overpredicts mean earnings across the reported groups, however, and does not reproduce the ordering of Types 3–5. Panel (b) shows a broadly similar rise in mean earnings over ages 25–54, with simulated earnings below the data over much of this age range.

The largest discrepancy in the age profiles concerns dispersion. In the data, the standard deviation of earnings generally increases with age; in the model, it remains approximately flat. The model therefore overpredicts dispersion at younger ages and is closer to the data at ages 50–54 (panel c). The age panels are restricted to 25–54 because the model omits partial retirement and adjustments in hours worked, limiting its ability to represent late-career earnings patterns.

Employment, treatment, and saving. Panel (d) shows that the model reproduces the ordering of employment rates across mental and physical health states but overpredicts their levels. The largest discrepancy is among men in the Severe mental health state: simulated employment is 82.6 percent, compared with 78.0 percent in the data. Panel (e) shows a close match for therapy take-up by employment status and the aggregate saving ratio.

Figure 4: Model Fit: Targeted Moments



(a) Mean earnings by genetic risk, health, and permanent type

(b) Mean earnings by age

(c) Earnings dispersion by age

(d) Employment by health

(e) Therapy take-up and saving

Notes: Red bars and solid lines show data for men in the UK Household Longitudinal Study; blue bars and dashed lines show model simulations. All moments are estimation targets. PGI denotes the depression polygenic index, with higher values indicating greater genetic risk; MH and PH denote mental and physical health. Types 1–5 are permanent groups based on wage and health histories. Earnings are gross labor income in 2015 £10,000s; dispersion is the standard deviation. Employment and therapy take-up are rates within the reported groups (emp.: employed; non-emp.: non-employed). Rates are proportions. The model closely matches therapy use and saving, but overpredicts employment and understates the increase in earnings dispersion with age.

Mental health transitions. Appendix Figure 15 compares the targeted one-year mental health transition probabilities with their simulated counterparts. The model reproduces these probabilities almost exactly.

5.5 Untargeted Model Fit

We next assess the model's implications for health prevalence by age, the genetic gradients and the event-study profiles documented in empirical Section 3. These comparisons evaluate fit beyond the moments directly targeted in SMM.

Health prevalence. Panels (a)–(b) of Figure 5 compare empirical and simulated health-state shares at ages 25–64. The model closely tracks the age profile of physical health, including the rise in the share in the Severe state (panel b). Mental health prevalence is also matched equally well. These age-specific prevalence moments are not directly targeted in SMM and provide an additional check on the model's health dynamics.

Gradients by genetic risk. Table 3 documents higher rates of depression onset, employment exit, and entry into the depression trap among high-PGI men, together with lower rates of recovery, re-employment, and trap exit. We use this reduced-form evidence as a benchmark for the structural model, with particular attention to the genetic gradient in trap persistence.

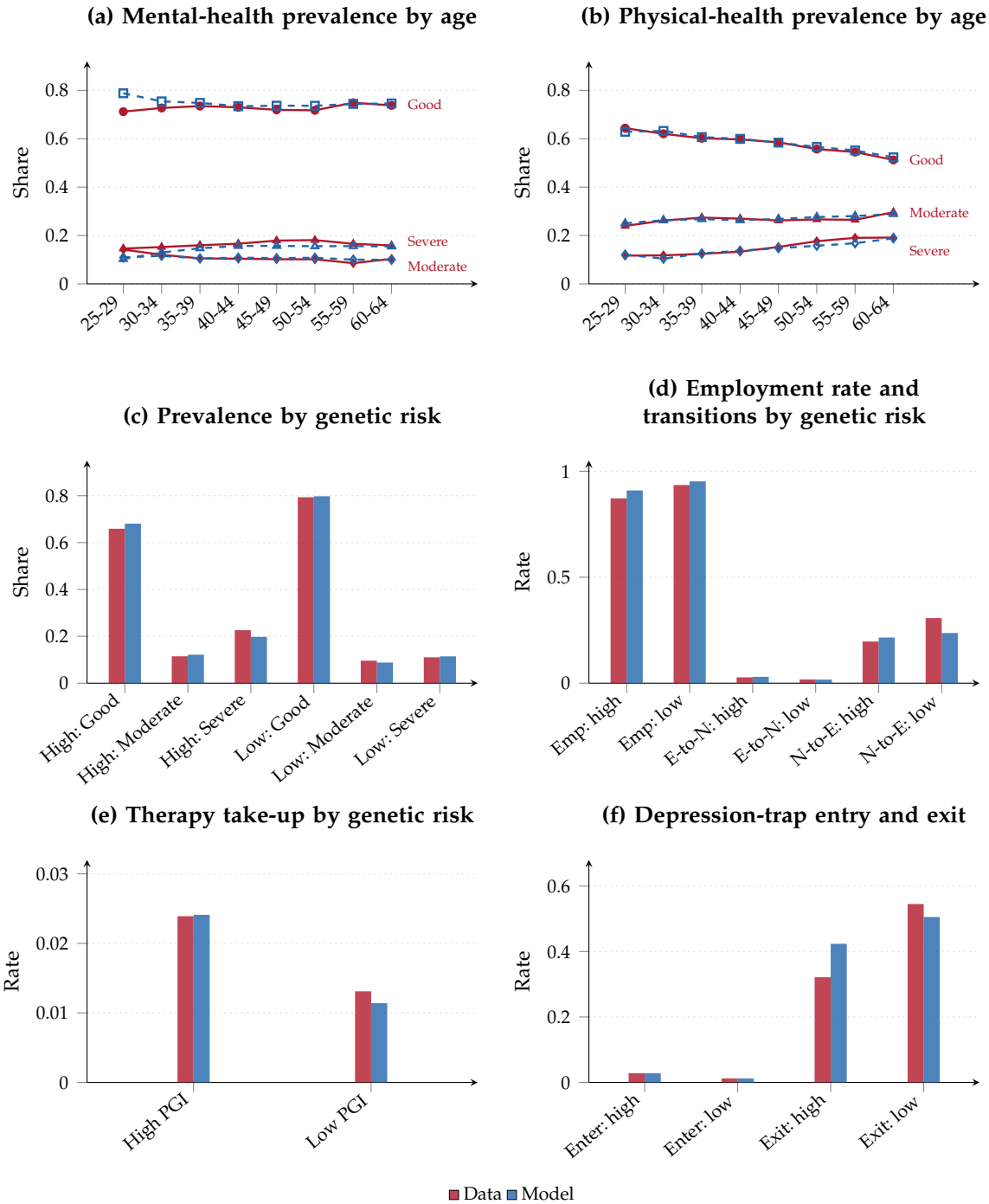
Panels (c)–(f) of Figure 5 compare mental health prevalence, employment, therapy use, and depression-trap transitions across genetic-risk groups. The model reproduces the higher share in the Severe mental health state among high-PGI men, although it underpredicts its level (panel c). It also reproduces the lower employment rate of high-PGI men but understates the gap between genetic groups (panel d). Therapy take-up is close to the data (panel e), and the model matches trap entry rates almost exactly for both groups (panel f).

Replication of the reduced-form event studies. We apply the event-study specifications of Section 3.5 to a panel simulated at the estimated parameters, using the same event windows and age range as in the data. Recurrence is defined in both samples as a return to the Severe mental health state after recovery. For job loss, the empirical event is involuntary redundancy, whereas employment exits in the model are endogenous. We use simulated exits following at least two years of employment as the model counterpart. This difference in event selection limits the interpretation of the comparison as evidence about responses to involuntary job loss.

The model reproduces both legs of the trap out of sample: on the employment side it matches the depth of the redundancy-driven fall and the pace of the recovery that follows, together with the smaller contemporaneous dip after a recurrence; on the mental-health side it reproduces the sharp rise in severe depression at a recurrence and its subsequent decay. That a model disciplined only by static moments recovers the magnitude and timing of two dynamic causal responses, identified from independent sources of variation, is strong evidence that its behavioural mechanisms are the ones operating in the data.[19]

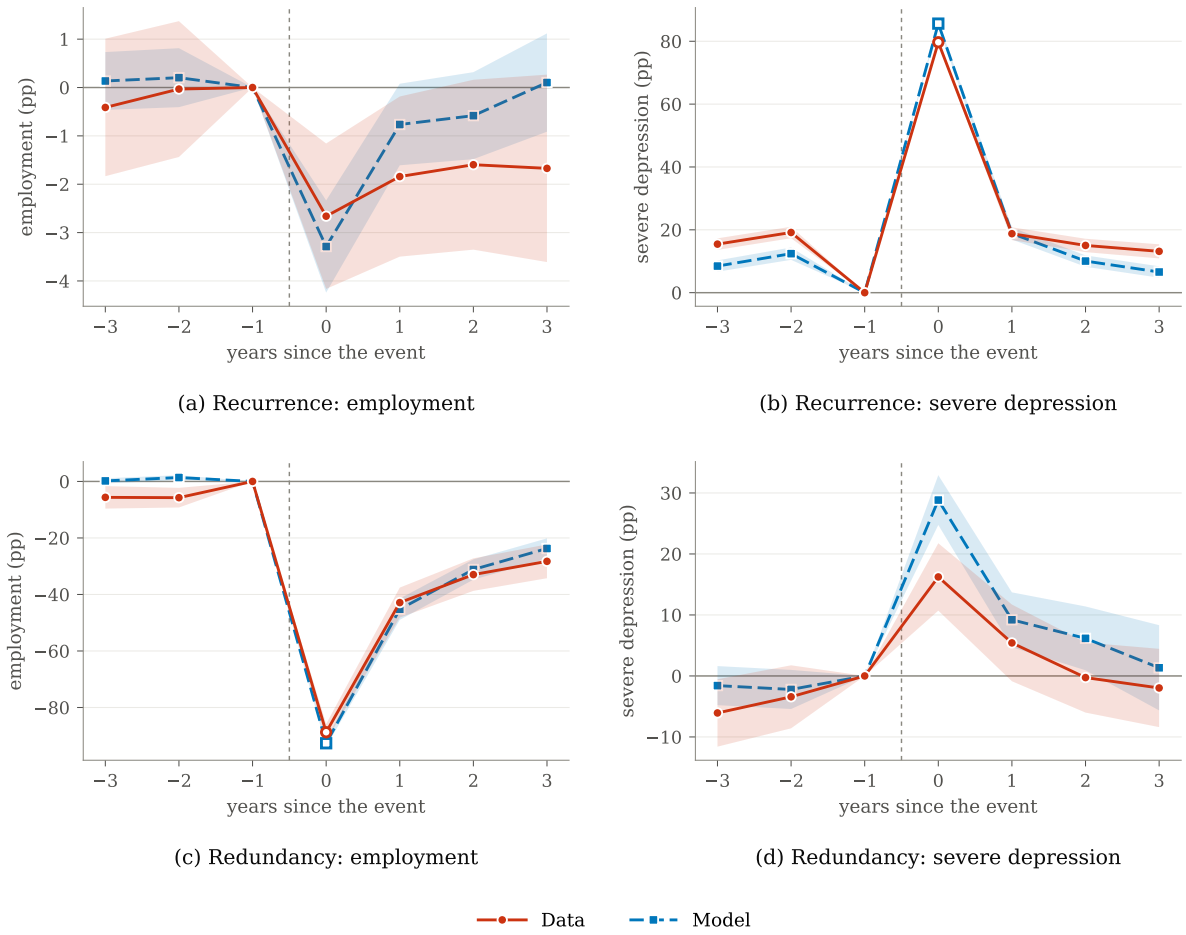
[19]The one feature the model cannot match is the decline in employment that begins before a redundancy in the data, which reflects the anticipation of an announced job loss that a voluntary separation cannot generate; the model correspondingly overstates the mental-health response to redundancy on impact.

Figure 5: Model fit: untargeted moments



Notes: Data are for men in the UK Household Longitudinal Study. Red solid lines with filled markers and red bars show the data; blue dashed lines with open markers and blue bars show the model. Panels (a)–(b) report mental and physical health shares by age among non-retired men aged 25–64. Both health measures have three states: Good, Moderate, and Severe. Panels (c)–(f) report mental health shares, employment rates and transitions, therapy take-up, and depression-trap entry and exit by genetic risk. PGI denotes the depression polygenic index; high and low PGI indicate greater and lower genetic risk. In panel (d), Emp and E denote employment, and N denotes non-employment. The trap denotes joint depression and non-employment. Rates are proportions; transition probabilities are annual and conditional on initial status. These moments are not directly targeted in SMM.

Figure 6: Event Study: Recurrence & Redundancy (Model vs Data)



(a) Recurrence: employment

(b) Recurrence: severe depression

(c) Redundancy: employment

(d) Redundancy: severe depression

Notes: Each panel plots an event study estimated on the data (red) against the identical event study run on data simulated from the model (blue). Shaded areas are 95 percent confidence intervals and the year before the event is the omitted reference, so every line is normalised to zero at that point. The same specification, window and age range are applied on both sides, so the two series are directly comparable. The sample is restricted to individuals aged 25 to 55, and the model is simulated at the estimated parameters. Recurrence is a return to severe depression after a period of recovery, an event the model generates on its own. Redundancy in the data is an involuntary job loss. The model has no involuntary displacement, since leaving work is always a choice, so the closest available counterpart is used: a move out of work by someone who had been employed for at least two years. The model matches the depth of the employment loss and the pace of the recovery that follows it, but not the decline that begins before the event in the data, which reflects the anticipation of job loss that a voluntary separation cannot reproduce.

Recovery of Permanent Types. The permanent types of Table 6 are treated as observed throughout estimation, so we verify that the classification procedure can recover the model's own type structure. We apply the k-means of Appendix F to the model's simulated panel, constructing the same person-level moments as in the data, namely health histories over the full observed lifespan and the wage predicted at age 40 from the estimated wage equation. Table 7 reports the resulting confusion matrix: 95.4 percent of simulated individuals are assigned to their true type, with Types 3, 4 and 5 recovered essentially exactly and the only substantive confusion a 14 percent leak from Type 2 into Type 1, the two types whose permanent wage gap is smallest.

Table 7: Recovery of Permanent Types from Model-Simulated Data

	Recovered Type (% of row)				
True type	1	2	3	4	5
Type 1	**95.5**	4.5	0.0	0.0	0.0
Type 2	14.4	**85.3**	0.1	0.1	0.2
Type 3	0.0	0.0	**98.9**	0.6	0.5
Type 4	0.0	0.5	0.1	**98.6**	0.9
Type 5	0.0	0.1	0.2	1.4	**98.3**

Notes: Rows identify true types; columns report the percentages assigned to each recovered type, with correct assignments in bold. Classification uses health histories over simulated ages 25–80 and wages predicted at age 40; Appendix F describes the full procedure. Overall accuracy is 95.4 percent.

6 Genetic Risk and the Design of Precision Psychiatry

We use the model estimated in Section 5 to examine how genetic risk shapes economic outcomes and whether information about this risk can improve the allocation of therapy. In the model, high-risk men experience larger economic losses and more persistent non-employment following an episode of the depression trap. Giving high-risk applicants priority access to therapy improves health outcomes and the fiscal balance while holding the total number of treatment courses fixed. These gains depend on the information available to the planner: imperfect risk classification substantially reduces them.

6.1 Genetic Risk and Economic Inequality

We first quantify the contribution of genetic risk to economic inequality within the model. This exercise tries to understand the causal genetic effect by asking how an individual's outcomes would differ under an alternative genetic endowment at conception, holding background conditions fixed (Biroli et al., 2022). For individuals first observed between ages 25 and 45, we simulate outcomes under both high and low genetic risk, holding all other initial characteristics fixed.[20] Outcomes are measured over the following ten years, except assets, which are measured immediately before retirement at age 65. Figure 7 reports the results.

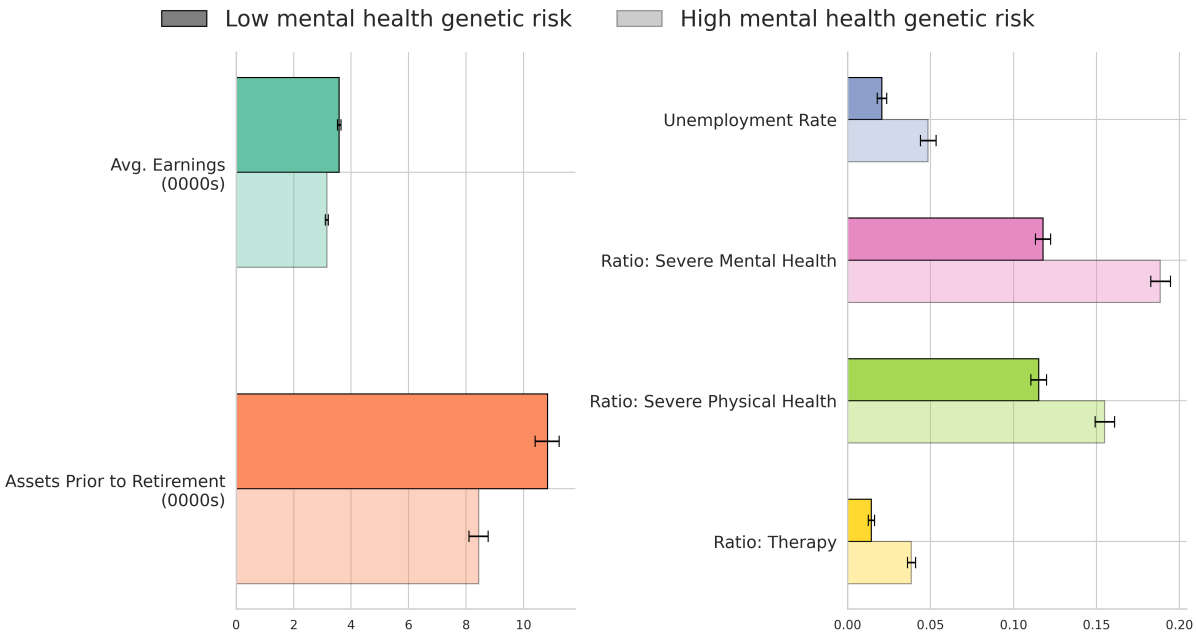
Switching from low to high genetic risk raises non-employment from 2.0 to 4.8 percent, lowers average earnings by 11 percent, and reduces assets at retirement by 22 percent. The share of person-years in the Severe mental health state is 60 percent higher, and therapy use more than doubles. Thus, even with the same initial health and resources, genetic risk generates substantial differences in subsequent health, employment, and wealth.

6.2 Genetic Risk and Recovery from the Depression Trap

Beyond predicting economic outcomes over the life cycle, genetic risk may also be informative about recovery from the depression trap. Table 3 shows that high-risk men are less likely to

[20] Our counterfactual changes genetic-risk group conditional on adult initial states. It therefore measures subsequent model-implied effects, excluding differences already embodied in initial health, human capital, and assets.

Figure 7: Genetic Risk and Economic and Health Outcomes



Notes: Model simulations assign each individual both high and low genetic risk for depression, holding all other initial characteristics fixed. Individuals enter at ages 25–45; outcomes cover the next ten years, except assets, measured at age 65. Earnings and assets are in £10,000s at 2015 prices; other outcomes are shares of person-years.

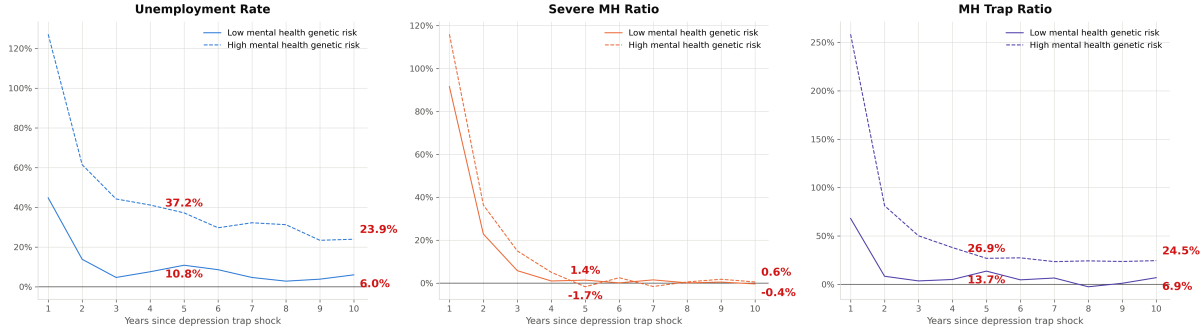
escape the trap. We use the structural model to examine the dynamics behind this pattern by tracing recovery from a joint episode of Severe mental health and non-employment, complementing the separate analyses of job loss and mental health deterioration in Section 3.5.

Starting from the empirical initial distribution, we place all individuals first observed between ages 25 and 45 in the Severe mental health state and in non-employment in the first year. All other initial characteristics remain at their benchmark values. We follow outcomes for ten years and compare each genetic group with its own benchmark path. Figure 8 expresses these differences as percentages of the corresponding benchmark levels.

Mental health prevalence returns close to its benchmark, while the economic losses persist. By year 4, the share in Severe mental health is within 2 percent of its benchmark in both groups. Non-employment remains elevated for longer, particularly among high-risk men. At year 10, their proportional increase in non-employment is about four times that of low-risk men, and their proportional increase in trap prevalence is about three and a half times as large. A recovery in aggregate mental health prevalence therefore coexists with persistent economic losses.

In the model, high-risk men are more likely to remain in the Severe mental health state or return to it after recovery. These prolonged or repeated episodes increase their time out of work. During non-employment, human capital depreciates, lowering subsequent wage offers and making a return to work less attractive even after mental health improves. Being out of work also increases the risk of further mental health deterioration. This feedback between health and employment makes the depression trap more persistent among high-risk men.

Figure 8: Persistence Following a Depression-Trap Episode by Genetic Risk



Notes: Individuals initially aged 25–45 are placed in Severe mental health and non-employment in year 1; all other initial characteristics are unchanged. Subsequent choices and transitions follow the estimated model. Each series reports $(Y_{g,t}^{\text{shock}}/Y_{g,t}^{\text{benchmark}} - 1)$, where $Y_{g,t}$ is the relevant outcome value for genetic-risk group g in year t. The depression trap is joint Severe mental health and non-employment. Labels report years 5 and 10.

Appendix Figure 16 repeats the experiment among individuals employed before the shock and, separately, among those initially in Good mental health. The genetic differences in subsequent outcomes persist under both restrictions. Genetic risk therefore provides information about economic persistence beyond either of these initial states, motivating its use as an additional signal for allocating access to treatment.

6.3 Priority Access to Therapy

The preceding experiments show that genetic risk predicts both economic disadvantage and difficulty escaping the depression trap. We therefore ask whether giving high-risk applicants priority access to therapy can improve health and fiscal outcomes while holding aggregate treatment capacity at its benchmark level. In the benchmark, every applicant receives a treatment offer with probability $\lambda_\tau = 0.689$, calibrated to the ratio of treatment starts to referrals in NHS Talking Therapies (NHS England, 2024). The policy replaces this common probability with risk-specific probabilities $\lambda(L)$ and $\lambda(H)$.

Let X denote the proportional reduction in high-risk applicants' expected waiting time relative to low-risk applicants. Under continued annual applications and a constant offer probability $\lambda(g)$, expected waiting time is $(1-\lambda(g))/\lambda(g)$, measured in completed years before treatment. The priority rule is therefore

$$\underbrace{\frac{1-\lambda(H)}{\lambda(H)}}_{\substack{\text{Expected waiting time}\\ \text{for the high-risk group}}} = (1-X) \underbrace{\frac{1-\lambda(L)}{\lambda(L)}}_{\substack{\text{Expected waiting time}\\ \text{for the low-risk group}}}. \tag{21}$$

We consider $X \in \{25\%, 50\%, 75\%\}$, so high-risk applicants wait 25, 50, or 75 percent less than low-risk applicants.

For each X, $\lambda(L)$ and $\lambda(H)$ adjust so that the total number of therapy courses remains at its benchmark level, allowing application decisions to respond to access probabilities. Priority therefore reallocates existing treatment capacity towards high-risk applicants.

Table 8 reports the results. Fiscal gains increase with priority intensity, reaching £25.3 per person in present value at age 25 under $X = 75\%$. Of this gain, £8.5 comes from higher tax revenue and £16.8 from lower benefit payments. The policy thus improves the public balance through both earnings and benefit receipt, despite holding treatment capacity fixed. This fiscal gain is comparable in magnitude to the cost of obtaining genetic information: SNP-array genotyping costs about US$20 per person (Raben et al., 2022).

The same reallocation improves the clinical return to treatment. Years in the Severe mental health state averted per course rise from 0.415 to 0.534, an increase of about 29 percent. The policy's main contribution is to increase the effectiveness of therapy courses and the fiscal returns to the available courses. However, whether these gains can be realized depends on the planner's ability to identify high-risk applicants.

Table 8: Priority Access to Therapy

	Treated (%)		Fiscal effect (£ per person)			Years in severe depression
Priority (X)	Low risk	High risk	Tax revenue	Benefit savings	Net balance	averted per course
benchmark	1.22	2.51	—	—	—	0.415
25%	1.03	2.72	4.8	7.9	12.8	0.444
50%	0.79	2.98	5.7	9.6	15.3	0.482
75%	0.45	3.33	8.5	16.8	25.3	0.534

Notes: Total therapy provision is held at its benchmark level. Priority X reduces high-risk applicants' expected waiting time by X relative to low-risk applicants. With continued annual applications and annual offer probability $\lambda(g)$, expected waiting time is $(1-\lambda(g))/\lambda(g)$ completed years before treatment. Treated shares are percentages of working-age person-years within each genetic-risk group. Fiscal effects are changes from the no-priority benchmark, in pounds per person discounted to age 25; net balance equals tax revenue plus benefit savings. The final column reports years in the Severe mental health state averted per therapy course.

6.4 The Value of Genetic Information for Treatment Priority

The preceding experiment assumes that the planner observes genetic risk. We now examine how much the policy depends on this information. We first ask whether observed histories can predict genetic-risk group, using simulated data and survey data linked to respondents' genetic information. The predictors summarize mental and physical health, employment, earnings, and therapy receipt up to a given age.

Table 9 reports cross-validated prediction performance. In the simulated data, a gradient-boosting classifier achieves an area under the receiver operating characteristic curve (AUC) of 0.974 by age 30 and approximately 0.99 thereafter. Prediction is substantially weaker in the survey: AUC ranges from 0.589 to 0.667. The near-perfect predictability within the model therefore does not carry over to the observed panel. Health, employment, earnings, and treatment

histories help predict an individual's genetic-risk group, but substantial classification errors remain when genetic information is unavailable.

Table 9: Predicting Genetic Risk from Observable Histories

History observed to age	30	35	40	50	60
AUC (simulated data)	0.974	0.992	0.993	0.994	0.992
AUC (survey data)	0.589	0.649	0.667	0.650	0.591

Notes: Entries are cross-validated areas under the receiver operating characteristic curve (AUC) for predicting high versus low genetic risk from health, employment, earnings, and therapy histories through the stated age. An AUC of 0.5 indicates chance discrimination and 1 perfect discrimination. Genotype is excluded from predictors.

To assess the value of genetic information, we repeat the priority-access experiment under imperfect risk classification. We take an AUC of 0.65, consistent with the survey evidence in Table 9, as a proxy for realistic information quality. In the model, the individual knows his true genetic-risk group g, but the planner assigns him to the correct group with probability q and to the other group with probability $1-q$. This represents a setting in which a clinician observes a patient's health, employment, earnings, and treatment histories, but has no genetic information. The clinician must use these histories to infer the patient's risk group and assign priority access to therapy. The individual understands these and accounts for them when deciding whether to apply for therapy. His value function is

$$
\begin{aligned}
V_t(S_t) &= \max_{d_t \in \{0,1\}} \left\{ (1-d_t) V_t^0(S_t) + d_t \left[\widetilde{\lambda}(g) V_t^1(S_t) + \left(1-\widetilde{\lambda}(g)\right) V_t^0(S_t) \right] \right\}, \\
\widetilde{\lambda}(g) &= q\lambda(g) + (1-q)\lambda(g'), \qquad g' \neq g.
\end{aligned}
\tag{22}
$$

Here, $V_t^1(S_t)$ and $V_t^0(S_t)$ are the conditional value functions in Equation (15) for receiving and not receiving therapy, respectively, and g' denotes the other genetic-risk group. These values and the health transition probabilities continue to depend on the individual's true risk; classification errors affect his access to therapy. Setting $q = 1$ recovers the full-information policy.

Table 10 shows that imperfect risk classification substantially reduces the returns to priority access. Under the strongest priority rule, the fiscal gain falls from £25.3 to £7.4 per person, a reduction of about 71 percent. Years in the Severe mental health state averted per course fall from 0.534 to 0.437, compared with 0.415 in the no-priority benchmark. Imperfect classification therefore removes about 82 percent of the incremental clinical gain. Much of the value of targeting therapy by genetic risk thus depends on identifying high-risk applicants accurately.

Under the strongest priority rule ($X = 75\%$), observing genetic risk directly rather than relying on imperfect classification raises the fiscal return by £17.9 per person in present value at age 25. Given the reported SNP-array genotyping cost of about US$20 per person (Raben et al.,

Table 10: Information Quality and the Returns to Priority Access

Priority (X)	Information	Fiscal effect (£ per person)			Years in severe depression averted per course
		Tax revenue	Benefit savings	Net balance	
benchmark		—	—	—	0.415
25%	Perfect	4.8	7.9	12.8	0.444
	Imperfect	0.6	0.1	0.7	0.421
50%	Perfect	5.7	9.6	15.3	0.482
	Imperfect	2.7	4.1	6.8	0.429
75%	Perfect	8.5	16.8	25.3	0.534
	Imperfect	2.7	4.6	7.4	0.437

Notes: Total therapy provision is held at its benchmark level. Priority X reduces expected waiting time for applicants assigned to the high-risk group by X relative to those assigned to the low-risk group. Expected waiting time is $(1 - \lambda(g))/\lambda(g)$ completed years under continued annual applications with offer probability $\lambda(g)$. Perfect information reveals true genetic risk. Imperfect information assigns the correct risk group with $AUC = 0.65$ in each group. Fiscal effects are changes from the no-priority benchmark, in pounds per person discounted to age 25; net balance equals tax revenue plus benefit savings and excludes the costs of obtaining risk information and administering priority access. The final column reports years in the Severe mental health state averted per therapy course.

2022), the additional fiscal returns from better targeting of mental health treatment alone could finance genotyping of the working-age male population represented by the model.[21]

7 Post-Pandemic Mental Health and Therapy Expansion

Section 6 examines how genetic information can improve the allocation of a fixed treatment capacity. We now use the same framework to assess the economic consequences of the post-pandemic shift in mental health and the returns to expanding therapy provision.

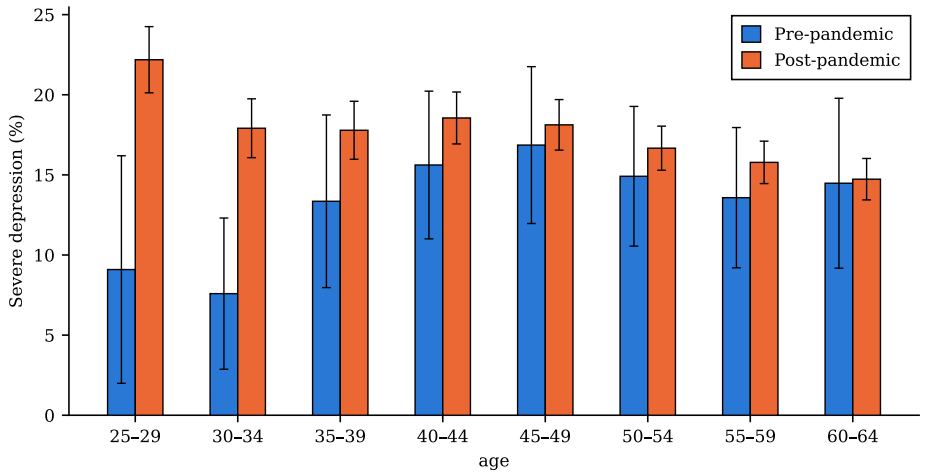
7.1 The Fact

Figure 9 compares the share of people in severe depression before the pandemic with the share in the most recent wave, by age, in the survey data the model is estimated on. Across working ages the share rises from 14.0 to 17.3 percent. The aggregate change is not the striking part. What is striking is where it falls. Among people aged 25 to 29 the severe share more than doubles, from 9.1 to 22.2 percent, and the increase then declines monotonically with age until it is essentially absent among those aged 60 to 64. The pandemic did not shift the population's mental health uniformly. It rotated the age profile, replacing a gradient that rose with age with one that is close to flat.

This age pattern matters because a deterioration early in working life can affect employment, human capital, and saving over a longer remaining horizon. We use the model to quantify these consequences. The comparison describes an observed post-pandemic change; it does not separately identify the pandemic's causal effect on mental health.

[21] This comparison concerns the cost of genotyping and excludes additional implementation costs.

Figure 9: Severe depression by age, before and after the pandemic

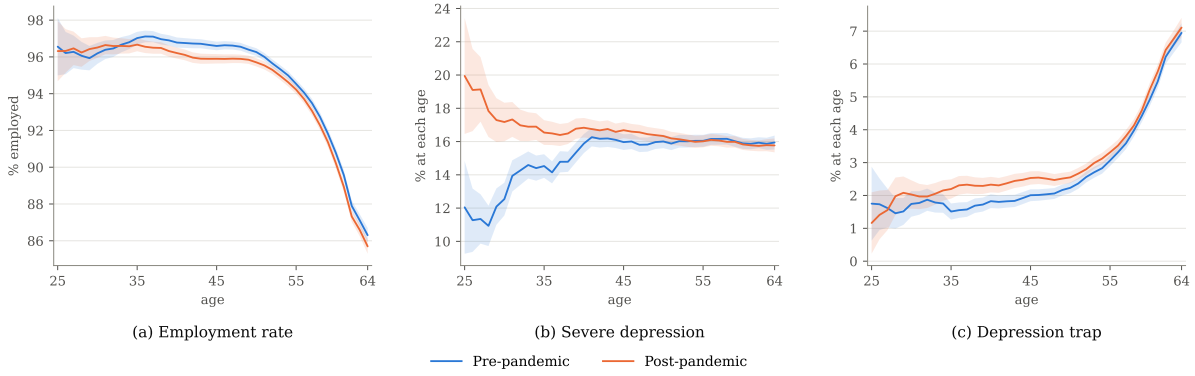


Notes: Survey-weighted share of each age band reporting severe depressive symptoms, in the pre-pandemic estimation sample and in the most recent post-pandemic wave. Bars show 95 percent confidence intervals computed from the effective sample size implied by the survey weights, which is why the pre-pandemic bands, drawn from a much smaller sample, are visibly less precise.

7.2 Employment and Fiscal Consequences

We feed this shift into the estimated model. The post-pandemic population is constructed by taking the pre-pandemic population and moving only the mental-health margin until its age profile matches the post-pandemic data, holding genotype, permanent type, physical health, assets and human capital fixed. Everything the model then produces is attributable to mental health.[22]

Figure 10: Simulated life-cycle profiles before and after the mental-health shift



Notes: Simulated shares by age in the pre-pandemic economy and in an economy identical to it except that the age profile of mental health has been shifted to match the post-pandemic data. The depression trap is the share who are both depressed and out of work. Each series is a centred five-year moving average with a 95 percent confidence band built from the sample inside the window, which is why the bands are wide at the youngest ages, where few of the simulated agents have yet entered.

[22]The reassignment is rank-preserving: within each age band, agents are ordered by observed mental-health state first and by a fitted severity index second, and the required number are moved. This keeps the correlation between mental health and its observable predictors intact, which random reassignment would destroy and reweighting would exaggerate.

Figure 10 shows the resulting life-cycle profiles. Employment falls, severe depression rises, and the share of people in the depression trap, rises with it. As in the data, the damage is concentrated at the young ages, and the three panels show it propagating: the mental-health gap opens first, the employment gap follows, and the trap accumulates over the life cycle rather than at the moment of the shock.

Table 11 reports the fiscal consequences over the working life. The mental health shift reduces net fiscal contributions by £804 per person in present value at age 25, or 1.38 percent of the pre-pandemic contribution. Tax revenue falls by £529 and benefit payments rise by £275. Although benefits increase more in percentage terms, lower labor income tax revenue accounts for approximately two-thirds of the fiscal loss. Worse mental health therefore places pressure on both sides of the public budget, reducing labor income tax revenue while also substantially increasing spending on benefits.

Table 11: Fiscal cost of the mental-health shift, per person

Present value at age 25	Pre-pandemic	Post-pandemic	Change
Tax paid	£64,381	£63,852	−0.82%
Benefits received	£5,932	£6,207	+4.63%
Net contribution	£58,449	£57,645	−1.38%

Notes: Present value at age 25, discounted at the model's interest rate, of labour income tax paid and of benefits received over the working life, per person. Tax follows the estimated progressive schedule and applies to labour income only; benefits and asset income are untaxed and pension income is excluded. The two economies differ only in the age profile of mental health.

7.3 Expanding the Therapy Supply

To address the post-pandemic deterioration in mental health, we evaluate an expansion in therapy supply, a policy already being pursued by the NHS.[23] NHS England's medium-term planning framework targets an increase from 805,000 courses in 2026/27 to 915,000 in 2028/29, approximately 14 percent (NHS England, 2025). The 2023 workforce plan committed over £600 million to training approximately 15,000 additional psychological therapists and practitioners, implying average expenditure of roughly £40,000 per trainee (NHS England, 2023). We simulate expansions by increasing the probability of being offered a course by 5 to 20 percent, holding the psychological cost of applying fixed.

Table 12 reports the fiscal and health returns per additional therapist. The health gains are substantial: each additional therapist averts 557, 482, and 437 person-years in the Severe mental health state under the 5, 10, and 20 percent expansions, respectively. The declining return per therapist is consistent with diminishing returns as unmet treatment needs are progressively addressed: smaller expansions deliver larger gains per additional therapist, while

[23] Appendix H.4 and H.5 present two complementary theoretical experiments: a subsidy that offsets the mental health wage penalty and a reduction in the psychological cost of taking therapy. These experiments examine how employment incentives and the cost of seeking treatment affect therapy demand and mental health.

further increases in capacity yield smaller additional benefits. To put these gains in perspective, individuals who experience Severe mental health spend, on average, 4.1 years in that state over the working years observed in our data. The person-years averted are therefore equivalent to the entire observed Severe mental health burden of approximately 136, 118, and 107 such individuals, respectively.

These health improvements also generate fiscal gains through higher tax revenue and lower benefit payments. At the 10 percent expansion, each additional therapist generates £226,000 in tax revenue and reduces benefit payments by £146,000. Against these gains, the government spends £316,000 on therapist wages, leaving a net fiscal surplus of £56,000 per additional therapist. The expansion is therefore self-financing, with a surplus that also exceeds the roughly £40,000 expenditure per trainee implied by the workforce plan. As the expansion increases, diminishing fiscal returns become decisive: tax revenue gains and benefit savings per therapist fall, while wage costs remain roughly constant. At the 20 percent expansion, the net fiscal balance becomes a deficit of £103,000 per additional therapist, although the health gains remain substantial.

Table 12: Fiscal and Health Returns per Additional Therapist

	Fiscal effects (£1,000 per therapist)				Severe mental health
Increase in therapist workforce	Tax revenue	Benefit savings	Therapist wages	Net fiscal balance	person-years averted
5%	266	188	351	+103	557
10%	226	146	316	+56	482
20%	160	57	321	−103	437

Notes: Workforce expansion is modeled by increasing the annual treatment-offer probability, holding the psychological cost of applying fixed. Results are changes from the post-pandemic benchmark divided by the implied number of additional therapists, using the assumed working-age population served per therapist. Fiscal effects are in thousands of pounds per additional therapist, in present value at age 25. Therapist wages are paid by the government; net fiscal balance equals additional tax revenue plus benefit savings less these wage costs, before training costs. The final column reports person-years in the Severe mental health state averted per additional therapist.

8 Conclusion

In this paper, we show that genetic predisposition to depressive symptoms strongly predicts lifetime mental health and economic outcomes. A one standard deviation increase in genetic risk for depression raises the probability of probable major depression by 6.9 p.p., lowers the probability of employment by 3.1 p.p. and lowers labor income by 0.12 s.d. These results are robust to the inclusion of controls for socioeconomic background and broadly consistent with within-family estimates. We identify the "depression trap", a self-reinforcing cycle of depression and non-employment, as a key mechanism explaining this disparity in labor market outcomes. Men with a high genetic risk for depression are more likely to enter the trap and less likely to leave it. Each leg of that cycle is identified from a shock of its own, involuntary job loss and

the recurrence of a depressive episode, and both shocks cost high-risk men more employment despite doing them no more clinical damage.

We then estimate a life-cycle model of mental health, benefits, treatment and labor supply, in which therapy is rationed rather than priced. Holding initial conditions fixed, genetic endowment drives inequalities in lifetime earnings, employment, asset accumulation and health. Its most consequential role, though, is not in how ill people become but in whether they come back. Placing every agent in the depression trap and simulating recovery, severe depression returns to baseline within four years at nearly the same speed for both genetic groups, while a decade later non-employment among high-risk men remains four times as far above its own baseline. Because talking therapy in England is free at the point of use and rationed by a queue, the instrument a planner holds is the order of that queue. Giving high-risk individuals priority raises employment, lowers severe depression and improves the fiscal balance, and under fixed capacity it does so without treating anyone extra and so without costing the exchequer anything. The gain comes from who is treated rather than from how quickly the queue moves: accelerating everyone equally returns exactly zero. The policy is nonetheless a reallocation and not a Pareto improvement: at high intensities the wait facing lower-risk patients lengthens from months to years, and that cost has to be weighed against the gain rather than netted out of it.

Our analysis builds on recent advances in psychiatric genetics (Baselmans et al., 2019). It highlights the untapped potential of Polygenic Indices to inform cost-effective precision public health strategies. Currently, PGIs explain only about 5% of the variation in depressive symptoms within our sample. Because of this, our estimates likely represent a lower bound of the true impact of genetic risk. Yet, even at this lower bound, the life-cycle effects and policy responses we uncover are non-trivial.

Furthermore, the predictive power of GWAS is rapidly increasing. As larger-scale databases and higher-quality sequencing become available, genetic predictors could theoretically capture up to 40% of the variation in depression (Sullivan et al., 2000; Wainschtein et al., 2025). The UK National Health Service already plans to offer whole-genome sequencing to all newborns over the next decade (Davies, 2025). As these predictive tools mature, using ex-ante genetic screening to keep vulnerable individuals out of the "depression trap" will become a highly practical policy option.

The data and scope of this paper limits the applicability of our findings: we focus on the working-age men as we have very few observations below age 25 and to avoid modelling fertility decisions. Future research could use data focussed on early childhood development to explore in more detail the mechanisms by which genetic risk for depression affects the development of mental health and human capital. Early-life, targeted mental health interventions could plausibly be even more cost-effective, insofar as they could improve educational and occupational attainment. Follow-up research could also examine how genetic risk for depression

shapes the career and mental health outcomes of women: women suffer from depression at higher rates than men (IHME, 2025) and have more elastic labor supply (Keane, 2011). Appendix B shows that the descriptive evidence of Section 3 replicates for women, and often more strongly, so what stands in the way is not the reduced form but the modelling; Appendix C sets out what would be required. Genetic risk predicts women's completed fertility as well as their health and labor supply, about a third of their employment penalty runs into caring for home and family rather than into disability, and the recurrence design that identifies the mental-health-to-employment leg of the trap fails for them because episodes coincide with the employment interruptions around childbearing. Extending the model to women therefore means treating fertility, labor supply and health as jointly determined, rather than re-estimating the present model on a different sample. Furthermore, our analysis focusses on white men with european-like ancestry, despite the fact that ethnic minorities suffer from mental health issues at higher rates in the UK (Nazroo and Iley, 2011). This is due to data limitations, both in our discovery and target samples and due to the limited portability of PGIs across ancestry groups (Kim and Milliken, 2018). As more data becomes available, and better methods are developed for improving the portability of PGIs, exploring the role of genetic risk for depression in shaping the mental health and labor market outcomes of minority ethnic groups will be of increasing research interest.

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Appendices

A Genetic Data and Polygenic Indices

We summarize genetic predisposition using polygenic indices (PGIs) constructed from common genetic variants. The raw genotype data record variation at single-nucleotide polymorphisms (SNPs), which are genomic locations where individuals differ by a single DNA base. At each SNP j, an individual carries two alleles, one inherited from each biological parent. We code the genotype as an allele count $SNP_{i,j} \in \{0, 1, 2\}$, which equals the number of effect alleles carried by individual i at SNP j.

GWAS provide the weights used to aggregate SNPs into a scalar index. In a GWAS discovery sample, indexed by k, the outcome of interest Y_k is regressed on each SNP separately:

$$Y_k = \beta_j^{GWAS} SNP_{k,j} + \gamma \mathbf{X}_k + u_{k,j}, \tag{23}$$

where $\mathbf{X}_k$ includes standard controls such as sex, age polynomials, and genetic principal components to address population structure. The estimated SNP coefficients $\widehat{\beta}_j^{GWAS}$ are then applied to individuals in our UKHLS analysis (target) sample, indexed by i, to form the PGI:

$$PGI_i = \sum_{j=1}^{J} \widehat{\beta}_j^{GWAS} SNP_{i,j}. \tag{24}$$

Consistent with evidence that predictive performance improves when many variants are included, we construct the index using genome-wide SNP information rather than restricting to genome-wide significant hits (Okbay et al., 2016).

The UKHLS genetic subsample includes blood and saliva genotypes for 9,961 respondents. The released data contain 24,214,238 SNPs (hg38 build). After standard quality control, 9,920 individuals remain for analysis. We construct PGIs using PLINK2 and a standard workflow following Choi et al. (2020), combining UKHLS genotypes with publicly available GWAS summary statistics. In addition to the depressive-symptoms PGI used in the main analysis, we also construct PGIs for a broader set of traits for robustness and controls, including educational attainment, health, and related outcomes.

A.1 UKHLS Genetic Sample

We use data from Understanding Society: the UK Household Longitudinal Study (UKHLS). UKHLS is a nationally representative panel with rich information on income, employment, education, health, and well-being. A genetic subsample collected blood and saliva from 9,961 respondents. Genotypes are available on 24,214,238 SNPs (hg38 build). After standard quality control, 9,920 individuals remain.

We construct PGIs using PLINK2 and a standard pipeline following Choi et al. (2020). We combine UKHLS genotypes with publicly available GWAS summary statistics. In addition to the depressive-symptoms PGI used in the main analysis, we construct PGIs for a wider set of traits for robustness and controls, including education, health, and related outcomes.

The genetic data can be linked to family members within UKHLS. This feature allows us to characterize genetic relationships within households and, when needed, study patterns consistent with assortative mating or intergenerational links. Table 13 reports sample sizes for the main variables and the linked-relative subsamples after quality control.

Table 13: Sample Size by Information Availability

Sample Category	N
Core Analysis Sample	
Full genetic sample (post-QC)	9,920
Physical Health	9,920
Mental Health	9,888
Educational attainment	9,915
labor force status	9,920
Benefits	9,920
Saving information	7,178
Therapy	7,299
Linked Biological Relatives	
Biological parent genotype linked	550
Biological child genotype linked	392
Biological sibling genotype linked	223

Notes: This table reports sample sizes for key variables after applying standard genetic quality control (QC) protocols. "Full genetic sample (post-QC)" refers to individuals with valid genomic data. "Linked" categories indicate successful matching of genotype records between the primary respondent and their biological relatives within the dataset.

B Additional Empirical Results

B.1 Descriptive Statistics

Table 14 compares men below and above the median of the depression polygenic index on health, labor market and background characteristics. The health and labor market gaps are large and precisely estimated, while the parental background variables are balanced, which is the comparison Section 3 relies on.

Table 14: Descriptive Statistics and Tests of Differences by Genetic Risk for Depressive Symptoms

	High Genetic Risk		Low Genetic Risk		
Variable	Mean	(Std. Dev.)	Mean	(Std. Dev.)	Diff. (p-value)
Health Outcomes					
Depression (Prob.)	0.23	(0.42)	0.12	(0.32)	0.00
Poor Physical Health (Prob.)	0.18	(0.38)	0.13	(0.33)	0.00
Disability (Prob.)	0.05	(0.22)	0.02	(0.13)	0.00
Therapy (Prob.)	0.02	(0.15)	0.01	(0.11)	0.00
Economic Outcomes					
Annual Pay (£)	33,888	(18,942)	36,493	(21,051)	0.00
High Occ. (Prob.)	0.19	(0.39)	0.24	(0.43)	0.00
Total Assets (£)	18,825	(51,119)	24,342	(101,924)	0.10
Employed (Prob.)	0.89	(0.32)	0.94	(0.24)	0.00
Benefit Receipt (Prob.)	0.28	(0.45)	0.22	(0.42)	0.00
Annual Benefits (£)	3,197	(3,455)	2,765	(3,041)	0.00
Demographics & Human Capital					
College (Prob.)	0.27	(0.45)	0.32	(0.47)	0.00
Father High Occ. (Prob.)	0.18	(0.38)	0.20	(0.40)	0.31
Mother's Education (Yrs)	10.90	(1.92)	11.00	(2.01)	0.26
Age	42.81	(8.33)	42.74	(8.19)	0.55
Observations	9,770		10,574		

Notes: Standard deviations are reported in parentheses. The final column reports p-values from a two-sided t-test for the difference in means between individuals with an depressive symptoms PGI above (High Genetic Risk) and below (Low Genetic Risk) the median. Parental education is mapped to years based on highest qualification. High occupation indicators are binary for managerial roles (SOC90 < 20). Sample sizes for parental variables are smaller ($N \approx 1,500$–$2,500$) due to retrospective reporting. All variables are for UKHLS men aged 25–55.

B.2 Descriptive Employment and Mental-Health Transitions

Table 15 shows that both legs of this relationship are present in our own data. We estimate a dynamic logit for each transition on genotyped men aged 25–55, controlling for lagged earnings, demographics, survey wave, the genetic principal components, and within-person time averages of the time-varying regressors, so that the comparisons are net of permanent individual heterogeneity as well as of income. Panel A reports the effect of last year's employment on this year's mental health. Men who were employed are 9.2 percentage points less likely to become depressed (s.e. 1.8) and 18.0 percentage points more likely to recover if they were depressed (s.e. 2.8). Panel B reports the reverse. Men who were depressed are 1.2 percentage points more likely to leave employment (s.e. 0.3) and 10.2 percentage points less likely to re-enter it (s.e. 2.1).

These magnitudes are associations, not treatment effects, and the recovery and re-entry rows in particular are estimated on the smaller at-risk samples. They nonetheless describe a feedback loop of the size the literature above would lead one to expect, and they do so holding both permanent heterogeneity and past income fixed. Section 3.5 returns to each leg with event-study evidence.

Table 15: The Two-Way Relationship Between Employment and Mental Health

Transition	Effect on annual transition prob. (pp)	Observations at risk
Panel A: Effect of being employed at $t-1$ on mental health transitions		
Onset (healthy → depressed)	-2.15***	67,789
	(0.71)	
Recovery (depressed → healthy)	3.97***	13,787
	(1.24)	
Panel B: Effect of being depressed at $t-1$ on employment transitions		
Exit (employed → non-employed)	1.09***	73,534
	(0.17)	
Entry (non-employed → employed)	-3.79***	8,042
	(0.50)	

Notes: Each entry is an average marginal effect, in percentage points, of the lagged state named in the panel heading on the transition named in the row, from a dynamic logit on UKHLS men aged 25–55 observed for at least five waves. Every row is signed to read as the effect on the transition named, so a positive entry in Panel B row one means depression raises the probability of leaving employment. Marginal effects are averaged over the men at risk of each transition. Controls: lagged earnings, so these are comparisons at a given past income, plus number of children, a quadratic in age and wave fixed effects, and within-person time averages of the time-varying regressors, so the comparisons are net of permanent heterogeneity. Unlike Table 3 this table is estimated on the full panel rather than the genotyped subsample, since it has no genetic regressor; the genotype restriction leaves only 544 within-person employment switches against 2,772 depression switches, too few to sign Panel A. Depression is GHQ-12 caseness ≥ 4. Standard errors, clustered at the individual level, are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

B.3 Dynamics by Genetic Risk: Alternative Specifications

Table 16 repeats the main transition table at an alternative GHQ-12 caseness threshold of 2, which counts mild symptoms and above, instead of the threshold of 4 used throughout the paper. The estimates are almost unchanged. Depression onset is 3.64 percentage points higher for high-risk men at the lower threshold against 3.51 percentage points in the main table, and the trap exit differential is -26.14 percentage points against -24.44. The choice of caseness threshold is therefore not driving the genetic gradient in either the entry or the persistence margin.

Table 16: Dynamics by Genetic Risk: Alternative GHQ-12 Threshold (Caseness ≥ 2)

Annual transition probability (%)	Low genetic risk (1)	High genetic risk (2)	Difference (2)−(1)
Panel A: Mental health dynamics			
Onset (healthy → depressed)	16.29	19.93	3.64***
	(0.70)	(0.83)	(1.09)
Recovery (depressed → healthy)	55.61	42.51	-13.10***
	(1.62)	(1.37)	(2.12)
Panel B: Employment dynamics			
Exit (employed → non-employed)	1.58	2.26	0.68**
	(0.18)	(0.21)	(0.28)
Entry (non-employed → employed)	38.40	25.07	-13.33***
	(3.83)	(2.21)	(4.42)
Panel C: "Depression trap" dynamics			
Entry into trap	1.22	1.86	0.64***
	(0.16)	(0.18)	(0.24)
Exit from trap	61.09	34.95	-26.14***
	(5.78)	(3.18)	(6.60)

Notes: Each entry is an average predicted annual transition probability, in percentage points, from a dynamic logit on genotyped UKHLS men aged 25–55 observed for at least five waves. Genetic risk is a median split of the depression polygenic index. The model is estimated separately within each risk group and both fits averaged over the same pooled covariate distribution, so column (3) is a difference in behaviour, not composition; its standard error is the square root of the sum of the two squared standard errors, valid because the subsamples are disjoint. Each cell conditions on the origin state only, so entries are marginal with respect to the other lagged state. Controls: lagged earnings, number of children, a quadratic in age, wave fixed effects and the ten genetic principal components. Depression is GHQ-12 caseness ≥ 2; the "depression trap" is non-employment jointly with caseness ≥ 2. Standard errors clustered on the individual. All panels use 15,206 person-waves on 1,534 individuals. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Conditioning on Permanent Types

The main table conditions on lagged earnings, demographics, wave and the genetic principal components, but not on the permanent heterogeneity the structural model carries as the permanent type x_i. Since types govern productivity and health trajectories, a natural question is whether the genetic gradient is instead a gradient in type. Table 17 adds indicators for the five types of Appendix F to the specification, holding the sample fixed at the 1,534 men who have a type assignment.

Column (4) is the specification to read. The types are estimated on moments residualized on a depression polygenic index, so conditioning on them is only clean when the sample is split on that same index; the check is otherwise contaminated by construction, since the type dummies then absorb part of the genetic signal they were built to exclude. On that split the type dummies change essentially nothing. Five of the six margins land between 92 and 121 percent of their uncontrolled values, only trap exit moves materially, from -21.30 to -17.90 percentage points, and every estimate remains significant at the five percent level or better. The implied trap spell is 1.80 years at low genetic risk against 2.66 at high, compared with 1.72 against 2.72 without the type dummies.

Column (2) shows what the contaminated version of the same check looks like: the persistence margins shed about a fifth, with recovery falling from -15.31 to -12.11 and trap exit from -24.44 to -19.06. That attenuation is the mechanical consequence of conditioning on a variable built from person-average mental health that has been purged of a correlate of the split variable rather than of the split variable itself. It is reported here because the distinction between the two columns is the substance of the exercise, not because the column (2) attenuation should be read as evidence of confounding.

The residual caveat applies to both columns. Types are a five-point discretization of person-mean mental health, physical health and wages, and only the linear projection on the polygenic index is removed when they are constructed, so a nonlinear residue of "conditioning on a consequence of genetic risk" survives. That residue biases against finding a gradient, so the estimates in columns (2) and (4) are if anything conservative. A correlated random effects version of the same table is the limiting case of this logic and is uninformative for exactly that reason: within-person time averages absorb all between-person variation, which is the only variation a fixed genetic endowment has, and every one of the six transitions is nulled. Five type dummies do not, which is what makes them informative.

Table 17: Dynamics by Genetic Risk, Conditioning on Latent Types

Difference in annual transition probability, high − low risk (pp)	Main index (Table 3)		Type-clustering index (alternative depression PGI)	
	Baseline (1)	+ types (2)	Baseline (3)	+ types (4)
Panel A: Mental health dynamics				
Onset (healthy → depressed)	3.51***	3.75***	6.28***	6.56***
	(0.78)	(0.70)	(0.75)	(0.65)
Recovery (depressed → healthy)	-15.31***	-12.11***	-14.54***	-13.36***
	(2.72)	(2.20)	(2.79)	(2.19)
Panel B: Employment dynamics				
Exit (employed → non-employed)	0.71**	0.76***	0.81***	0.98***
	(0.28)	(0.28)	(0.26)	(0.27)
Entry (non-employed → employed)	-13.70***	-11.65***	-9.12*	-9.20**
	(4.47)	(4.11)	(4.67)	(4.22)
Panel C: "Depression trap" dynamics				
Entry into trap	0.69***	0.63***	0.98***	1.02***
	(0.23)	(0.23)	(0.21)	(0.21)
Exit from trap	-24.44***	-19.06***	-21.30***	-17.90**
	(7.68)	(6.96)	(8.19)	(6.99)
Expected trap spell, low risk (years)	1.66	1.77	1.72	1.80
Expected trap spell, high risk (years)	2.80	2.68	2.72	2.66
Latent type dummies	No	Yes	No	Yes

Notes: Each entry is the difference in the average predicted annual transition probability between high- and low-genetic-risk men, in percentage points, constructed exactly as in Table 3: a dynamic logit estimated separately within each risk group, both fits averaged over the same pooled at-risk covariate distribution, with SE(diff) $= \sqrt{se_0^2 + se_1^2}$. Columns (2) and (4) add indicators for the five latent types x_i estimated by the clustering procedure of Appendix F and treated as a fixed state variable in the structural model. Columns (1)–(2) split on the depressive-symptoms index used throughout the paper; columns (3)–(4) split on the alternative depression index that the type clustering residualises its moments against, so that in column (4) the type dummies are orthogonal to the split variable by construction. Column (4) is therefore the cleaner comparison: the type moments are purged of the column (3)–(4) index but not of the column (1)–(2) index, with which they correlate 0.79, so column (2) mechanically absorbs part of the genetic signal that column (4) does not. All four columns use the same 15,206 person-waves on 1,534 men, namely the Table 3 sample restricted to men with a type assignment; the two median splits agree on 78.9 percent of them. Expected trap spells are the reciprocal of the exit probability. Controls otherwise as in Table 3. Standard errors clustered on the individual. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Women

Table 18 repeats the same table for women. The construction is identical except for the sex filter and the labor-force states admitted: women's sample adds looking after home and family, because on the men's state set those person-waves are deleted rather than counted as non-employment, and that is where most of women's non-employment sits.

Every one of the six gradients holds with the same sign and at conventional significance. Women at high genetic risk are 4.7 percentage points more likely to become depressed in a given year and 8.7 points less likely to recover; 0.9 points more likely to leave employment and 3.8 points less likely to enter it; and 1.7 points more likely to enter the depression trap. Entry into the trap is in fact more strongly graded for women than for men, at 1.65 percentage points against 0.69.

Exit is not. The women's exit differential is −7.66 points against the men's −24.44, and it is the one estimate in this appendix that moves materially with a specification choice: on the men's narrower state set it falls to −1.65 (s.e. 5.62) and is indistinguishable from zero. Since the paper's mechanism is about persistence rather than incidence, this is the most consequential difference between the sexes, and the sensitivity is itself informative. For men, leaving employment while depressed is close to unambiguous, and the state the model calls the depression trap is the state the data record. For women a large share of the same transitions run into and out of family care, which for many is a response to a birth rather than a symptom of illness. Appendix C shows that this is not a hypothetical concern: the arrival of a first child moves women's employment by more than thirty percentage points while moving their mental health no more than it moves men's.

Table 18: Mental Health, Employment and "Depression Trap" Dynamics by Genetic Risk: Women

	Women			Men
Annual transition probability (%)	Low genetic risk (1)	High genetic risk (2)	Difference (2)−(1)	Difference (3)
Panel A: Mental health dynamics				
Onset (healthy → depressed)	14.03	18.73	4.71***	3.51***
	(0.51)	(0.60)	(0.78)	(0.78)
Recovery (depressed → healthy)	54.85	46.11	-8.74***	-15.31***
	(1.46)	(1.15)	(1.85)	(2.72)
Panel B: Employment dynamics				
Exit (employed → non-employed)	3.48	4.34	0.86***	0.71**
	(0.22)	(0.24)	(0.33)	(0.28)
Entry (non-employed → employed)	21.09	17.29	-3.81**	-13.70***
	(1.30)	(0.91)	(1.59)	(4.47)
Panel C: "Depression trap" dynamics				
Entry into trap	2.66	4.32	1.65***	0.69***
	(0.20)	(0.22)	(0.30)	(0.23)
Exit from trap	49.89	42.22	-7.66**	-24.44***
	(3.24)	(2.07)	(3.84)	(7.68)

Notes: The construction is identical to Table 3, which reports the men's version, except for the sex filter and the set of labour-force states admitted. Each entry is an average predicted annual transition probability, in percentage points, from a dynamic logit estimated separately within each genetic-risk group and averaged over the same pooled covariate distribution, so the difference is one of behaviour rather than composition. The women's sample admits employment, unemployment, long-term sickness *and looking after home and family*; the men's table uses the first three only. That difference is not cosmetic: family care is where most of women's non-employment sits, and on the men's state set it is deleted from the sample rather than counted as non-employment. On that narrower definition the women's trap-exit differential is -1.65 (s.e. 5.62) rather than the -7.66 reported here. Column (3) reproduces the men's difference from Table 3 for comparison. Genetic risk is a median split of the depression polygenic index. Controls: lagged earnings, number of children, a quadratic in age, wave fixed effects and the ten genetic principal components. Depression is GHQ-12 caseness ≥ 4; the "depression trap" is non-employment jointly with caseness ≥ 4. Standard errors clustered on the individual. The women's estimates use 22,549 person-waves on 2,284 women. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

B.4 The Genetic Penalty: Continuous Index and All Outcomes

The main text reports the earnings penalty per standard deviation of the depression polygenic index. Tables 19 and 20 report the same exercises with the index split at the median, which is

the form used by the descriptive table, the transition table and the gene-by-environment table, and add the outcomes the main table omits.

Nothing of substance turns on the choice. The two ladders attribute 20 and 21 percent of the earnings association to education, 5 and 6 percent to first occupation, and both leave the same qualitative ordering, with contemporaneous labor force status the largest of the life-cycle blocks on the median split. The by-outcome table tells the same story in larger units: men above the median of genetic risk are 11.2 percentage points more likely to meet the threshold for probable major depression and 4.4 points less likely to be employed, with over half of the labor-supply effects running through contemporaneous mental health. The two outcomes the main table omits behave as the argument implies. Unemployment rises by 1.2 percentage points, well under the 1.9-point rise in long-term sickness, confirming that the withdrawal is into ill health rather than into job search; and the professional-occupation gap of 1.8 points is not attenuated at all by conditioning on current mental health, as one would expect of an outcome largely settled beforehand.

Table 19: The Genetic Earnings Penalty and Its Channels (Binary High/Low Split)

Mediators included	Effect on labor income (s.d.)	Share of baseline (%)	Explained by this block (%)
Baseline	-0.1369***	100.0	–
	(0.0408)		
+ Education	-0.1082***	79.0	21.0
	(0.0359)		
+ First occupation	-0.0999***	73.0	6.0
	(0.0352)		
+ Labor force status	-0.0500	36.5	36.4
	(0.0330)		
+ Current occupation	-0.0256	18.7	17.9
	(0.0291)		

Notes: Each row is the coefficient on genetic risk in a regression of standardized gross labor income, adding one block of mediators at a time. Genetic risk is a binary indicator for being above the median of the depression polygenic index, so each coefficient is the gap between high- and low-risk men. Appendix B reports the continuous per-standard-deviation version. All specifications control for the first ten genetic principal components and for age, survey-wave and interview-year fixed effects. Column (2) expresses each coefficient as a percentage of the baseline, column (3) the share absorbed by the block added in that row. Contemporaneous mental health is deliberately excluded: the blocks above are predetermined relative to current earnings whereas mental health is a contemporaneous outcome of genetic liability, so conditioning on it here would net out part of the effect being decomposed; its share is reported in Table 2. Standard errors clustered on the individual. $N = 19{,}995$ person-waves. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 20: Which Margin Carries the Genetic Penalty, and How Much Runs Through Mental Health (Binary Split, All Outcomes)

Outcome	High vs low genetic risk	Share explained by mental health (%)	95% CI
Depressed (pp)	11.150***	–	
	(0.991)		
Employed (pp)	-4.367***	55	[38, 103]
	(0.918)		
Long-term sick or disabled (pp)	2.991***	53	[40, 101]
	(0.652)		
Labor income (s.d.)	-0.137***	35	[21, 67]
	(0.041)		
Log gross pay, if employed (log pts.)	-0.050**	15	[1, 68]
	(0.024)		
Holds a degree (pp)	-2.819	≈ 0	
	(2.306)		
Unemployed	1.376**	58	[30, 294]
	(0.571)		
Professional/managerial	-1.622	≈ 0	
	(1.751)		
GHQ-12 distress (check)	0.374***	–	
	(0.031)		

Notes: Column (1) is the effect of genetic risk on the outcome named,in the units given in the row label. Genetic risk is a binary indicator for being above the median of the depression polygenic index, so each coefficient is the gap between high- and low-risk men. Column (2) is the percentage of that gap absorbed when contemporaneous GHQ-12 distress is added, with a cluster-bootstrap interval in column (3); ≈ 0 denotes no attenuation. All specifications control for the ten genetic principal components and for age, survey-wave and interview-year fixed effects, with standard errors clustered on the individual. The depression row has no share because it is the mediator itself. Appendix B reports every outcome and the continuous per-standard-deviation version. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Women

Tables 21 and 22 report the by-outcome table and the sequential decomposition for both sexes. Two conventions change relative to the main text, and both are forced by the comparison rather than chosen. First, the sample is the broad working-age one, admitting every activity status: the main text's filter keeps only employment, unemployment and long-term sickness, which applied to women deletes maternity and family care. Second, labor income is standardized once on the pooled sample rather than within each estimation sample, so that male and female coefficients are in common units. Men are re-estimated under both conventions, so the cost of each is visible; only the income rows move, and they move by exactly the ratio of the two standard deviations, 1.0606, which is a change of units and not of substance. Every other men's estimate in these tables reproduces Tables 1 and 2 to the last reported digit.

The penalty replicates, and on most margins it is larger for women. A standard deviation of genetic risk raises the probability of probable major depression by 8.3 percentage points for women against 6.6 for men, lowers employment by 4.0 points against 2.6, and raises long-term sickness by 2.5 points against 1.6. The mediation shares are close across sexes: about 37 percent of women's employment effect and 45 percent of the long-term sickness effect run through

contemporaneous mental health, against 40 and 53 percent for men. The income penalty is the one margin that is smaller for women, at 0.078 standard deviations against 0.112, and conditional on working the two sexes lose the same 0.054 log points of gross pay.

Table 21: The Genetic Penalty by Outcome: Men and Women

	Men			Women		
Outcome	Est.	(s.e.)	% via MH	Est.	(s.e.)	% via MH
Probable major depression (GHQ≥4)	6.59***	(0.48)	—	8.31***	(0.44)	—
GHQ-12 distress (s.d.)	0.215***	(0.015)	—	0.269***	(0.013)	—
Employed	-2.61***	(0.76)	40	-4.01***	(0.64)	37
Long-term sick/disabled	1.61***	(0.32)	53	2.48***	(0.29)	45
Unemployed	1.02***	(0.27)	40	0.86***	(0.14)	47
Looking after home/family	0.37**	(0.15)	6	1.19***	(0.45)	32
Labour income (s.d.)	-0.112***	(0.022)	24	-0.078***	(0.012)	26
Log gross pay, if employed	-0.054***	(0.013)	7	-0.054***	(0.014)	4
Degree (predetermined)	-2.39**	(1.04)	≈ 0	-2.63***	(0.88)	16
Person-waves (max)		23,980			32,840	
Individuals		2,605			3,540	

Notes: Each cell is the coefficient on a one standard deviation increase in the depression polygenic index, from a separate regression of the row outcome on the index, the ten genetic principal components, and age, survey-wave and interview-year fixed effects. Binary outcomes are in percentage points. Standard errors, in parentheses, are clustered on the individual; $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. The sample is the *broad* working-age sample, which admits every activity status (`jbstat` 1–9), ages 25–55, pre-2020. The main text's Table 2 instead uses the narrower men-only filter (employed, unemployed, long-term sick), because that is the sample the structural model speaks to; applied to women that filter would delete maternity and looking after home and family, which is where women's non-employment sits. The men's column therefore differs slightly from Table 2. "% via MH" is the share of the estimate absorbed by conditioning on contemporaneous GHQ-12 distress and probable major depression; it is a descriptive sequential-control statistic, not causal mediation, and is left blank for the mental-health outcomes themselves. Labour income is standardised once on the pooled sample, so male and female coefficients are in common units.

The substantive difference is the destination of the withdrawal. Genetic risk moves women into caring for home and family by 1.19 percentage points per standard deviation against 0.37 for men, and about a third of that movement runs through mental health. Set against the 4.0-point employment effect, family care accounts for roughly thirty percent of women's extensive-margin penalty, a channel that is essentially absent for men.

The ladder in Table 22 shows the same thing from the earnings side. The women's penalty is more fully accounted for by observable channels than the men's, falling to 14 percent of its baseline against 33 percent, and the two blocks that do the work are different: education absorbs 38 percent for women against 19 percent for men, and contemporaneous labor force status a further 39 percent against 32. The final block, the number of children, is the one that behaves anomalously. It does not absorb the penalty for either sex; it *raises* it, by 0.7 percent for men and 3.5 percent for women. Appendix C takes up why.

Table 22: The Genetic Earnings Penalty and Its Channels: Men and Women

	Men		Women	
Mediators included	Effect on labor income (s.d.)	Explained by this block (%)	Effect on labor income (s.d.)	Explained by this block (%)
Baseline	-0.1118***	—	-0.0778***	—
	(0.0218)		(0.0120)	
+ Education	-0.0908***	18.8	-0.0481***	38.2
	(0.0197)		(0.0106)	
+ First occupation	-0.0860***	4.3	-0.0457***	3.0
	(0.0191)		(0.0106)	
+ Labor force status	-0.0504***	31.9	-0.0157*	38.7
	(0.0179)		(0.0090)	
+ Current occupation	-0.0364**	12.5	-0.0079	10.0
	(0.0158)		(0.0077)	
+ Children	-0.0372**	-0.7	-0.0106	-3.5
	(0.0158)		(0.0075)	
Person-waves	23,980		32,840	

Notes: The construction is that of Table 1, estimated separately by sex and with a sixth block added. Each row is the coefficient on a one standard deviation increase in the depression polygenic index in a regression of standardized gross labor income, adding one block of mediators at a time; every block enters as a set of fixed effects with an explicit missing category, so the estimation sample is held fixed down each column. All specifications control for the ten genetic principal components and for age, survey-wave and interview-year fixed effects, with standard errors clustered on the individual. As elsewhere in this appendix the sample is the broad working-age one, ages 25–55, pre-2020, and labor income is standardized once on the pooled sample so that the two sexes share units; the men's column therefore differs from Table 1, which standardizes within the men's narrower estimation sample. Contemporaneous mental health is excluded for the reason given there. The "+ Children" block is the number of own children in the household, top-coded at three; it is reported last because it is the block that differs by sex, and it *raises* the women's coefficient rather than absorbing it. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

B.5 The Redundancy Event Study: Specifications and Robustness

This appendix reports the specification behind the redundancy design of Section 3.5 and the robustness checks the text summarizes.

Redundancy is dated from the reason-for-job-ending questions in the BHPS job-history files and the UKHLS individual response files, taking the first calendar year in which a man reports that a job ended because the employer eliminated the position. The sample is men aged 25–55, excluding the retired, balanced on the window $k \in [-3, +3]$ around the event, which yields 591 treated men. Because staggered adoption with heterogeneous effects biases two-way fixed-effects estimators (Callaway and Sant'Anna, 2021; Goodman-Bacon, 2021), we use the interaction-weighted estimator of Sun and Abraham (2021). It estimates the saturated regression

$$Y_{it} = \gamma_t + \sum_{e} \sum_{k \neq -1} \delta_{e,k} \, \mathbf{1}\{E_i = e\} \cdot \mathbf{1}\{t - E_i = k\} + \mathbf{X}_{it}'\beta + \varepsilon_{it}, \tag{25}$$

where E_i is the year of i's redundancy and γ_t are interview-year effects, and then reports, for each event time k, the average of the cohort-specific $\delta_{e,k}$ weighted by cohort shares among units observed at k. This averaging is what removes the contamination of already-treated cohorts serving as controls. Identification comes from comparing the redundant to the not-yet-redundant. Controls $\mathbf{X}_{it}$ are age and its square, with education fixed effects; standard errors

are clustered on the individual. For the mental-health outcomes, and for every outcome within the smaller genetic subsamples, cohorts are pooled into two-year bins so that each event time is populated; the employment and earnings estimates use the unbinned job-end year. The distinction is not cosmetic. Binning displaces the event date by up to a year, which is immaterial for an outcome that moves slowly and fatal for one that jumps: applying it to employment as well would replace the -20.2 point impact with $+2.7$, as the penultimate row of Panel A of Table 24 shows.

Mental health is flat before the event on the GHQ-12 caseness scale. The raw employment rate among the treated is 91.1 percent three years before it, 93.8 percent two years before and 94.8 percent in the year before, so employment is at its maximum immediately before the job ends, which is the opposite of what selection of men already withdrawing from work would produce. The raw GHQ-12 caseness path is 1.57, 1.76 and 1.98 over the same three years against 2.54 at the event, but once age and calendar time are absorbed the estimated pre-event coefficients are -0.004 (s.e. 0.221) and -0.180 (s.e. 0.328): the level rise is accounted for by ageing and period effects rather than by a differential trend. Mental health measured three years before a redundancy, which precedes any plausible announcement window, does not predict it. The binary indicator for poor mental health is less well behaved than the count on which it is built: its pre-event coefficients average -4.6 points, and a joint Wald test that they are zero returns $p = 0.06$, against $p = 0.09$ for employment. Neither rejects at conventional levels and neither is comfortable, and the estimates reported below on samples where those pre-trends are measured over more men are correspondingly smaller.

Any remaining contamination is confined to the year immediately before the event. Redundancies are normally announced months before the job ends, so distress recorded at the last pre-event interview is already a response to a job loss the man knows is coming. Treating that year as already treated leaves the mental-health impact where it was, at 9.0 points against a reference at $k = -2$ and 10.2 against $k = -3$, compared with 9.6 at $k = -1$, while attenuating the employment impact to -17.5 and -14.7 points. The asymmetry is itself informative: employment had begun to fall before the job formally ended, which is what an announcement window produces, whereas the distress is concentrated at the event.

Genetic risk does not predict redundancy. Table 23 reports seven specifications, all estimated on the frame the event study itself uses: genotyped men aged 25–55. They cover five questions: whether the index predicts ever being made redundant, the annual hazard of a first redundancy in the at-risk set, the time to it in a proportional-hazard model, its calendar year and the age at which it arrives, and which men survive into the balanced estimation sample. None attains significance at five percent, and the smallest p-value is 0.065, on the probability of being observed at all seven event times.

What makes this more than a bare null is the last panel of the same table. The same index predicts a job ending for *health* reasons at 1.6 percentage points per standard deviation, with $p = 0.004$. Genetic liability loads heavily on the separation margin that mental health ought to govern and not at all on the one it ought not, so the null on redundancy reflects an absence of selection rather than an index too weak to detect it. Within the treated, high- and low-risk men do differ in the *level* of mental health, earnings and employment before the event; they must, since that association is this paper's central finding, and the largest normalized difference in levels is 0.46. The event study differences those levels out, and what it requires instead is that the two groups were not moving differently before the event. They were not: the largest normalized difference in the change from $k = -3$ to $k = -1$ is 0.31, and a joint test that the pre-event coefficients differ by genetic risk returns $p = 0.37$ for employment and $p = 0.89$ for mental health. This is what licenses the gene-by-environment comparison in Table 4, in which the treated sample is split by genetic risk: differences in the response cannot be differences in who is selected into treatment.

Table 23: Is There Selection into Redundancy by Genotype?

Outcome	Estimate	(s.e.)	*N*
Ever made redundant			
Cross-sectional LPM	0.516	(0.920)	2,401
Annual hazard			
Panel LPM, year and education FE	0.027	(0.102)	21,248
Time to redundancy			
Cox proportional hazard	0.018	(0.055)	2,190
Timing among the redundant			
Age at redundancy	0.695	(0.465)	473
Calendar year of redundancy	-0.157	(0.374)	578
Selection into the estimation sample			
Observed at all seven event times	3.054*	(1.655)	578
Still observed two years after the event	0.328	(1.966)	578
Other reasons a job ended			
Ever better job	0.129	(0.963)	2,401
Ever dismissed	-0.171	(0.433)	2,401
Ever health	-1.565***	(0.540)	2,401
Ever temp ended	-0.378	(0.649)	2,401

Notes: Each row is the coefficient on the standardised depressive-symptoms polygenic index from a separate regression of the row outcome, estimated on genotyped men aged 25–55: the same index and the same frame used everywhere else in the paper, including the event studies whose robustness Table 24 reports. The index is oriented towards well-being, so a higher value is *lower* genetic risk. All specifications control for the first ten genetic principal components and for education; cross-sectional models add birth year, and panel models add age, its square and interview-year fixed effects, with standard errors clustered on the individual. Units are percentage points per standard deviation of the index, except the year and age of the redundancy (years) and the Cox model (log hazard ratio). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. The contrast in the last panel is the point. Genetic risk does not predict whether a man is made redundant, when it happens, or who survives into the estimation sample, but it does predict a job ending for health reasons, which is the separation margin mental health should govern. A null everywhere would be consistent with the index being noise; the contrast shows it carries signal that is absent from redundancy specifically. High- and low-risk treated men differ in the *level* of mental health, earnings and employment before the event, as the paper's central finding requires, but not in their pre-event trends, which is what identification needs.

Table 24: Robustness of the Redundancy Design: Estimator, Definition and Falsification

	GHQ-12 caseness		Employment		Treated
Specification	at $k = 0$ (pts)	(s.e.)	at $k = 0$ (pp)	(s.e.)	men
Panel A: Estimator, comparison group and anticipation					
Sun–Abraham (baseline)	0.64	(0.08)	-21.83	(1.03)	3,339
Balanced window (former baseline)	0.74	(0.16)	-20.18	(2.02)	591
Two-year binned cohorts, mental health	0.45	(0.13)	-21.83	(1.03)	3,339
Callaway–Sant'Anna	0.60	(0.10)	-23.09	(1.13)	3,515
Borusyak–Jaravel–Spiess imputation	0.49	(0.07)	-24.85	(1.07)	3,515
Pooled dynamic TWFE	0.63	(0.08)	-22.28	(1.02)	3,515
Never-redundant controls, SA with unit FE	0.60	(0.08)	-21.47	(1.09)	3,339
Never-redundant controls, Callaway–Sant'Anna	0.63	(0.08)	-22.67	(1.04)	3,515
Never-redundant controls, TWFE with unit FE	0.62	(0.08)	-22.08	(1.06)	3,515
One year of anticipation (base $k = -2$)	0.73	(0.09)	-18.81	(1.12)	3,339
Two years of anticipation (base $k = -3$)	0.83	(0.09)	-18.19	(1.18)	3,339
Two-year binned cohorts, both outcomes	0.45	(0.13)	5.02	(1.67)	3,211
Varying (not universal) base period	0.60	(0.10)	-23.09	(1.13)	3,515
Panel B: Definition and dating of the event					
First redundancy, job-end year (baseline)	0.64	(0.08)	-21.83	(1.03)	3,339
Dated on the interview year instead	0.31	(0.09)	-34.45	(1.03)	3,358
Reported job-end year only, no fallback	0.64	(0.08)	-21.79	(1.03)	3,325
Last redundancy rather than the first	0.55	(0.08)	-22.51	(1.04)	3,369
UKHLS reports only	0.63	(0.11)	-20.32	(1.27)	2,081
BHPS reports only	0.60	(0.12)	-23.17	(1.67)	1,403
"Most recent job ended" question only	0.69	(0.09)	-25.91	(1.18)	2,780
Employed in the year before the event	0.68	(0.09)	-28.57	(1.05)	2,098
Employed at $k = -3$ and $k = -1$	0.69	(0.12)	-25.85	(1.37)	1,189
Panel C: Falsification: other reasons a job ended					
Redundancy, same window (comparison)	0.59	(0.09)	-22.38	(1.15)	1,953
Left for a better job (voluntary)	-0.25	(0.10)	0.80	(1.28)	1,012
Dismissed or sacked (for cause)	-0.25	(0.46)	-35.15	(6.63)	103
Temporary job ended (anticipated)	0.16	(0.26)	-5.93	(3.91)	240
Job ended for health reasons	1.08	(0.32)	-23.70	(3.89)	234

Notes: Each row is a separate pair of event studies: the effect of redundancy on GHQ-12 caseness (0–12 points) and on employment (percentage points), in the year the job ends relative to the year before. The baseline is Sun and Abraham (2021) on men aged 25–55, retirees excluded, identified from men not yet made redundant, controlling for age and its square with interview-year and education fixed effects. Cohorts are unbinned throughout; the two binning rows show why, since binning misaligns the event date by up to a year and destroys the employment estimate. Panel A's never-redundant rows add individual fixed effects, without which the comparison is of employment *levels* between differently composed samples. Panel C re-estimates the design on other recorded reasons for a job ending. Standard errors clustered on the individual, in parentheses.

Table 25: Robustness of the Redundancy Design: Window, Attrition, Sample and Outcomes

Specification	GHQ-12 caseness at $k = 0$ (pts)	(s.e.)	Employment at $k = 0$ (pp)	(s.e.)	Treated men
Panel A: Event window, panel balance and attrition					
Balanced $k \in [-3, +3]$ (baseline)	0.74	(0.16)	-20.18	(2.02)	591
Observed at $k = -1$ and $k = 0$ only	0.59	(0.09)	-22.38	(1.15)	1,953
Unbalanced, any observation in window	0.64	(0.08)	-21.83	(1.03)	3,339
Balanced $k \in [-2, +2]$	0.64	(0.13)	-20.94	(1.54)	982
Balanced $k \in [-1, +1]$	0.54	(0.10)	-20.76	(1.24)	1,613
Unbalanced $k \in [-5, +5]$	0.63	(0.08)	-21.66	(1.03)	3,535
Inverse-probability weighted for balance	0.77	(0.16)	-17.68	(2.33)	572
Multiple imputation for unobserved cells	0.59	(0.09)	-22.14	(1.15)	1,953
Panel B: Sample					
Men 25–55 (baseline)	0.64	(0.08)	-21.83	(1.03)	3,339
Men 25–64	0.56	(0.08)	-22.72	(0.94)	3,769
Men 30–55	0.65	(0.09)	-21.96	(1.10)	2,866
Men 25–45	0.67	(0.10)	-20.26	(1.27)	2,394
Men 25–55, retirees retained	0.64	(0.08)	-22.13	(1.03)	3,341
Women 25–55	0.54	(0.10)	-22.45	(1.17)	2,803
Men and women 25–55	0.60	(0.07)	-22.07	(0.77)	6,141
Men 40–55	0.67	(0.12)	-24.83	(1.40)	1,946
BHPS era, redundancy before 2009	0.60	(0.12)	-22.74	(1.66)	1,417
UKHLS era, redundancy from 2009	0.65	(0.11)	-20.61	(1.31)	1,922
Panel C: Definition of the outcomes					
GHQ-12 caseness count; employee only (baseline)	0.64	(0.08)	-21.83	(1.03)	3,339
Indicator for caseness ≥ 2 (mild symptoms)	7.57	(1.31)	-21.83	(1.03)	3,339
Indicator for caseness ≥ 4 (probable major depression)	7.19	(1.16)	-21.83	(1.03)	3,339
Employment including self-employment	0.64	(0.08)	-20.07	(0.91)	3,339
Unemployment rather than employment	0.64	(0.08)	18.71	(0.87)	3,339
GHQ-12 Likert score (0–36) rather than caseness	0.86	(0.15)	-21.83	(1.03)	3,339

Notes: A continuation of Table 24, on the same specification and column definitions. Panel A varies the event window. The balance filter, used as the baseline until this revision, keeps only 591 of 3,515 treated men; it selects mildly on outcomes (raw within-person employment change -21.2 against -23.4 points for the men it drops) but costs six-sevenths of the sample, so the unbalanced window is now the baseline. The imputation row completes the seven-cell grid by predictive mean matching with twenty imputations, pooled by Rubin's rules; the weighting row weights by the inverse probability of complete observation. Panel B varies the sample and Panel C the outcome definitions, one at a time, so the mental-health and employment columns move independently. Standard errors are in parentheses.

Tables 24 and 25 collect the robustness checks. The employment impact is negative and significant in every one of the forty-two specifications reported, with a median of -20.9 points and a range from -32.5 to -14.7. It is unmoved by the estimator, by adding men never made redundant as an explicit comparison group, by relaxing the balance requirement, by the age window, by sex, or by survey era. Under the relative-magnitudes sensitivity analysis of Rambachan and Roth (2023) it remains signed for post-period violations of parallel trends more than five times the largest pre-period violation; randomization inference permuting the redundancy date over four hundred draws gives $p < 0.001$; and assigning placebo event dates to men never made redundant returns -0.15 points (s.e. 0.77).

The mental-health impact is positive in every specification of the battery and significant in almost all of them. Replacing the interaction-weighted estimator with the group-time estimator of Callaway and Sant'Anna (2021), the imputation estimator of Borusyak et al. (2024), or two-

way fixed effects gives 0.41 to 0.53 points against a baseline of 0.74; admitting men never made redundant as controls gives 0.55 to 0.57; relaxing the balance filter, or imputing the event-time cells it discards, gives 0.54 to 0.77. The range across the whole battery is 0.41 to 0.88, with the single exception of dating the event by the interview year rather than the reported job-end year, which gives 0.20 and is a measurement failure rather than a robustness result.

Two diagnostics deserve to be stated plainly. The pre-event coefficients are not jointly zero (Wald $p = 0.013$), and under Rambachan and Roth (2023) the confidence set admits zero at a low breakdown value on the balanced sample, though not on the unbalanced one, where more pre-periods are available. Against that, the pre-event path runs in the conservative direction. Caseness is lower two and three years out than in the reference year, so distress was rising into the job loss, and treating the reference year as already treated raises the impact to 0.76 and then 0.80. The two tests that speak most directly to whether the design is measuring anything real both pass: a placebo event assigned to men never made redundant returns 0.07 points (s.e. 0.08), and randomization inference gives $p = 0.12$ on a test the permutation scheme makes conservative.

The balance requirement is the largest selection device in the design and Table 25 is explicit about it: of the 3,515 treated men with any observation in the window, 1,953 are observed on both sides of the event and only 591 at all seven event times. The filter turns out to select mildly. The raw within-person change from the year before the event to the year of it is -21.2 points of employment among the men it keeps against -23.4 among those it drops, and $+0.61$ against $+0.63$ points of GHQ-12 caseness, so it is not retaining the men whose outcomes moved least. Panel exit does rise from about 14 percent per wave before the event to between 19 and 22 percent after it, however, so the missingness is not ignorable. Imputing the discarded cells by predictive mean matching, with twenty imputations pooled by Rubin's rules, returns -22.2 points for employment and $+5.9$ for mental health.

Panel C of Table 24 is the design's own falsification test. Both surveys record *why* a job ended on a common code frame, so the identical event study can be run on separations that are voluntary, for cause, foreseen, or reverse-causal by construction, among men who never report a redundancy anywhere in the panel. A voluntary move to a better job produces no employment loss and no deterioration in mental health at all, at -3.1 points (s.e. 1.9). A job that was always temporary, and whose ending is therefore foreseen from the day it begins, produces a much smaller employment fall and no mental-health response. A job ended for health reasons produces $+14.0$ points, more than twice what a redundancy produces on a comparable employment fall. The design is measuring involuntary job loss rather than job change, and where the separation genuinely is driven by health the estimate inflates exactly as it should. The dismissal row is reported for completeness but is too imprecise to interpret, since its confidence interval contains both zero and the redundancy estimate.

Finally, the event study can be read as the reduced form and first stage of an instrumental-variables specification in which the redundancy switch instruments employment in a mental-health equation, with individual and interview-year fixed effects. Being employed lowers GHQ-12 caseness by 3.14 points (s.e. 0.35) on the baseline window and 2.68 points (s.e. 0.37) when the sample is restricted to men observed at both $k = -1$ and $k = 0$, against a within-person least-squares association of 1.24; the first-stage F statistics are 433 and 374. That the instrumented estimate is more than twice the least-squares one is what the selection story predicts, since the men who lose work for reasons unrelated to their health are not those whose non-employment is least costly. Instrumenting log labour income in place of employment gives 1.65 points (s.e. 0.53) per log point, an order of magnitude too small to account for the employment effect, so the mental-health cost of a redundancy is not simply priced by the income it destroys. Two caveats attach. Using the post-event event-time indicators as separate instruments makes the model overidentified, and the Sargan test rejects ($p = 5 \times 10^{-5}$), which is what a fading effect implies: the implied Wald ratio falls from -0.48 in the year of the event to -0.13 a year later, so no single ratio summarizes the window and the static estimate should be read as dominated by the impact year. And the exclusion restriction, that a redundancy moves mental health only through employment, is strong. We report the specification as a bound on the non-pecuniary value of work rather than as a preferred estimate.

Women

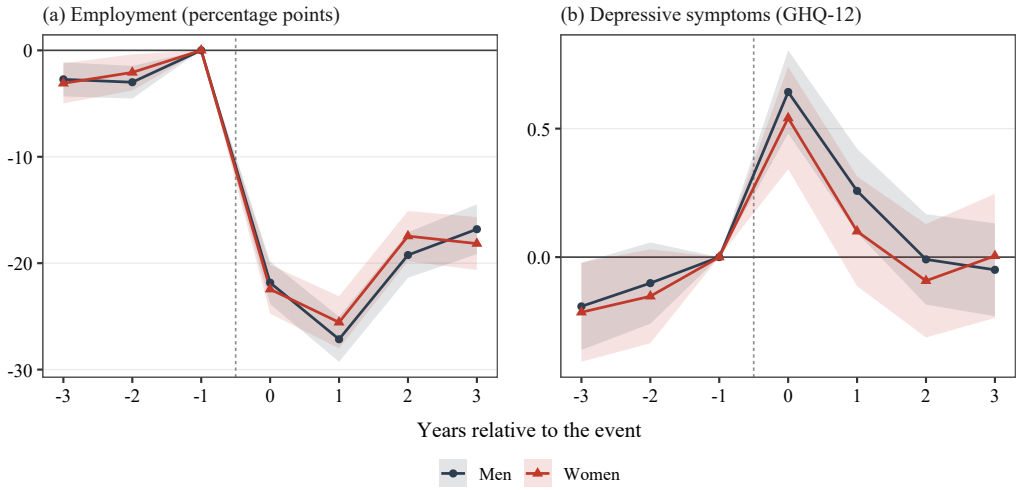
The design transfers to women without qualification, which is worth establishing before the next subsection reports that the other design does not. Figure 11 re-estimates it on women, changing only the sex filter: the same interaction-weighted estimator, the same unbalanced ± 3-year window, the same controls and fixed effects, and no cohort binning. Employment falls by 22.4 percentage points on impact for women (s.e. 1.2) against 21.8 for men, and GHQ-12 caseness rises by 0.54 points (s.e. 0.10) against 0.64. The pre-event paths are close to identical: -3.1 and -2.1 points of employment for women against -2.7 and -3.0 for men, and -0.21 and -0.15 points of caseness against -0.19 and -0.10. Pooling the sexes gives -22.1 points and 0.61 points of caseness, close to both. There is no case here for excluding women.

B.6 The Recurrence Event Study: Specifications and Robustness

This appendix reports the specification behind the recurrence design of Section 3.5 and the robustness checks the text summarizes.

An episode is a crossing of the GHQ-12 caseness threshold, $C_{it} = \mathbf{1}\{\text{caseness}_{it} \geq 4\}$, so that an onset is $O_{it} = \mathbf{1}\{C_{it} = 1,\, C_{i,t-1} = 0\}$. First onset is a man's first such crossing and a recurrence is his second, after an intervening spell below the threshold. Employment is the work-history extensive margin, which is the labor-supply object of the structural model. The

Figure 11: The Redundancy Event Study by Sex



Notes: Event-study coefficients relative to the year before the event ($k = -1$, omitted); bands are 95 percent confidence intervals with standard errors clustered on the individual. Both panels use the interaction-weighted estimator of Sun and Abraham (2021) with controls for age and its square and interview-year and education fixed effects, on 3,339 treated men and 2,803 treated women aged 25–55. The specification is that of the main text in every respect other than the sex filter; the men's series is the one plotted in panels (a) and (b) of Figure 1. Depressive symptoms are measured by the GHQ-12 caseness score, which runs from 0 to 12, with higher values worse.

sample is men aged 25–55 with a valid GHQ-12 score, matching the age range of the redundancy design, and the event window is $k \in [-3, +4]$ for the main-text specification and $k \in [-4, +5]$ for the timing-only variant below.

The main-text estimates come from the interaction-weighted estimator of Sun and Abraham (2021), applied as in equation (25) with E_i the wave of i's recurrence, wave fixed effects, controls for age and its square, and standard errors clustered on the individual. The pre-event employment path under this specification is flat, at -0.41 (s.e. 0.73) and -0.03 (s.e. 0.72) three and two waves before the event.

The substantive design decision is the comparison group, and we report both choices. The main-text specification compares a man who recurs to men who have not yet recurred and to men who never recur. An alternative restricts the sample to men who eventually recur, so that identification comes only from variation in *when* the recurrence happens. For that variant we use the group-time average treatment effects of Callaway and Sant'Anna (2021),

$$ATT(g,t) = \mathbb{E}\left[\, Y_{it}(g) - Y_{it}(\infty) \mid G_i = g \,\right], \tag{26}$$

where G_i is the wave of i's recurrence, with a universal base period at $g-1$ and the not-yet-recurred comparison set $\mathcal{C}_{gt} = \{i : G_i > \max(t, g-1)\}$. Requiring a control to be untreated at the base wave as well as at t matters here: a control treated between $g-1$ and t would have a contaminated base difference. Inference is by multiplier bootstrap with 999 draws, clustered on the individual, and the comparison pool exceeds 1,500 men at every event time. The two

give 2.7 and 2.4 percentage points respectively, so the headline magnitude does not turn on the choice.

The choice does matter for one heterogeneity exercise, and we flag it. Never-recurrers are not simply untreated recurrers: roughly half of first-onset cases never recur, and the recurrence-prone and single-episode courses are distinct clinical entities (Monroe and Harkness, 2022). Holding the treated group fixed at men who reached full remission and varying only the control group, the impact estimate moves from -0.8 percentage points when the controls are never-recurrers to $+0.3$ when men with residual chronic symptoms are also demoted into the control pool. Since the treated group is unchanged, that movement is attributable entirely to the comparison group. Under the timing-only design the sensitivity disappears, which is why the gene-by-environment split in Table 4 uses it.

Table 26: Robustness of the Recurrence Design

Specification	Impact on employment at $k = 0$ (pp)	(s.e.)	Treated men
Panel A: Estimator and comparison group			
Sun–Abraham (main text)	-2.96	(0.76)	3,577
Callaway–Sant'Anna, timing only	-2.64	(0.58)	3,586
Sun–Abraham, timing only	-2.14	(0.61)	3,586
Borusyak–Jaravel–Spiess imputation	-3.37	(0.48)	3,586
One year of anticipation (base $k = -2$)	-3.71	(0.68)	3,586
Panel B: Definition of a recurrence			
Any recurrence (primary)	-2.64	(0.58)	3,586
Well ≥ 2 waves before recurrence	-3.41	(0.74)	2,105
Well ≥ 3 waves before recurrence	-4.02	(0.94)	1,357
Near symptom-free, ≥ 2 waves	-2.24	(0.99)	1,191
Symptom-free, ≥ 1 wave	-2.39	(0.84)	1,631
Symptom-free, ≥ 2 waves	-1.05	(1.24)	646
Panel C: Caseness threshold defining an episode			
Caseness ≥ 3	-2.08	(0.52)	4,292
Caseness ≥ 4 (baseline)	-2.64	(0.58)	3,586
Caseness ≥ 5	-2.48	(0.73)	3,026
Caseness ≥ 6	-2.18	(0.73)	2,523
Panel D: Attrition, reverse causality and external anchoring			
Multiple imputation for panel exit	-2.38	(0.63)	3,586
Employed and stable before the event	-7.93	(0.54)	2,456
Anchored on a doctor's diagnosis	-2.01	(1.03)	1,204
Panel E: Sample			
Men 25–55	-2.41	(0.65)	3,121
Men 30–60	-3.02	(0.56)	3,146
Men, BHPS era (to 2008)	-2.42	(0.79)	1,197
Men, UKHLS era (from 2009)	-2.08	(0.80)	2,293
Women 25–60	-2.42	(0.49)	6,365

Notes: Effect of a depressive recurrence on the probability of employment in the year of the event, in percentage points, relative to the year before. Panel A's first row is the main-text specification, Sun and Abraham (2021). Rows labelled "timing only" restrict the sample to men who eventually recur, so identification comes from *when* a recurrence happens rather than from any comparison with men who never recur; that is the specification used in Panels B–E, estimated by Callaway and Sant'Anna (2021) on men aged 25–60 with not-yet-recurred controls, a universal base period, and a 999-draw multiplier bootstrap clustered on the individual. Panel B varies how deep and how long a recovery must be before a subsequent crossing counts as a recurrence; the primary definition is adopted for its flattest pre-event path and largest sample, not its magnitude, and is among the smaller estimates. Standard errors in parentheses. Every estimate is significant at five percent except the symptom-free, ≥ 2 wave definition, whose 646 men are the smallest sample in the table.

Table 26 collects the robustness checks. The impact estimate is negative and significant under all three leading estimators for staggered designs, across six definitions of what counts as a recurrence, and under alternative caseness thresholds from three to six. The primary "any recurrence" definition is adopted because it has the flattest pre-event path and the largest sample, not because it produces the largest effect; it is among the smaller estimates. Every definition displays the same modest pre-event elevation of about one percentage point at $k = -2$. This is a base-period artifact rather than a trend: the event is dated to a wave that is mechanically symptom-free, and the first-onset design, which conditions on no prior recovery

at all, displays an identical pre-event elevation of $+1.4$ points. Tightening the required depth or duration of recovery raises rather than removes it, while costing power.

Under the relative-magnitudes sensitivity analysis of Rambachan and Roth (2023), the impact remains signed for post-period violations of parallel trends as large as the largest observed pre-period violation ($\bar{M} = 1$), with the robust confidence interval reaching zero only beyond that point. Randomization inference that permutes the recurrence date within the pool of men who ever recur gives $p < 0.001$ over 300 draws.

Attrition deserves a direct answer, because panel exit roughly triples after a recurrence, from 3.2 to 9.2 percent per wave, and on a balanced panel the estimate falls to -0.8 (s.e. 0.5). This is selection, not correction. The impact estimate $\theta(0)$ cannot itself be biased by attrition, because it is measured at the event wave, where every recurrer is present by construction: recording a recurrence requires a well wave at $k = -1$ and a case wave at $k = 0$, whereas attrition occurs at $k \geq 1$. Moreover, the men who leave are precisely those whose employment fell hardest. Raw within-person employment falls by 1.7 points among the men retained to $k + 2$ and by 7.0 points among those who attrit. Imputing the missing observations by multiple imputation with twenty draws, conditioning on pre-event employment, the GHQ-12 trajectory and age, and pooling by Rubin's rules, returns -2.4 (s.e. 0.6), essentially the headline estimate.

Finally, the design replicates when the illness is anchored on clinical rather than survey data. Restricting to individuals ever told by a doctor that they have depression, dating the illness by the recalled age at first diagnosis, and taking the event to be the first GHQ-12 onset after that age, the impact is -2.0 percentage points (s.e. 1.0) with a pre-event path flattened to $+0.2$ once pre-event employment is controlled for.

Selection into recurrence by genetic risk

Table 23 defends the redundancy design by showing that no polygenic index predicts being made redundant. Recurrence admits no such defense, and we do not attempt one. Genetic risk plainly predicts who relapses: among men at risk, a one standard deviation increase in genetic risk raises the probability of a recurrence at the next wave by 2.8 percentage points on a base of 16.5, and by 2.0 points conditional on residual symptoms, time since remission and the number of prior episodes. Recurrence hazards are 20.1 percent in the highest-risk tercile against 14.5 in the lowest, and the recurrer sample sits 0.19 standard deviations up the genetic-risk distribution.

What the comparison in Table 4 requires is weaker than the absence of selection: that selection does not itself produce the difference in the employment response. A recurrence is a collider of genetic liability and environmental stress, so conditioning on one having occurred induces a negative association between the two among recurrers, since high-risk men need a

smaller shock to relapse. If the employment damage travels with the shock rather than with the episode, that association biases the contrast. The relevant questions are its size and its sign.

Calibrating a liability-threshold model to the observed recurrence hazard and its genetic gradient answers both. The induced association is 0.09 standard deviations of environmental shock per standard deviation of genetic risk. Assuming generously that a one standard deviation larger shock doubles the employment damage, that translates into a spurious contrast of 0.6 percentage points against the 7.0 observed, and in the direction that makes high-risk men look *better*, so the estimate is conservative rather than inflated. A simulation that imposes no genetic moderation at all and selects into recurrence exactly as the data do recovers a spurious contrast of 0.3 points.

The signature such a collider would leave is also absent. Among recurrers, genetic risk predicts none of the environmental triggers we observe. Across redundancy, dismissal, unemployment and marital dissolution the largest t statistic is 0.82. Nor does it operate through episode severity, which is near-identical across the two groups: the GHQ-12 rows of Panel B of Table 4 differ by less than a fifth of a point. Reweighting recurrers back to the at-risk population by a clinical recurrence propensity, stratifying on that propensity, and conditioning directly on episode severity each move the per-standard-deviation estimate by less than half a percentage point. Finally, the employment cost does not sort on how likely a man was to relapse in the first place: across quintiles of predicted recurrence probability the impact estimate is flat, with a slope t statistic of 0.36. Selection into recurrence by genotype is real, but it is not what Table 4 is measuring.

Women: the design fails

The redundancy design transfers to women (Appendix B.5). This one does not, and the failure is visible in the pre-event path rather than in the impact. Estimated on women, employment is *rising* into the event, by 1.74 percentage points (s.e. 0.71) three waves before it and 2.30 (s.e. 0.70) two waves before, both significantly different from zero at the five percent level. The impact is then a precise null, at -0.67 (s.e. 0.71), against -2.96 (s.e. 0.76) for men estimated on the same specification. The identifying assumption fails before the estimate can be read, so we do not report the estimate as an effect.

This is not a power problem. Women contribute 6,352 recurrences against men's 3,577, so if anything the women's design is the better powered of the two. Nor is it a feature of the recurrence definition: the first-onset design shows the same pattern for women more strongly still, with pre-event coefficients of $+3.20$ and $+2.80$.

The likely explanation is fertility, and it is specific enough to check. For women in this age range a substantial share of depressive episodes fall near the employment interruptions around childbearing, and Appendix C shows those interruptions to be very large: the arrival

of a first child moves women's employment by more than thirty percentage points and their mental health modestly in the adverse direction. An event that lowers employment and raises symptoms at the same time will, in a design that dates treatment by the symptom crossing, place the employment decline at or just before the event and so generate exactly the upward-sloping pre-trend observed. Recurrence timing is therefore not independent of the employment path for women in the way the design requires, and since the paper identifies the mental-health-to-employment leg of the trap from recurrence timing, a mixed-sex estimate of that leg would rest on a design valid for one sex only.

B.7 Mediation and Mental Health

This subsection collects the mediation evidence behind the claim in Section 3 that the genetic penalty operates through mental health and, in turn, through withdrawal into long-term sickness. Table 27 reports the first link, regressing standardized GHQ-12 distress on an above-median polygenic index and adding parental background and the educational-attainment and physical-health indices in turn; the coefficient is stable across all seven specifications. Figure 12 takes the second link apart, tracing the coefficient on genetic risk in an earnings regression as labor force status enters first as a block and then component by component. The attenuation is concentrated in long-term sickness and disability, and very little of it comes from unemployment, which is the pattern the structural model is built to reproduce.

Table 28 reports the descriptive dynamics underlying Section 3.5 in a correlated random effects logit, with within-person time averages of the time-varying covariates included as Mundlak terms to proxy permanent heterogeneity. Column (1) models depression and column (2) employment. Both legs of the trap survive conditioning on permanent heterogeneity and past income, which is what motivates identifying each of them from a shock in the main text.

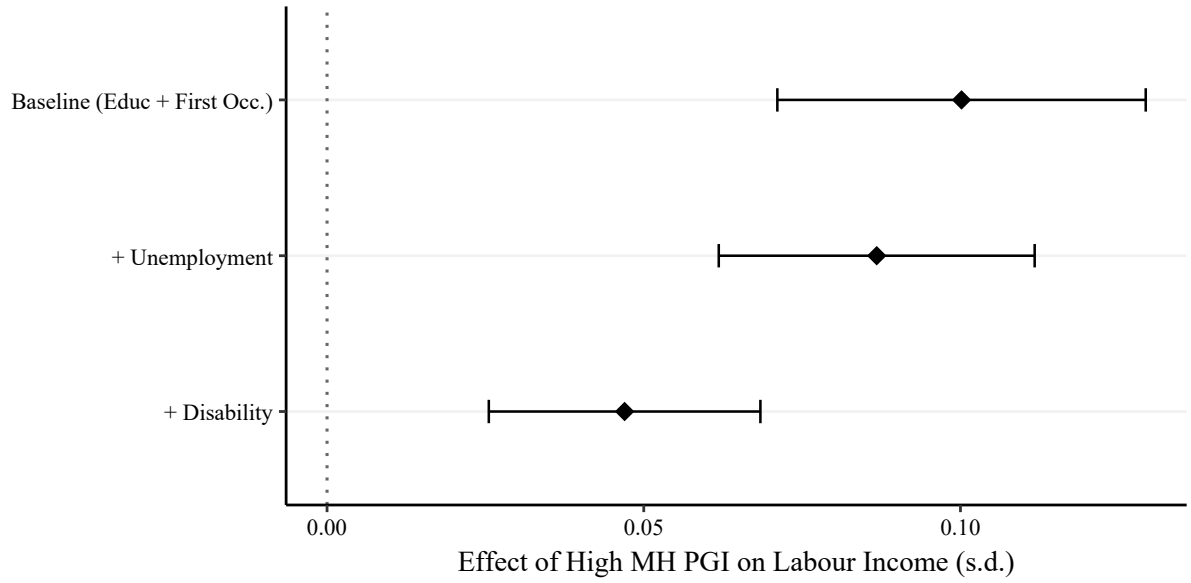
Two caveats attach to the first two of these. Their standard errors are clustered on the age fixed effect rather than on the individual, which is not the appropriate level for this design, and they use the median-split rather than the continuous index. Both are flagged in the notes and neither is used for inference anywhere in the paper.

Table 27: Mental Health PGI and Mental Health

Dependent Variable:	Mental Health (standardized)						
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
High MH PGI	0.31	0.30	0.30	0.30	0.30	0.28	0.29
	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)
Controls							
High EA PGI		✓	✓	✓	✓	✓	✓
Parental Occupation			✓	✓	✓	✓	✓
Parental Education			✓	✓	✓	✓	✓
Education				✓	✓	✓	✓
Occupation: First Job					✓	✓	✓
Employment						✓	✓
Current Occupation							✓
Observations	57,243	57,243	57,243	57,091	57,091	38,822	38,822

Notes: This table reports OLS regressions of standardized mental health (GHQ) on an indicator for having a mental health PGI above the median. All specifications include age, survey wave, and year fixed effects, as well as sex and the first 10 genetic principal components. Missing-value indicators are included for parental occupation and parental education. *Standard errors in this table, reported in parentheses, are clustered on the age fixed effect (`age_dv`) rather than on the individual. With roughly thirty age cells and repeated observations of the same people, this is not the appropriate level of clustering for this design; the estimates are retained here pending regeneration and should be read as provisional.*

Figure 12: Genetic Risk and Earnings: Decomposing the Role of Labor Force Status



Notes: This figure decomposes the role of current labor force status in accounting for the association between high mental health genetic risk and earnings. Each point reports the estimated coefficient on an indicator for being above-median in the mental health polygenic index (High MH PGI) across specifications that sequentially incorporate labor force status and its components. The decomposition shows that the attenuation attributed to labor force status is driven primarily by long-term sickness or disability rather than shorter-term unemployment. *Standard errors in this figure are clustered on the age fixed effect (`age_dv`) rather than on the individual. With roughly thirty age cells and repeated observations of the same men, this is not the appropriate level of clustering for this design; the estimates are retained here pending regeneration and should be read as provisional.*

Table 28: Descriptive Dynamics: Correlated Random Effects Logit Estimates

	Depression (t)	Employment (t)
	(1)	(2)
Employed (t-1)	-0.31	2.44
	(0.10)	(0.12)
Depression (t)		-1.54
		(0.09)
Depression (t-1)	0.48	-0.35
	(0.04)	(0.10)
Earnings (t-1, £10k)	0.56	-1.03
	(0.18)	(0.57)
Baseline Mental Health (Mean)	0.76	0.20
	(0.01)	(0.02)
Baseline Employment (Mean)	0.43	7.10
	(0.12)	(0.22)
Baseline Earnings (Mean)	-0.33	3.13
	(0.21)	(0.76)
Age	0.04	0.16
	(0.03)	(0.06)
Age^2	-0.00	-0.00
	(0.00)	(0.00)
Number of Children	-0.01	-0.01
	(0.01)	(0.03)
Fixed Effects		
Survey Wave	✓	✓
Year	✓	✓
Observations	46,681	46,681
Pseudo R^2	0.27	0.69

Notes: This table reports coefficients from a Mundlak-style correlated random effects (CRE) logit model. "Baseline" terms are within-person time averages of time-varying covariates, included to proxy individual permanent heterogeneity. Column (1) models depression at time t and column (2) models employment at time t. Standard errors clustered at the individual level are in parentheses.

C Fertility and the Restriction to Men

The preceding appendix reports each of the paper's empirical exercises for women alongside men. The results divide cleanly. The descriptive evidence generalizes and is often stronger for women: the genetic penalty is larger on every extensive margin (Appendix B.4), every transition gradient replicates (Appendix B.3), and the redundancy design transfers with almost identical coefficients (Appendix B.5). But the recurrence design fails for women (Appendix B.6); the trap-persistence gradient, which is the object the structural model is built around, is a third the size and sensitive to whether family care counts as non-employment; and about thirty percent of women's employment penalty runs into family care rather than into disability.

Those three problems have one common cause, and this appendix sets it out, because it is also the reason we do not simply re-estimate the model of Section 4 on women. Fertility is not a nuisance to be conditioned out of the women's regressions. It is predicted by the same

genetic risk that drives health and labor supply, it is the proximate cause of the labor-supply transitions the trap is meant to describe, and conditioning on it makes the penalty larger rather than smaller. A model of women would therefore have to treat fertility as a choice determined jointly with health and employment, which is a strictly larger object than the one estimated here.

C.1 Genetic Risk Predicts Fertility

Table 29 reports three margins. The timing of a first birth is not detectably affected: the annual hazard of a first birth among childless women moves by -0.18 percentage points per standard deviation of genetic risk on a base of 6.8, a precise null, and the age at first birth is unmoved. Completed fertility is a different matter. By ages 40–45, a standard deviation of genetic risk is associated with 0.09 fewer children in the household (s.e. 0.026) and a 3.2 percentage point higher probability of childlessness (s.e. 1.06), on a base of 26.7 percent. Neither association is present for men, whose corresponding estimates are 0.02 children (s.e. 0.030) and 1.5 points (s.e. 1.32). Genetic risk for depression predicts how many children a woman has and does not predict how many a man has.

Table 29: Genetic Risk for Depression and Fertility

		Women		Men	
Outcome	Mean	Est.	(s.e.)	Est.	(s.e.)
First-birth hazard, childless 20–45 (pp)	6.82	-0.182	(0.309)	-0.229	(0.309)
First birth observed in window (pp)	19.47	-1.156*	(0.695)	-1.148	(0.816)
Children in household, max over window	1.45	-0.009	(0.023)	0.023	(0.025)
Children in household at ages 40–45	1.32	-0.094***	(0.026)	-0.020	(0.030)
Childless at ages 40–45 (pp)	26.70	3.201***	(1.063)	1.520	(1.315)

Notes: Each cell is the coefficient on a one standard deviation increase in the depression polygenic index, from a separate regression. "Mean" is the sample mean of the outcome for women. The first row is a person-wave linear probability model for the annual hazard of a first birth, estimated on those currently childless and aged 20–45, with age and survey-wave fixed effects and standard errors clustered on the individual. The remaining rows are person-level regressions with heteroskedasticity-robust standard errors, controlling for the ages at which the individual enters and leaves the observation window where applicable. Every specification controls for the ten genetic principal components. Fertility is measured by `nchild_dv`, the number of own children in the household, so a transition from zero to one is a proxy for a first birth rather than a birth record, and the ages 40–45 rows are a proxy for completed fertility rather than a measure of it. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

C.2 Why It Cannot Be Conditioned Away

That fertility is itself an outcome of genetic risk is what makes it impossible to handle as a control. Adding the number of children to the women's specifications does not attenuate the penalty; it slightly amplifies it, moving the employment estimate from -4.01 to -4.21 percentage points and the family-care estimate from 1.19 to 1.39. The same is visible as the final rung of the earnings ladder in Table 22, where the children block raises the coefficient by 3.5 percent for women and 0.7 percent for men rather than absorbing any of it.

A confounder is absorbed on conditioning. A variable that is partly a consequence of the treatment need not be, and this one is not. The direction is what one would expect if higher genetic risk lowers completed fertility and lower fertility raises employment: conditioning on children then removes a channel that was partially *offsetting* the penalty, so the remaining estimate grows. Whatever the exact accounting, the implication for modelling is the same in either direction. Fertility cannot be held fixed without conditioning on an outcome.

C.3 The Size of the Interference

Figure 13 shows why this matters for the trap in particular, by applying the paper's own event-study machinery to the arrival of a first child. Women's employment falls by 35.7 percentage points on impact (s.e. 1.0) and is still 15.9 points below baseline four years later. Men's does not move, at -1.8 points (s.e. 0.6). The result is not an artifact of the estimator: the person-fixed-effects form and the Kleven et al. (2019) child-penalty form give -36.1 and -37.6 points for women at impact. Women's employment does rise into the first birth, by about ten points over the two preceding years, which is a familiar feature of that literature and reflects the life-cycle stage at which first births occur; the fall at the event is more than three times that movement, and the men's series provides the within-design control.

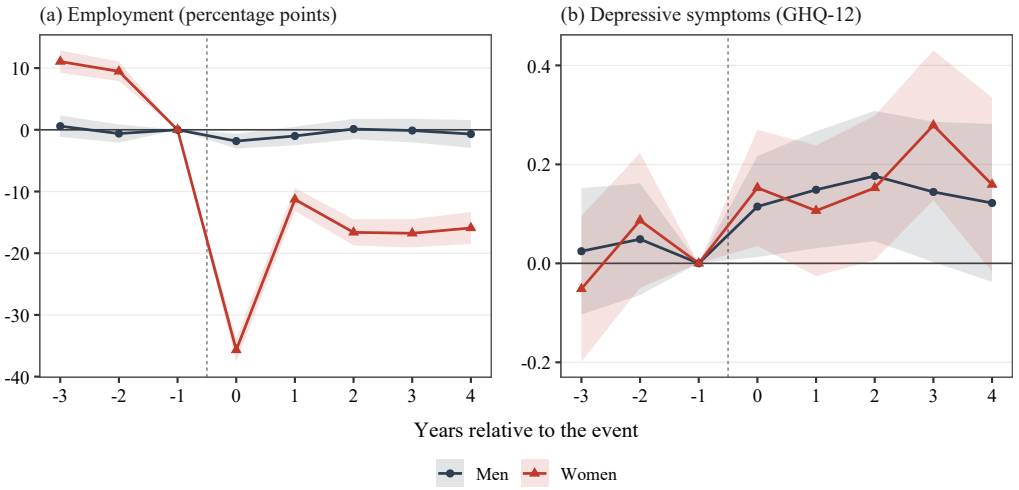
Panel (b) is the part that bears directly on the depression trap. Mental health deteriorates after a first birth for *both* sexes, and by almost the same amount: GHQ-12 caseness rises by 0.15 points for women and 0.12 for men at the event, and the two series are within a tenth of a point of one another throughout. The birth is therefore an event that moves women's employment by thirty-six points and their mental health by about as much as it moves men's.

That combination is exactly what makes the trap hard to read for women. A woman observed non-employed and above the caseness threshold in the years after a birth enters the data in the same cell as a man who has lost his job and become depressed, and the transition tables cannot tell them apart. For the man the joint state is the mechanism the model describes. For the woman a large part of the employment component was produced by the birth, and the modest accompanying rise in symptoms is no larger than the one men experience without leaving work. This is why the women's trap-exit gradient moves by a factor of five depending on whether family-care spells are counted (Appendix B.3), and why we do not read the women's transition estimates as establishing that the trap operates for them as it does for men.

C.4 What Would Be Required

Extending the model to women is therefore not a matter of re-estimating it on a different sample. It would need a fertility decision entering the state space, with the timing and number of births chosen jointly with labor supply and treatment; a home-production or family-care state distinct from both employment and disability, since that is where the genetic penalty sends women; and a reconsidered identification strategy for the mental-health-to-employment leg, since recurrence

Figure 13: Employment and Mental Health Around a First Birth, by Sex



Notes: Event-study coefficients relative to the year before the event ($k = -1$, omitted); bands are 95 percent confidence intervals with standard errors clustered on the individual. Both panels use the interaction-weighted estimator of Sun and Abraham (2021) with controls for age and its square and survey-wave fixed effects, on 3,983 men and 4,281 women aged 20–50 in panel (a). A first birth is dated as the first transition from no own children in the household to one, so it is a proxy for a birth rather than a birth record. As in the redundancy design, identification comes from variation in the timing of the event among those who experience it: people who never have a first child inside the panel are not a counterfactual for those who do, and admitting them as controls produces pre-event coefficients of $+25$ points for women, which is composition rather than a pre-trend. Depressive symptoms are measured by the GHQ-12 caseness score, which runs from 0 to 12, with higher values worse.

timing is not available. Each is feasible and none is a variation on the present model. We regard this as the natural next paper rather than a robustness check on this one.

D Clinical Background: Depression and the Channels to Labor Supply

To understand why depression acts as such a potent barrier to labor supply, it is necessary to examine its clinical manifestations. The diagnostic criteria for Major Depressive Disorder require five or more symptoms over the same two-week period, at least one of them either persistently depressed mood or anhedonia; the remainder cover sleep, guilt, energy, concentration, appetite, psychomotor disturbance and suicidality (American Psychiatric Association, 2022). Our measure is not a diagnostic interview but the GHQ-12, whose twelve items and caseness scoring are set out in Table 30. The correspondence is close enough for our purposes: the items on concentration, sleep, strain, unhappiness, confidence and worthlessness map directly onto the diagnostic domains that the channels below turn on.

Table 30: The GHQ-12 Items and the Caseness Score

	Item: "Have you recently…"	Direction
1	been able to concentrate on whatever you're doing?	Positive
2	lost much sleep over worry?	Negative
3	felt that you were playing a useful part in things?	Positive
4	felt capable of making decisions about things?	Positive
5	felt constantly under strain?	Negative
6	felt you couldn't overcome your difficulties?	Negative
7	been able to enjoy your normal day-to-day activities?	Positive
8	been able to face up to your problems?	Positive
9	been feeling unhappy and depressed?	Negative
10	been losing confidence in yourself?	Negative
11	been thinking of yourself as a worthless person?	Negative
12	been feeling reasonably happy, all things considered?	Positive

Notes: Each item is answered on a four-point scale. Positively worded items run from "better than usual" to "much less than usual"; negatively worded items from "not at all" to "much more than usual". The *caseness* score (`scghq2_dv`) recodes each item to one if the respondent's answer falls in either of the two more distressed categories and zero otherwise, then sums the twelve indicators, giving an integer from 0 to 12. This is the measure used throughout the paper: a threshold of two or more denotes mild depressive symptoms and four or more probable major depressive disorder (Anjara et al., 2020). The alternative Likert scoring (`scghq1_dv`) instead sums the raw responses coded 0–3, giving a score from 0 to 36. The GHQ-12 is a screening instrument for common mental disorder rather than a diagnostic interview: it identifies probable cases, and its caseness threshold has been validated against clinical diagnosis of depression (Wojujutari et al., 2024).

D.1 Mechanisms of Labor Supply Reduction

The symptoms of depression are fundamentally inimical to productive labor, acting through three primary channels: reduced marginal productivity, increased disutility of effort, and the erosion of physical health capital.

Productivity and the wage channel. Depression impairs productivity directly through cognitive dysfunction. Clinical symptoms such as diminished concentration and excessive guilt capture attention and facilitate rumination, distorting an individual's beliefs regarding their own abilities (Ridley et al., 2020; Gotlib and Joormann, 2010). These impairments manifest as "presenteeism", where individuals remain physically present at work but exhibit diminished output per hour (Mall et al., 2015). In a structural framework, this cognitive tax acts as a negative shock to human capital, reducing the wage offer. Furthermore, external barriers such as employer stigma and discrimination exacerbate this channel, as others may underestimate the productivity of individuals with mental health conditions, hindering job retention and wage growth (Ridley et al., 2020).

Disutility and the participation channel. Beyond hourly productivity, depression reduces the overall desire and ability to engage in the labor market. Symptoms like fatigue and anhedonia significantly increase the non-pecuniary cost of effort. This is compounded by a "pessimism about the returns to effort" (De Quidt and Haushofer, 2016), where depressed individuals struggle with delayed gratification and reduced motivation. Job search requires high upfront effort for uncertain future rewards, which makes it particularly taxing in this psychological

state. Functional impairments like sleep disturbances and psychomotor retardation further restrict the ability to adhere to rigid work schedules, often precipitating the transitions into long-term sickness or disability observed in our descriptive analysis.

Physical health spillovers. Finally, depression reduces labor supply through its negative impact on physical health. Clinical evidence suggests that chronic mental distress triggers systemic inflammatory responses (Dantzer et al., 2008), which accelerate physiological ageing and cardiovascular decay (Prince et al., 2007). This "weathering" effect leads to higher rates of chronic physical conditions (Katon, 2011), which in turn impairs productivity and labor supply (Currie and Madrian, 1999).

We incorporate these pathways in our structural model, allowing mental health to affect not only contemporaneous wage offers and the cost of working but also an individual's physical health trajectory. This multi-channel approach is essential for capturing the "depression trap," where initial psychological shocks lead to a self-reinforcing cycle of non-employment and physical-mental co-morbidity.

E Robustness: Within-Family Identification

A central challenge in identifying causal genetic effects in the genoeconomics literature is the correlation between an individual's genotype and their developmental environment. Genetic variants may have a 'direct genetic effect', a causal effect on their mental health. However, 'genetic risk for depression' may instead reflect their environmental upbringing. In the literature, this is referred to as an 'indirect genetic effect' and occurs for two main reasons (Young et al., 2022). First, genetic variants may be more or less concentrated in certain areas of social classes. Norman ancestry remains a strong predictor of wealth in the UK a millennium following the conquest (Clark and Cummins, 2015). While we mitigate this in our primary analysis by controlling for the first 10 principal components of the genome (Price et al., 2006), residual confounding ma persist. Second, 'genetic nurture' may occur as an individual's genotype is inherited from their parents and thus mechanically correlated with their genetic predispositions (Kong et al., 2018). A parent with a high genetic predisposition for depression may affect their child's risk of mental health issues through their interactions or the home environment they provide.

Many of the key conclusions of this paper do not require us to disentangle direct from indirect genetic effects. Even if our genetic associations purely capture indirect genetic effects, genetic variation remains a measurable, ex-ante predictor of depression risk, the persistence of the 'depression trap' and labor market outcomes. It may still be useful as a diagnostic tool in providing targeted mental health care, and doing so may be cost-effective in generating increased labor supply and reducing benefit payments. Nonetheless, it may suggest that rich socioeconomic indicators could be just as effective in targeting mental health interventions, and

may have different implications regarding the design of mental health policy. If our results are closer to direct genetic effects, individuals in the same family may differ significantly in their risk profile and need for mental health care. We therefore aim to assess whether our key results reflect direct genetic effects by exploiting variation within family lineages.

E.1 Identification Strategy: Family vs. Sibling Designs

While family fixed effects in a between-sibling comparison provide the standard "lottery" of alleles within a shared household, the molecular genetic sample in the UKHLS is subject to significant attrition when restricted to genotyped sibling pairs. We identify only 153 siblings observed during working age with valid labor force status. When restricted to male sibling pairs, the resulting sample size is insufficient for meaningful statistical inference.

To maintain statistical power while adhering to causal identification standards, we expand our strategy to include all biological pedigrees. We identify 998 individuals connected by direct parent-child or sibling links, grouped into 538 unique family units. When we filter on all-male pedigrees, we are left with 418 men in 334 families. We identify the genetic effect by including family fixed effects, which purge the estimates of shared dynastic shocks and persistent social background factors. While variation between generations in a pedigree is influenced by assortative mating and environmental shifts, we include birth year fixed effects to capture cohort effects, and the stability of coefficients across these units provides a robust diagnostic for whether genetic nurture is a first-order driver of our results.

In these restrictive models, we transition from the binary measure used in our structural model to a continuous PGI to capture the smaller identifying variation within lineages. Furthermore, as the identifying variation within families is limited, we pool the sample and control for sex-genotype interactions to maximize power while maintaining a focus on the male "genetic penalty" established in Section 3. For employment and labor income, we report within-family results using the all-male family units.

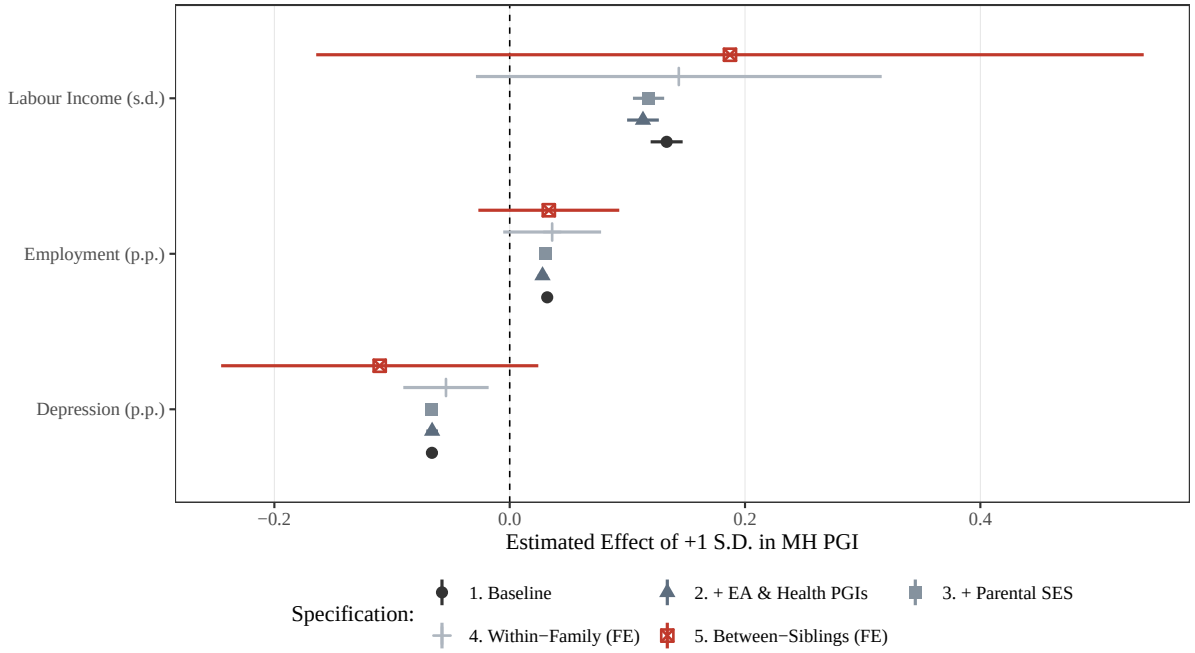
E.2 Results and Coefficient Stability

Figure 14 displays the stability of the point estimates for our primary outcomes: labor income, employment, and clinical depression. To formally evaluate this stability, we perform a Z-test for the equality of coefficients between our socioeconomic mediation models (β_{med}) and these within-family models (β_{WF}).

As shown in Table 31, we fail to reject the hypothesis that the coefficients are statistically equivalent across all outcomes ($p > 0.50$). The point estimates for employment and income are especially stable, which suggests that the genetic penalty is rooted in individual genetic liability rather than purely environmental transmission or dynastic sorting. The point estimates in Figure 14 are also generally quite robust to the inclusion of controls for socioeconomic background or EA or Health PGIs. For our main analyses, we therefore rely on between-family

variation in MH PGIs for identification, which enables more comprehensive mediation analysis and dynamics. Given the robustness across specification, we generally interpret these as direct genetic effects. Okbay et al. (2022) test wellbeing, neuroticism and depression-related PGIs, comparing population-based estimates to within-family estimates with much larger sample sizes, and generally find consistent results.

Figure 14: Stability of MH PGI Estimates Across Specifications



Notes: This figure compares point estimates and 95% confidence intervals for the effect of a one standard deviation increase in the mental health polygenic index (MH PGI) across specifications. The "Socioeconomic Controls" specification includes parental occupation and parental education indicators. The "Within-Family" specification includes family fixed effects, identifying the association within extended family networks. Income is standardized; employment, disability, and depression are in probability units, with depression defined using GHQ-12 caseness (≥ 4). All models control for the first 10 genetic principal components and include survey-wave fixed effects; age enters as a quadratic (age and age^2) rather than as fixed effects, and no calendar-year fixed effects are included. All between-sibling specification controls for sex and sex interacted with the MH PGI, and report the estimated coefficient for men. Within-family estimates are done comparing men for all outcomes except depression, which also includes sex and sex interacted with the MH PGI as a control. Standard errors are clustered at the household level in the socioeconomic-controls specifications and at the family level in the within-family specifications.

Table 31: Tests for Equality of Coefficients: Socioeconomic vs. Pedigree Specifications

Outcome	Socioeconomic Controls (β_{SES})	Family FE (β_{WF})	Z-stat	p-value (Diff)
Employment	0.03	0.04	0.12	0.91
Depression	-0.07	-0.05	0.67	0.50
Income (s.d.)	0.13	0.14	0.14	0.89

Notes: This table tests whether coefficient estimates differ across two specifications: one controlling for socioeconomic characteristics and one using family fixed effects (pedigree specification). The Z-statistic tests the null hypothesis $\beta_{SES} = \beta_{WF}$ for each outcome, and the final column reports the corresponding two-sided p-value.

F Estimation of individual heterogeneity with k-means clustering

Individuals have types x that take values on a discrete set $x \in \{1, ..., K\}$. We estimate these types by building upon the approach of Bonhomme et al. (2022) (BLM) (see also Bonhomme

et al. (2019). BLM estimate individual heterogeneity classes by k-means clustering on relevant moments of outcome variables, with moments calculated at an individual level. Worker have types $x \in \{1, ..., K\}$, there are individuals $i \in \{1, .., n\}$ and $\mathbf{m}_i$ is a vector of M moments of individual i's outcomes. Workers are partitioned into K classes by finding the partition that solves

$$\min_{\{x_i\} \in \{1,...,K\}^N} \sum_{i=1}^{N} \|\mathbf{m}_i - \widetilde{\mathbf{m}}(x_i)\|^2, \tag{27}$$

The consistency of BLM's approach relies on an injective map between individual permanent heterogeneity and the vector of moments m_i. In our application, productivity, mental health and physical health are the primary sources of individual heterogeneity. We therefore compute individual moments m using average wages, mental health status and physical health status.

In aiming to capture individual heterogeneity in productivity, we face an additional identification challenge. For a subset of individuals in our sample, we do not observe realized wages. These are the persistently non-employed, who we do not observe working, and retirees who we do not observe during their working lives. We address this challenge by generating a 'wage potential' (w_i^*) for each individual based on their ex-ante characteristics. We first estimate a log-wage regression on the subsample of employed workers[24]:

$$w_{it} = \gamma_{c(i)} + f(Age_{it}) + \delta_e Educ_i + \delta_o Occ_i + \delta_{pgi} PGI_i + \epsilon_{it} \tag{28}$$

where $\gamma_{c(i)}$ and $f(Age_{it})$ capture cohort fixed effects and a third-degree age polynomial, respectively. $Educ_i$ and Occ_i represent educational attainment and first recorded occupation, while PGI_i includes indices for educational attainment and depressive symptoms. We calculate w_i^* for all individuals by predicting their log-wage at a fixed reference age (40) and anchoring them to a common cohort (1970):

$$w^i = \hat{\gamma}1970 + \hat{f}(40) + \hat{\delta}_e Educ_i + \hat{\delta}oOcc_i + \hat{\delta}pgiPGI_i \tag{29}$$

As we do not want our types to absorb variation attributable to genetic risk for depression, we residualize our vector of moments m_i on the depressive symptoms PGI. As in Jolivet and Postel-Vinay (2025), the data requires us to make an additional modification to the original BLM technique. In the UKHLS, individuals start the data observation period with differing ages, and we observe individuals for differing lengths of time. Estimating Equation 27 would lead to clustering based on cohort and age effects in additional to individual heterogeneity. We resolve this by first residualising physical and mental health on birth-year dummies to remove cohort effects, and then adapting the partitioning to account for individual's ages:

[24] We filter out workers earning less than £1500 per year

$$\min_{\{x_i\}\in\{1,\dots,K\}^N} \sum_{i=1}^{N} \left[\sum_{H\in\{MH,PH\}} \|\bar{m}_{H,i} - g_H(x_i, Age_i)\|^2 + \|w_i^* - \bar{w}(x_i)\|^2 \right] \tag{30}$$

where Age_i is the average age of an individual through the periods observed and $\mathbf{g}(1,\cdot), ...g(K,\cdot)$ are second-degree polynomials in age, aimed at capturing within-class moment differences due to age. The age polynomial is not included for the productivity moment, as it already accounts for age-effects. This approach mirrors Jolivet and Postel-Vinay (2025), who do not include cohort effects as they use the same observation window for all individuals.

F.1 Algorithm for the classification step

We present a simple iterative algorithm based on the one that has been developed by Bonhomme and Manresa (2015), which was in turn adapted by Jolivet and Postel-Vinay (2025) to implement their group fixed effect approach. Since this setting shares many features with theirs, we can use a similar, though slightly modified, algorithm for our classification step. The number of classes is set to $K \geq 1$.

1. Set $s = 0$ and let $\hat{k}_i^{(0)}$ be an initial partition.
2. In each class $k^{(s)}$ with at least two elements, regress by OLS each of the $M-1$ individual moments (excluding w^i) on a constant, T, T^2. Set the $g^{(s)}$ function parameters for this class to the resulting estimates. In each class with only one individual, set the $g^{(s)}$ slope parameters to zero and the constant to the value of the individual's moment.
3. If at least one class is empty:
   (a) Compute the "distance"
   $$\sum_{H\in\{MH,PH\}} \|\bar{m}_{H,i} - g_H(x_i, Age_i)\|^2 + \|w_i^* - \bar{w}(x_i)\|^2$$
   for each observation.
   (b) For each $k \in \{1,\dots,K\}$, if class $k^{(s)}$ is empty, assign to $k^{(s)}$ the observation with the highest distance and reset this distance to 0.
   (c) Do step 2 and replace the g_H function parameters with the new values.
4. Update the partition as follows:
   $$k_i^{(s+1)} = \underset{k=1,\dots,K}{\operatorname{argmin}} \sum_{H\in\{MH,PH\}} \|\bar{m}_{H,i} - g_H(x_i, Age_i)\|^2 + \|w_i^* - \bar{w}(x_i)\|^2$$
5. If no observations changes class at Step 4 then stop. Otherwise, set $s = s+1$ and go to Step 2.

As explained by Jolivet and Postel-Vinay (2025), this algorithm is a variant of one of the first heuristic clustering procedures, referred to as H-means algorithms. More precisely, a H-means algorithm would not include Step 3 and could thus terminate with empty classes. The addition of Step 3 overcomes this issue and leads to what is sometimes labelled a H-means+ algorithm. See (Brusco and Steinley, 2007) for more information about clustering algorithms.

G Model Estimates

This appendix collects the estimates behind Sections 5. Table 33 reports the estimated physical health transition processes, Table 32 the survival probabilities and the parameters taken from outside the model, Table 34 the external calibration of benefits, pensions, taxes and therapy costs, and Table 35 the parameters estimated by simulated method of moments. Figure 15 shows the full set of one-year mental-health transition probabilities, all 45 cells, compared against the raw one-wave transition frequencies in the data.

Table 32: Survival Probability Parameters and Calibration Summary

Panel A: Logit Model (Pr(Death))			Panel B: UK Life Table Calibration		
Variable	Estimate	(S.E.)	Age	Shift (α_t)	Target (f)
ξ_0: Intercept	-28.55	(22.07)	65	0.10	1.25%
ξ_1: MH Good	49.64	(24.09)	70	0.36	2.15%
ξ_2: MH Moderate	27.67	(32.27)	75	0.38	3.46%
ξ_3: PH Good	-16.20	(25.15)	80	0.38	5.98%
ξ_4: PH Moderate	-7.64	(25.48)			
ξ_5: Age	0.68	(0.61)			
ξ_6: Age^2/100	-0.43	(0.42)			
Interactions					
MH Good $\times$ Age	-1.42	(0.66)			
MH Good $\times$ Age^2/100	0.99	(0.45)			

Notes: Panel A reports logit coefficients for the probability of death in the next wave, estimated on the retired UKHLS sample, with standard errors in parentheses. Panel B summarizes the life-table calibration: α_t is an age-specific shift applied to the latent index so that model-implied mortality matches the target rate f at each listed age.

Table 33: Estimated Health Transition Parameters

Parameter	Estimate
Physical Health Transition Process (π)	
$\kappa_{ph,1}$: Threshold 1 (Severe $\rightarrow$ Moderate)	1.18
$\kappa_{ph,2}$: Threshold 2 (Moderate $\rightarrow$ Good)	2.76
$\pi_{1,1}$: Lagged Physical Health (Moderate)	1.10
$\pi_{1,2}$: Lagged Physical Health (Good)	2.18
$\pi_{2,1}$: Lagged Mental Health (Moderate)	0.25
$\pi_{2,2}$: Lagged Mental Health (Good)	0.40
π_3: Employment (n_t)	0.12
$\pi_{4,g}$: Genetic risk (g_i)	-0.34
$\pi_{4,t}$: Age (linear)	-0.00
π_{4,t^2}: Age (squared/100)	0.02
$\pi_{4,x,1}$: Type 1	2.65
$\pi_{4,x,2}$: Type 2	0.86
$\pi_{4,x,3}$: Type 3	1.00
$\pi_{4,x,4}$: Type 4	1.74

Notes: This table reports point estimates from standard (proportional-odds) ordered logit health transition models: each covariate has a single coefficient, common across both thresholds, while $\kappa_{mh,1}, \kappa_{mh,2}$ and $\kappa_{ph,1}, \kappa_{ph,2}$ are the threshold-specific cutpoints. Coefficients enter the latent health indices for mental health (mh^*) and physical health (ph^*); positive values indicate a higher probability of transitioning to a better health state. Health type 5 is the omitted (reference) category for health-type indicators. Coefficient subscripts follow the numbering used in the model equations 4.

Table 34: External Calibration: Benefits, Pensions, Taxes, and Therapy Costs

Panel A: Benefit Schedule ($b(mh, ph)$)

MH State	PH: Good	Moderate	Severe
Good	5,128	6,179	8,589
Moderate	5,651	6,627	9,268
Severe	5,889	8,433	10,126

Values in £ per annum

Panel B: Retirement & Therapy

Variable	Value (£)
Pension (£)	
Type 1	23,438
Type 2	19,437
Type 3	15,067
Type 4	15,980
Type 5	14,951
Therapy Cost	
Unit Cost (£)	97.82
Sessions	8.60
Private Share	0.26
Total (P_τ)	218.63

Panel C: Progressive Income Tax Schedule ($\tau(w)$)

0% rate (personal allowance)	£0–£6,745
20% basic rate	£6,745–£37,400
40% higher rate	$>$ £37,400

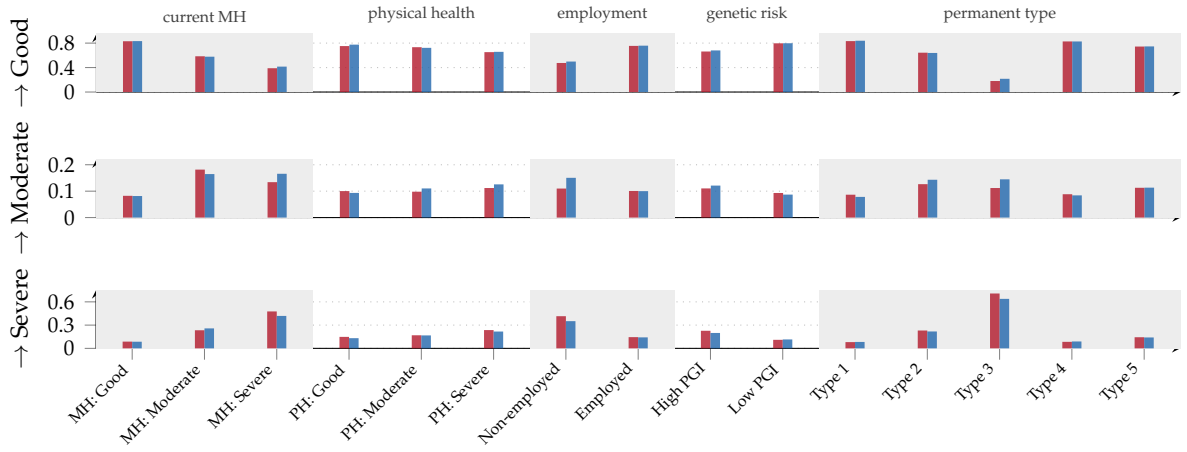
Notes: This table summarizes external inputs used to calibrate the model. Panel A reports mean annual benefit receipt by mental health state (mh) and physical health state (ph). Panel B reports mean annual pension income by permanent type and the components used to calibrate the out-of-pocket therapy cost P_τ. Panel C reports the progressive income tax schedule applied to labor earnings. All pecuniary amounts (benefits, pensions, and therapy costs) are expressed in 2015 £.

Table 35: Estimated Structural Parameters

Parameter	Estimate	Std. error
Preferences		
β: Discount factor	0.9682	(0.0013)
γ: Consumption share of utility	0.5970	(0.0033)
Wage Process		
β_0: Wage intercept	-0.0567	(0.0041)
β_1: Genetic return	0.0983	(0.0074)
$\beta_{2,1}$: Type 1 return	0.4297	(0.0111)
$\beta_{2,2}$: Type 2 return	0.3476	(0.0106)
$\beta_{2,3}$: Type 3 return	0.0000	(0.0186)
$\beta_{2,4}$: Type 4 return	0.0032	(0.0108)
$\beta_{3,1}$: Mental health return (Good)	0.0827	(0.0043)
$\beta_{3,2}$: Mental health return (Moderate)	0.0775	(0.0044)
$\beta_{4,1}$: Physical health return (Good)	0.1310	(0.0049)
$\beta_{4,2}$: Physical health return (Moderate)	0.0697	(0.0052)
β_5: Human capital (linear)	0.0616	(0.0008)
β_6: Human capital (squared)	-0.0544	(0.0022)
ρ: Wage persistence (AR1)	0.8500	(0.0031)
σ_η: Wage shock std. dev.	0.4216	(0.0046)
Time Cost of Moderate and Severe Health		
$\delta_{1,M}$: Time cost (Moderate MH)	0.0250	(0.0018)
$\delta_{1,S}$: Time cost (Severe MH)	0.0623	(0.0027)
$\delta_{2,M}$: Time cost (Moderate PH)	0.0014	(0.0104)
$\delta_{2,S}$: Time cost (Severe PH)	0.0318	(0.0031)
Treatment Costs		
$\psi(n{=}1)$: Therapy utility cost (Employed)	0.0095	(0.0005)
$\psi(n{=}0)$: Therapy utility cost (Non-employed)	0.0001	(0.0001)
Mental Health Transition		
$\mu_{1,G}$: Lagged mental health (Good)	1.4548	(0.0080)
$\mu_{1,M}$: Lagged mental health (Moderate)	0.4506	(0.0079)
μ_2: Therapy received *(calibrated)*	0.6530	—
μ_3: Employment	0.5948	(0.0128)
$\mu_{4,G}$: Lagged physical health (Good)	0.2634	(0.0092)
$\mu_{4,M}$: Lagged physical health (Moderate)	0.1645	(0.0085)
μ_{5,x_1}: Type 1	0.2982	(0.0228)
μ_{5,x_2}: Type 2	-0.6024	(0.0221)
μ_{5,x_3}: Type 3	-2.0544	(0.0160)
μ_{5,x_4}: Type 4	0.2581	(0.0181)
$\mu_{5,g}$: Genetic return	0.5659	(0.0181)
$\mu_{5,t}$: Age	-0.0318	(0.0008)
μ_{5,t^2}: Age (squared)	0.0407	(0.0017)
κ_1: First threshold	-0.4317	(0.0244)
$\kappa_2 - \kappa_1$: Threshold spacing	-0.2271	(0.0006)

Notes: Point estimates with standard errors beside them. The wage coefficients describe log wages. Type and health enter as sets of indicators measured against an omitted category, namely the lowest-paid permanent type and Severe mental and physical health, so a positive value means that group earns more than the omitted group. Moderate or Severe health is costly in time as well as in wages. The time-cost terms give the share of the time endowment it absorbs, measured against Good health, which absorbs none. The treatment parameters are the non-monetary burden of seeking therapy, allowed to differ for people in and out of work. The final panel governs how mental health evolves from one year to the next. A higher value means better mental health ahead, the health and type terms are again measured against an omitted category, and the two thresholds fix where the Good, Moderate and Severe categories begin.

Figure 15: Model Fit: One-Year Mental-Health Transition Probabilities



Notes: One-year mental-health transition probabilities, all 45 cells, grouped by conditioning variable and ordered within group by originating state. Red bars are raw one-wave transition frequencies in the UKHLS estimation sample (men aged 25–64, not retired); blue bars are the corresponding frequencies in the estimated model's simulated panel. Rows are destination states; each vertical pair of bars is one cell. Across the 45 cells the largest absolute error is 0.069, the mean absolute error 0.015, and the correlation between data and model cells 0.997.

H Additional Counterfactual Results

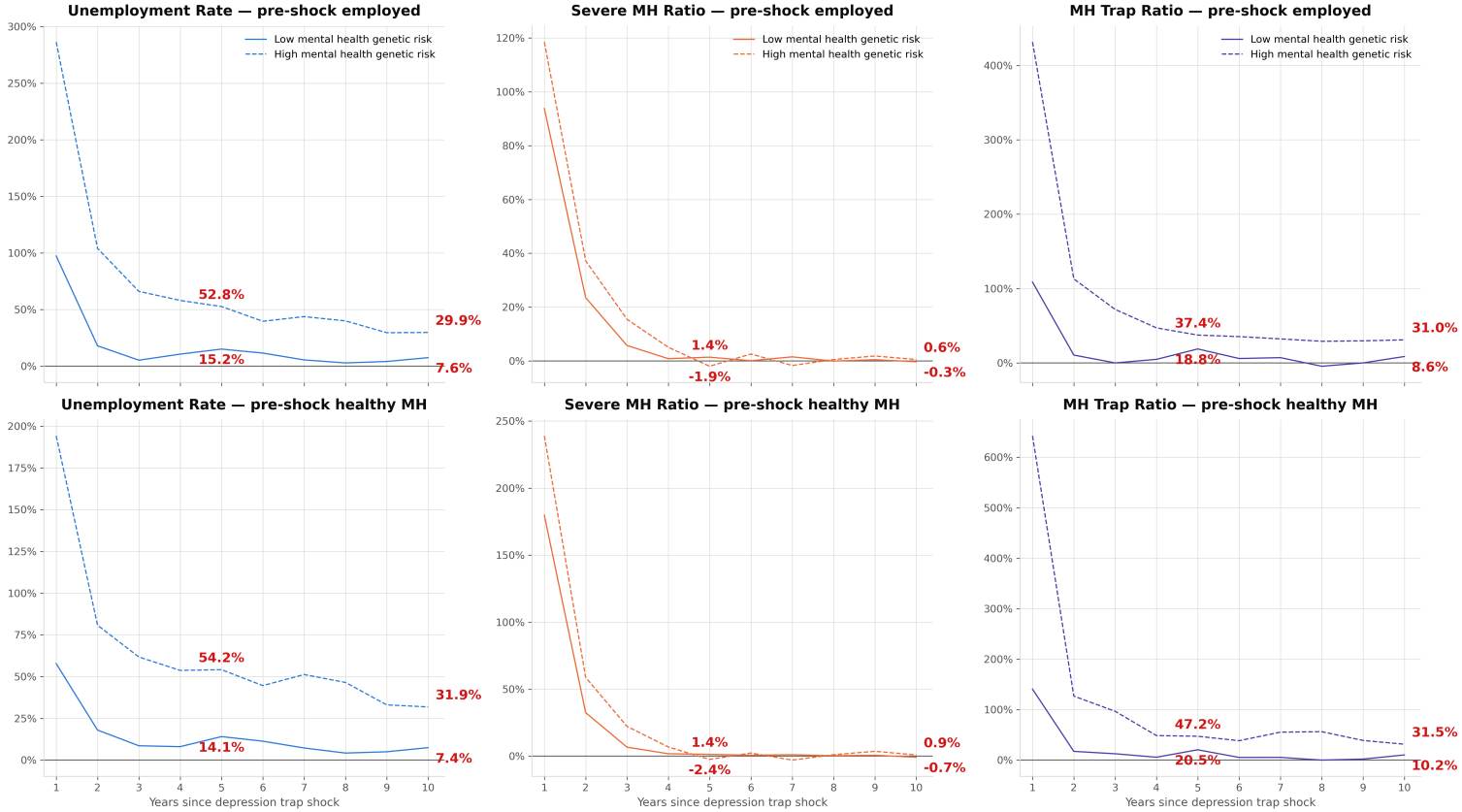
H.1 Recovery from the Depression Trap: Conditioning on the Pre-Shock State

Figure 16 repeats the experiment in Section 6.2 separately for men who were employed before the shock and men who were in Good mental health. These restrictions assess whether the genetic differences in recovery persist conditional on either pre-shock employment or mental health.

The differences persist in both samples. Ten years after the shock, non-employment among men initially employed is 29.9 percent above its benchmark for the high-risk group and 7.6 percent for the low-risk group. Among men initially in Good mental health, the corresponding increases are 31.9 and 7.4 percent. Both samples show a larger difference between the groups' proportional increases than the unrestricted sample, where the corresponding figures are 23.9 and 6.0 percent.

Trap prevalence follows a similar pattern. Among men initially employed, it remains 31.0 percent above benchmark for the high-risk group and 8.6 percent for the low-risk group after ten years; among men initially in Good mental health, the corresponding figures are 31.5 and 10.2 percent. In both samples and genetic groups, the share in Severe mental health returns to within a few percent of its benchmark by year 4 and within one percent by year 10. Employment losses therefore persist after mental health prevalence has returned close to its benchmark.

Figure 16: Recovery from the Depression Trap Conditional on the Pre-Shock State



Notes: Simulations begin with men first observed at ages 25–45. All are placed in Severe mental health and non-employment in year 1; subsequent choices and health transitions follow the estimated model for ten years. The top row restricts the sample to men employed before the shock; the bottom row separately restricts it to men in Good mental health. Values are percentage deviations from each genetic group's benchmark within the same restricted sample. The depression trap is joint Severe mental health and non-employment. High genetic risk denotes the lower polygenic-index group. Labels report years 5 and 10.

H.2 Priority Access to Therapy: Additional Detail

Solving for the market-clearing allocation

With treatment capacity fixed, offer probabilities must satisfy both the waiting-time priority rule and the aggregate capacity constraint. Let s_g denote the population share of genetic-risk group g and a_g its equilibrium application rate. Substituting the priority rule in Equation (21), the capacity constraint is

$$s_L a_L \lambda(L) + s_H a_H \frac{\lambda(L)}{1 - X + X\lambda(L)} = \bar{u}, \tag{31}$$

where $\bar{u}$ is the benchmark number of treatment courses per working-age person-year. Application rates are endogenous: they reflect individuals' decisions under the offer probabilities assigned to their genetic-risk group.

For each priority intensity X, the priority rule determines $\lambda(H)$ from $\lambda(L)$, leaving only one probability to choose. We adjust $\lambda(L)$ and update the simulated application rates under the corresponding pair of offer probabilities until total treatment provision matches the benchmark. This procedure holds capacity fixed while allowing individuals to respond to changes in access.

Fiscal accounting

We measure the fiscal balance per person in present value, discounting all flows to $t = 1$ at the model's interest rate, $r = 0.046$:

$$\text{NPV} = \sum_t \frac{\tau_t - b_t - c_t}{(1+r)^{t-1}}, \tag{32}$$

where τ_t is tax paid on gross labor earnings under the schedule reported in Table 34, b_t is benefit received, and c_t is the public cost of any therapy delivered in period t. Following the convention used elsewhere in the paper, c_t is the market price of a course less the patient's out-of-pocket contribution: a course costs $8.6 \times £97.8 = £841$ at 2015 prices, against out-of-pocket contributions of £353 for the employed and nil for the non-employed, so the state pays £488 and £841 respectively. Tax, benefits and provision are all discounted on the same schedule and over the same working-age window. Pensions are excluded: they depend only on permanent type and are therefore identical across scenarios, cancelling in every difference reported. Under fixed capacity total treatments are unchanged, so the provision term very nearly cancels; it is nonetheless computed from the simulated panels rather than assumed away, because the employed/non-employed composition of who is treated shifts and the two carry different unit costs.

Simulation detail

Every scenario draws 10,000 individuals with replacement from the initial distribution, weighted by the survey weight, with a common random seed across all scenarios so that comparisons are paired on identical agents. Each draw carries the individual's full entry bundle jointly, so that genetic risk, permanent type, entry age, mental and physical health, assets and human capital keep their empirical joint distribution; the persistent wage component is drawn independently from the stationary distribution of its AR(1). Value functions are re-solved for every parameter configuration, including at each iteration of the market-clearing loop.

H.3 Recovering Genetic Risk from Observables

Section 6.4 asks whether an individual's observed history is a sufficient statistic for the genetic risk on which the priority rule conditions. Table 36 reports the estimates behind the answer given there.

The predictor is a histogram gradient-boosting classifier trained on all observables available by the stated age: the number of observed years and the entry age; the employment rate, current employment status and number of employment transitions; therapy application and receipt rates and whether ever treated; shares of time in each mental-health state, the current state and the number of transitions into severe depression; the share of time in poor physical health and the current state; and the mean, most recent value and standard deviation of log earnings. The target is the binary depression polygenic-index group, whose population share is 0.47 to 0.48 across ages. All figures are five-fold cross-validated, with each individual contributing one observation, so the folds partition people and no individual appears in both training and test data.

Table 36: Recovering Genetic Risk from Observables: Simulated Panel

Age	AUC	(s.d. across folds)	Average precision
30	0.974	(0.013)	0.968
35	0.993	(0.003)	0.992
40	0.994	(0.002)	0.992
45	0.993	(0.003)	0.990
50	0.994	(0.001)	0.993
55	0.994	(0.002)	0.991
60	0.992	(0.001)	0.992

Notes: Simulated panel of 10,000 agents. A histogram gradient-boosting classifier predicts the binary depression polygenic-index group from the observables available by the stated age, listed in the text. All figures are five-fold cross-validated; each individual contributes one observation, so the folds partition people and no individual appears in both training and test data. The standard deviation is taken across folds. An individual enters the sample once observed at the evaluation age with at least three prior observations, so the number of individuals rises with age, from 862 at 30 to 9,198 at 60. The population share of the target group is 0.47 to 0.48 across ages, so a classifier with no information would score an AUC of 0.5 and an average precision of about 0.48.

An individual enters the sample once observed at the evaluation age with at least three prior observations, so the number of individuals rises with age, from 862 at age 30 to 9,198 at age 60;

the wider dispersion across folds at age 30 reflects that smaller sample. As Section 6.4 notes, the exercise speaks to the model and not to the data: genetic risk enters the simulation through the wage equation and the mental-health transition process alone, so a long history identifies it far more sharply than it could in a population where genetic risk competes with everything the model leaves out.

H.4 The Labour Margin: Subsidising the Mental Health Wage Penalty

The estimated model attributes a substantial wage penalty to mental health problems: the coefficient β_3 in Section 4 implies a penalty of 8.3 log points for Severe relative to Good mental health. The model does not distinguish whether this penalty reflects lower productivity or employer behaviour, but that distinction matters for its policy interpretation. Experimental evidence that individuals avoid coworkers with depressive or anxiety symptoms even when those coworkers perform equally well on the task provides a motivation for considering discrimination (Ridley, 2025). We therefore evaluate a theoretical subsidy that pays the difference between a worker's offered wage and the wage he would receive in Good mental health, holding other wage determinants fixed. If the penalty reflects discrimination, the subsidy offsets a distortion in the financial return to work. If it reflects lower productivity, the subsidy compensates the worker without restoring that productivity.

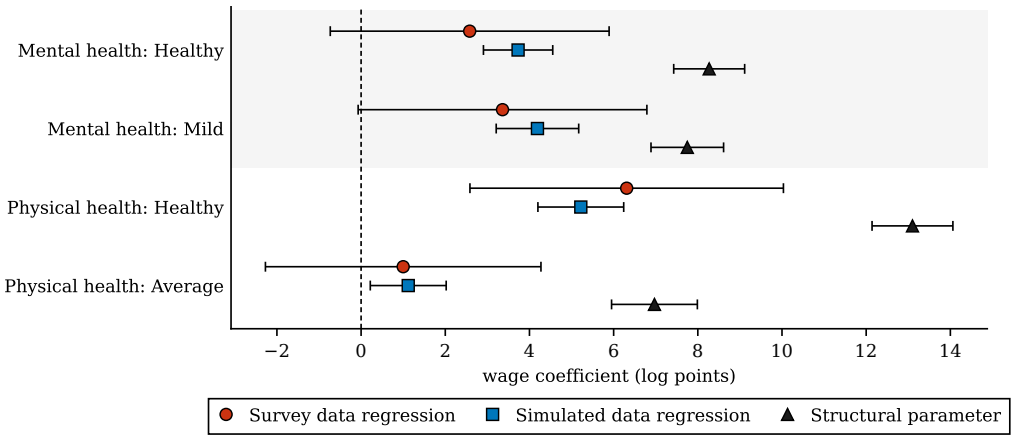
Before evaluating the subsidy, we examine what wage regressions reveal about the underlying penalty. Figure 17 compares three sets of coefficients: estimates from the survey data, estimates from the same regression applied to simulated data, and the structural parameters used to generate those data. The survey estimate of the mental health wage difference is small and cannot be distinguished from zero at conventional significance levels. The model reproduces these small regression coefficients closely.[25] Yet the structural effects underlying the simulated data are two to seven times as large. For mental health, the structural difference is 8.3 log points, compared with 2.6 log points in the survey regression.

Within the model, selection into employment explains this difference. Wages are observed only for individuals who work, and those whose earnings prospects are most adversely affected by mental health are disproportionately likely to leave employment. The wage regression therefore conditions on a decision that itself responds to the health penalty. A larger sample would estimate this conditional association more precisely but would not, by itself, remove the selection. Two implications follow. First, a small wage coefficient among employed individuals need not imply a small underlying wage penalty. Second, a subsidy calibrated to the observed wage difference could offset substantially less than the structural penalty in the model. The

[25] This agreement is partly by construction: estimation constrains all five simulated regression coefficients to lie within the corresponding survey confidence intervals. The comparison therefore illustrates how these regression coefficients can coexist with larger structural wage effects; their agreement is not an independent validation of the model.

comparison thus shows why the wage associations observed among workers can differ from the wage effects relevant to employment decisions and policy.

Figure 17: Health and Wages: Regression Estimates and Structural Effects



Notes: Circles show survey regression coefficients, squares show estimates from the same regression in simulated data, and triangles show the structural coefficients. Bars indicate 95 percent intervals. Both regressions use employed working-age person-years and regress log wages on Good and Moderate mental and physical health, with Severe omitted in each dimension, and a low-genetic-risk indicator. Controls include four permanent-type indicators and linear and quadratic human capital; individual fixed effects are not included. The comparison illustrates how small wage associations among employed individuals can coexist with larger structural health effects under employment selection.

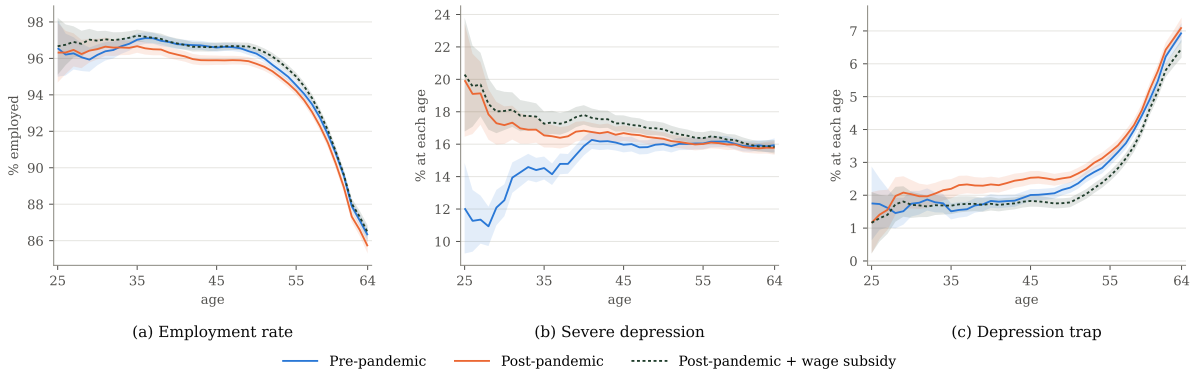
Figure 18 reports the subsidy experiment. Employment rises by 0.73 percentage points, more than offsetting the model-implied employment loss associated with the post-pandemic mental health shift, and the prevalence of the depression trap falls. The mental health response, however, moves in the opposite direction: therapy use falls sharply and Severe mental health becomes more prevalent. The mechanism operates through the psychological cost of seeking therapy, $\psi(n)$, which is much higher for employed than for non-employed individuals. By drawing more individuals into work, the subsidy also moves them into the state in which seeking treatment is more costly. In this experiment, the reduction in therapy use more than offsets the mental health benefits of greater employment.

Higher tax revenue, lower benefit payments, and lower expenditure on therapy recover approximately 60 percent of the subsidy cost. The subsidy therefore produces a larger employment gain than the other post-pandemic experiments considered here, but is not self-financing. Its net fiscal cost accompanies a further trade-off: stronger employment incentives reduce treatment use and worsen mental health in the model.

H.5 The Mental Health Margin: Reducing the Psychological Cost of Therapy

The employment difference in $\psi(n)$ that limits the subsidy's mental health benefits also points to an important determinant of therapy demand. The estimated psychological cost of applying is approximately 75 times higher for employed than for non-employed individuals. This difference reflects the model's explanation for the low treatment take-up among employed men with

Figure 18: Subsidising the Mental Health Wage Penalty



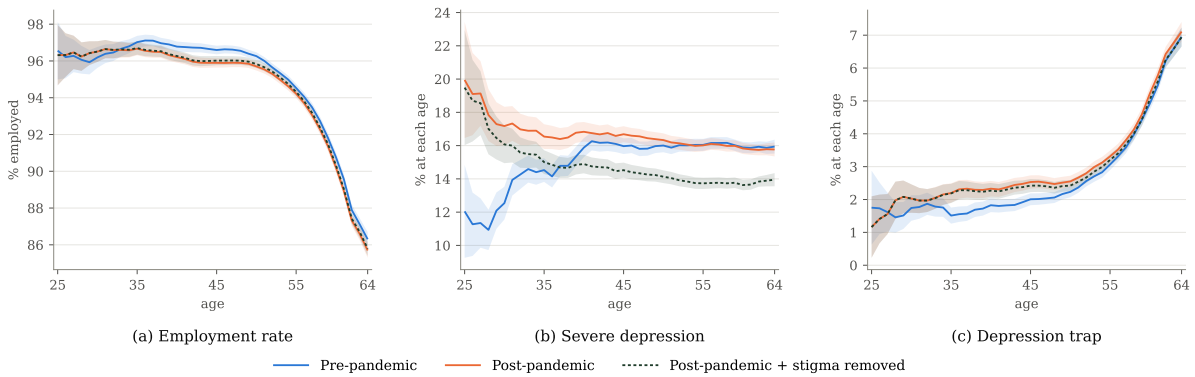
(a) Employment rate (b) Severe depression (c) Depression trap

Notes: Simulated age profiles of employment, Severe mental health, and the depression trap for working-age men. The series compare the pre-pandemic benchmark, the post-pandemic economy with the initial mental health distribution shifted, and the post-pandemic economy with the subsidy. The government pays the difference between each worker's offered wage and his wage in Good mental health, holding other wage determinants fixed. The depression trap is joint Severe mental health and non-employment. Lines are centred five-year moving averages with 95 percent confidence bands based on observations within each window. The subsidy raises employment and reduces trap prevalence, but increases Severe mental health.

Severe mental health; it is not a technological feature of therapy. A natural interpretation is the cost of disclosing a need for mental health treatment at work. The parameter can also capture other psychological barriers, however, so the estimated difference does not separately identify workplace stigma.

We examine this mechanism by progressively reducing the employed cost to 75, 50, and 25 percent of its estimated value, holding the non-employed cost fixed. A final experiment removes the psychological cost for both groups, setting $\psi(1) = \psi(0) = 0$. Figure 19 reports this final experiment. The share in Severe mental health falls by 2.062 percentage points, the largest reduction among the post-pandemic experiments considered here. From approximately age 40 onward, Severe mental health is less prevalent than in the pre-pandemic benchmark.

Figure 19: Removing the Psychological Cost of Seeking Therapy



(a) Employment rate (b) Severe depression (c) Depression trap

Notes: Simulated age profiles of employment, Severe mental health, and the depression trap for working-age men. The series compare the pre-pandemic benchmark, the post-pandemic economy with the initial mental health distribution shifted, and the post-pandemic economy with psychological therapy costs removed for both employed and non-employed individuals, $\psi(1) = \psi(0) = 0$. The depression trap is joint Severe mental health and non-employment. Lines are centred five-year moving averages with 95 percent confidence bands based on observations within each window. Removing these costs reduces Severe mental health below the pre-pandemic profile from approximately age 40 onward.

Table 37 reports the full sequence of reductions. The fiscal return per pound of additional therapy spending is highest when the employed psychological cost is halved. Halving this cost generates £1.20 in additional tax revenue and benefit savings per pound spent, while reducing it by three quarters generates £1.16. Removing both groups' costs entirely generates £0.96 per pound spent. Thus, partial reductions generate enough fiscal gains to cover the additional therapy expenditure, whereas complete removal falls slightly short despite delivering the largest mental health improvement.[26] Beyond the 50 percent reduction, the declining fiscal return is consistent with diminishing economic gains from the additional treatment delivered, while provision costs per course remain fixed. The highest fiscal return per pound therefore occurs at a partial reduction in psychological costs; this criterion does not determine the reduction that would maximise welfare.

Table 37: Reducing the Psychological Cost of Seeking Therapy

Disutility of taking therapy retained	Severe mental health	Employment rate	Courses per person	Additional therapists	Fiscal return per £1
100% (benchmark)	16.196	93.023	0.35	—	—
75%	−0.448	+0.008	0.61	7,400	1.14
50%	−0.963	+0.022	0.96	13,400	1.20
25%	−1.368	+0.046	1.32	17,800	1.16
0%	−2.062	+0.104	2.44	30,600	0.96

Notes: Experiments apply to the post-pandemic economy. The employed psychological cost is reduced to the stated fraction of its estimated value. The non-employed cost remains fixed except in the 0 percent row, which sets both costs to zero. For Severe mental health and employment, the benchmark reports percentages of working-age person-years; other rows report percentage-point changes. Courses per person are totals over working life. Additional therapists meet the extra treatment demand at the assumed caseload. Fiscal return is additional tax revenue plus benefit savings divided by additional therapy provision costs. Ratios above one cover those provision costs; the cost of reducing psychological barriers is not included.

The capacity requirements connect these experiments to the supply expansion in Section 7.3. Halving the employed psychological cost requires approximately 13,400 additional therapists, while removing both groups' costs requires 30,600. These requirements are substantial relative to the approximately 15,000 additional psychological therapists and practitioners included in the NHS workforce training commitment (NHS England, 2023), although that commitment covers a broader range of services. The supply experiment shows that expanding capacity can generate substantial health gains and cover its fiscal cost at moderate expansions. The present experiments show that reducing barriers to seeking treatment increases the capacity required to meet demand. Access and willingness to seek treatment therefore need to be considered together: achieving the projected gains from lower psychological costs requires enough therapists to accommodate the resulting increase in treatment use.

[26] These comparisons concern the cost of providing additional therapy. The experiments do not specify or cost an intervention that would reduce the psychological cost parameter itself.